
SPECIAL SENSES

Special Senses

1. **Vision**
2. **Hearing**
3. **Equilibrium**
4. **Olfaction**
5. **Taste**

<u>Modality</u>	<u>Stimulus</u>	<u>Receptor</u>	<u>Types of Receptors</u>
1. Vision	Light	Photoreceptor	Rods, Cones
2. Audition	Sound	Mechanoreceptor	Hair cells (Cochlear)
3. Balance	Head Motion	Mechanoreceptor (semicircular canals)	Hair cells
4. Taste	Chemical	Chemoreceptor	Taste Buds
5. Smell	Chemical	Chemoreceptor	Olfactory sensory N

PHYSIOLOGY OF VISION

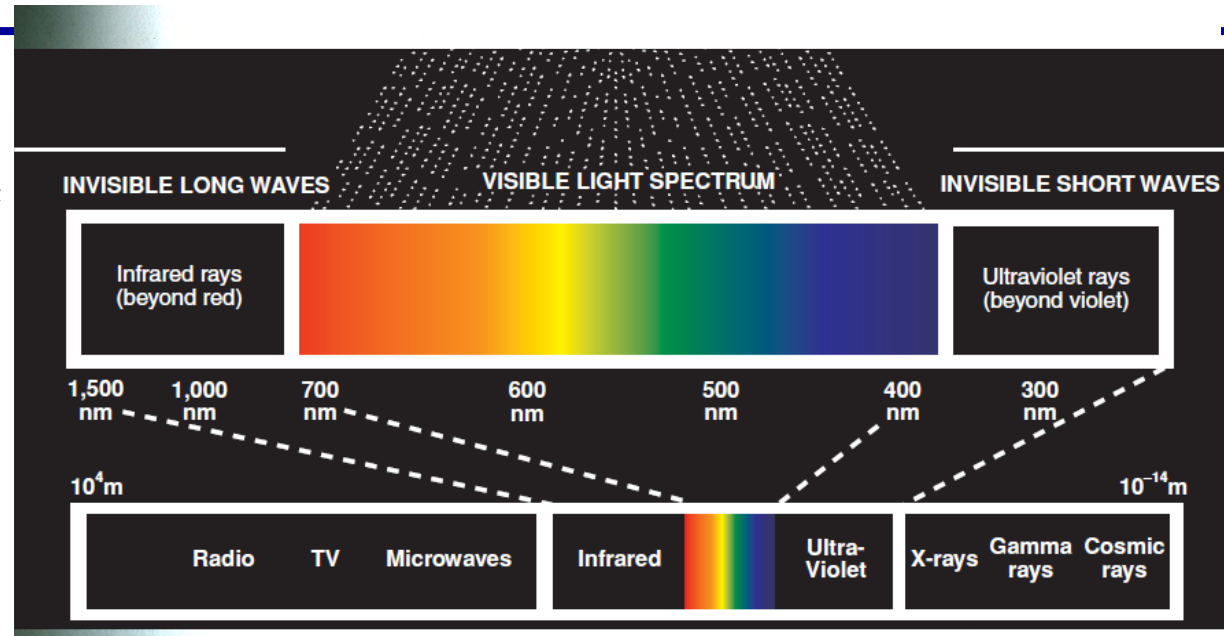
Physiological Significance

A mental image of the external world is created by the visual system.

- An image is formed on the retina by the **refractive surfaces** of the eye.
- The light energy is transduced into the an electrical signal by the **rods and the cones**.
- The information needed to create the mental image is encoded by the **neurons within the retina**.
- The information is used by the **visual cortex** to create the visual perception described as “seeing”

Electromagnetic spectrum

- Electromagnetic waves have
 - **Frequency**: Number of waves per second
 - **Wavelength**: Distance between successive peaks and troughs
- Light is the visible portion of the electromagnetic radiation spectrum- VIBGYOR

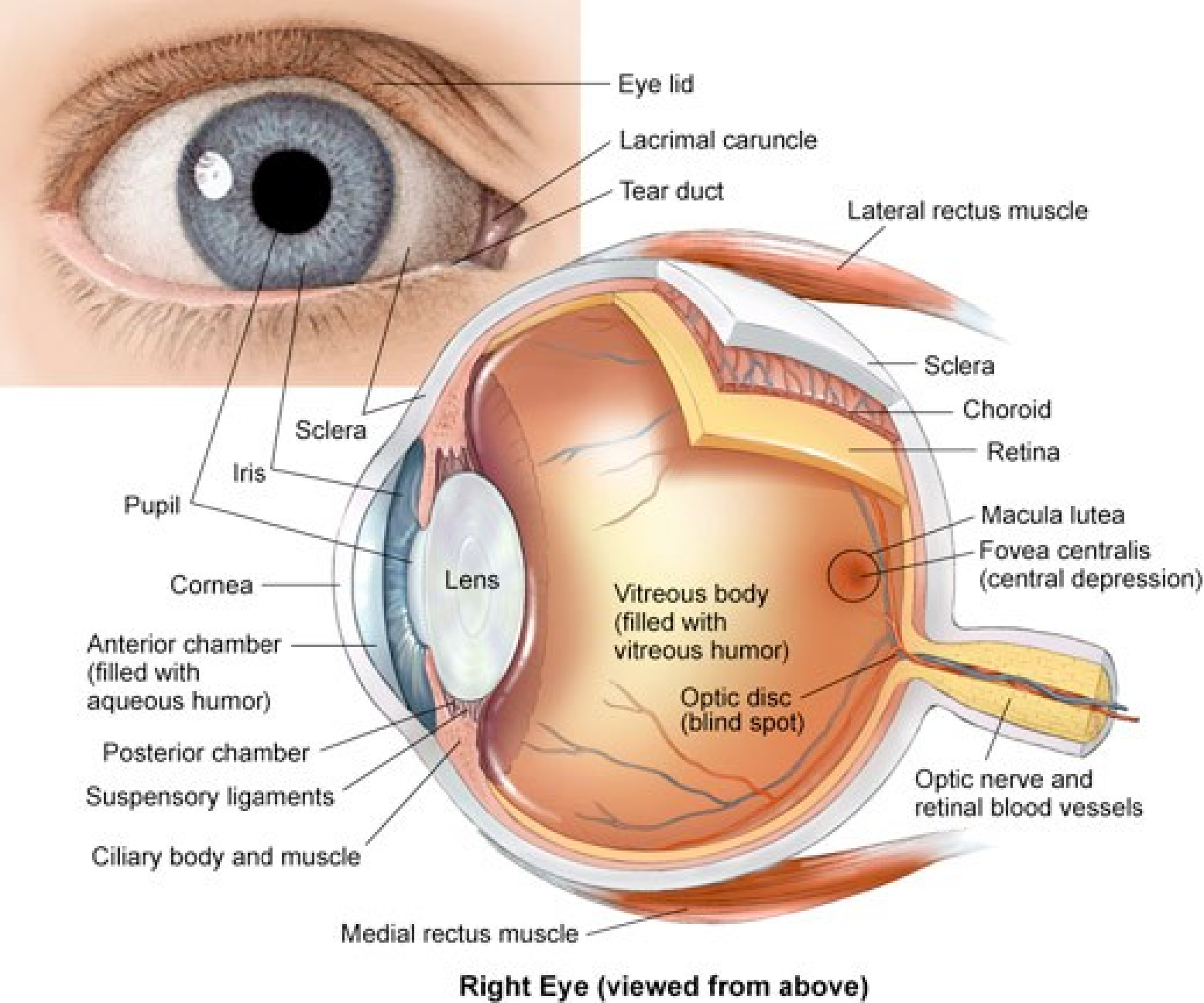


High energy gamma waves: $\lambda < 1 \text{ nm}$

Low energy radio waves: $\lambda > 1 \text{ mm}$

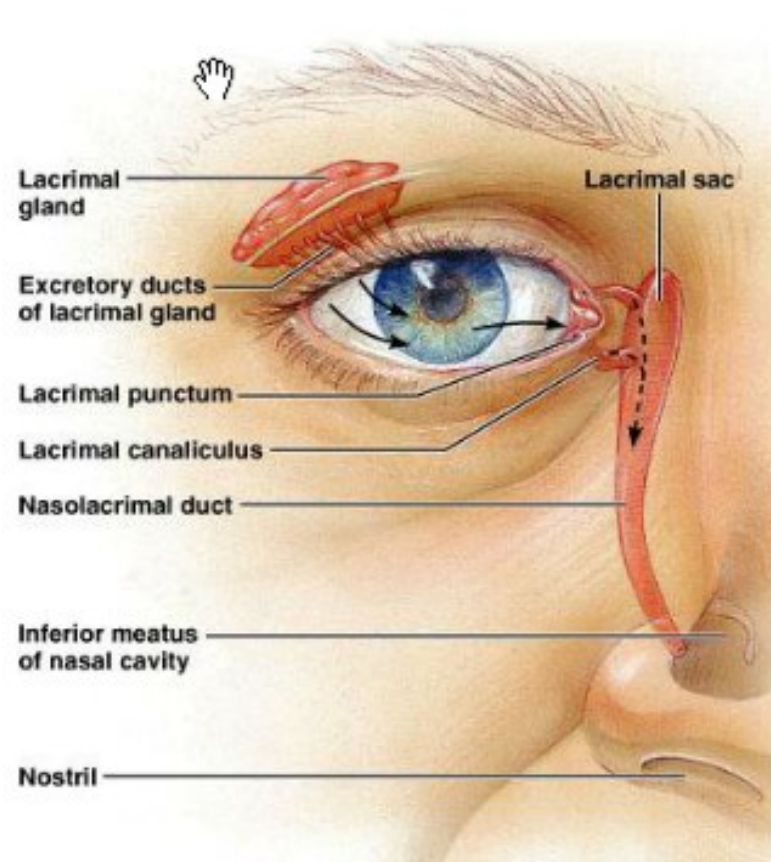
Amplitude: difference

Waves visible to the human eyes are **400-700nm:visible light spectrum**



- 70% of all sensory receptors are in the eyes
- 40% of the cerebral cortex is involved in processing visual information
- Each eye has over a million nerve fibers

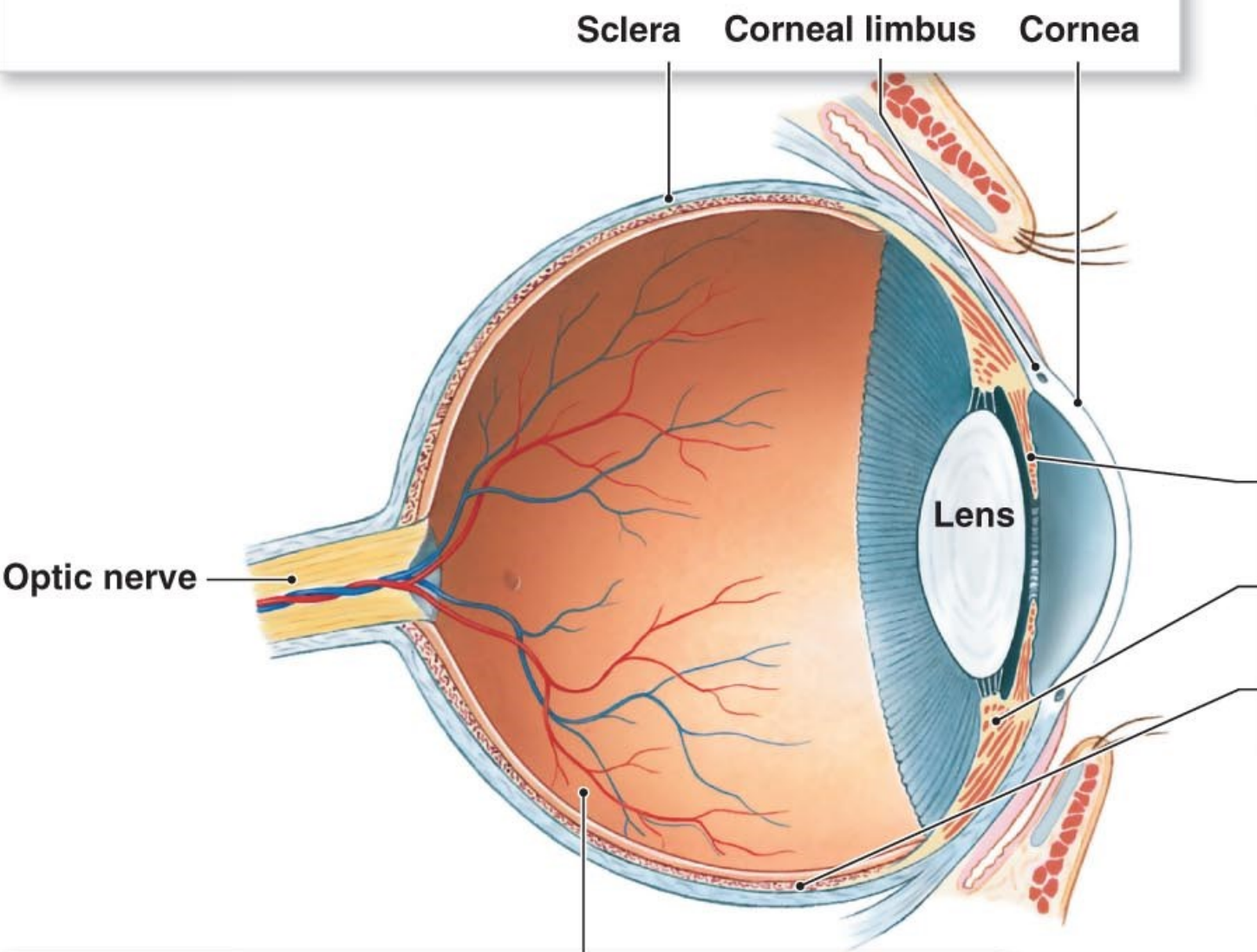
- Lacrimal gland produces the tears
- Tears contain mucous antibodies (anti-bacterial)
- Keeps the surface of the eye moist
- Lacrimal sac empties fluid into nasal cavity



The three tunics of the wall of the eye

Fibrous Tunic

The outermost layer of the eye, consisting of the cornea and the sclera



Vascular Tunic

The middle layer of the eye; also called the uvea; contains numerous blood vessels, lymphatic vessels, and the intrinsic (smooth) muscles of the eye /Ciliary body/iris

Iris

Ciliary body

Choroid

Neural Tunic

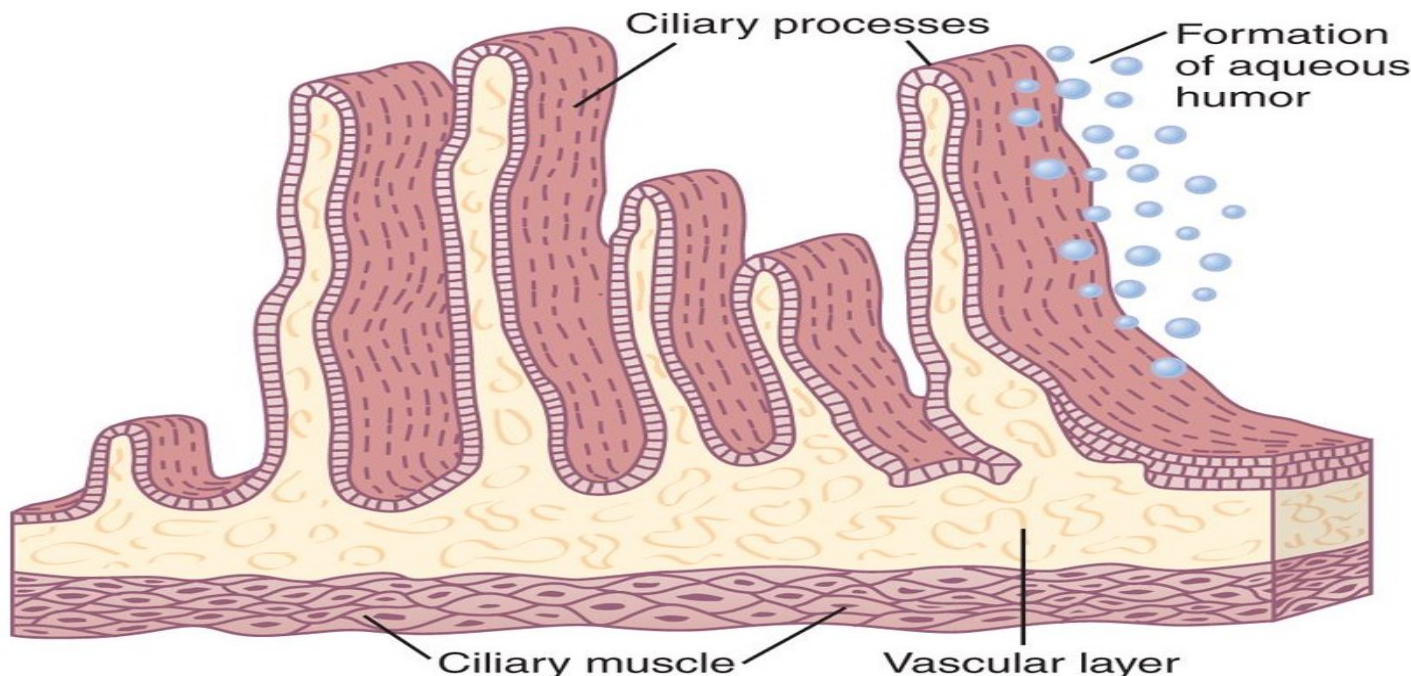
The innermost layer of the eye; also known as the retina; contains photoreceptors

Cornea

- Contains **dense connective tissue** (collagens fibers)
- **Avascular**: obtaining oxygen from air & glucose from aqueous humor
- Supplied by **sensory 'V' nerve** which rapidly regenerate if damaged
- **Functions**
 - Part of the **refractive power** of the eye
 - Produce protective **corneal reflex**
 - Absorb **UV light** to protect the retina
 - Its **transparency** contribute to sharp and **clear image**

Transparency

- the absence of blood vessels in the cornea allows light to pass through unhindered, but because it has **no blood vessels**, the cornea must receive oxygen and nutrients via diffusion through the aqueous humor.
- The aqueous humor is continuously secreted into the eye from the capillaries of the ciliary body, passes thru the pupil (~ **plasma**; **nourishes the cornea and the lens**)



Vitreous humor - clear gel that fills the space between the lens and the retina of the eyeball; 80% of the volume of the eyeball

- Increased formation of aqueous humor or blockage of the outflow can cause $\uparrow$ intraocular pressure above its normal value of *10-20mmHg*.
- Increased IOP is called *glaucoma* and can lead to blindness if not treated (IOP= Intra ocular pressure).
- High IOP above 20 $\rightarrow$ compression of optic nerve/ganglionic cells $\rightarrow$ lack of nutrition $\rightarrow$ death of neurons $\rightarrow$ blindness



The Lens

Zones of the lens

- Young *outer surface* contain *more water* and *less protein* than the compact inner layer
- In young, it is malleable (*accommodation*)
- Like the cornea, is *avascular and transparent*.

Lens formula

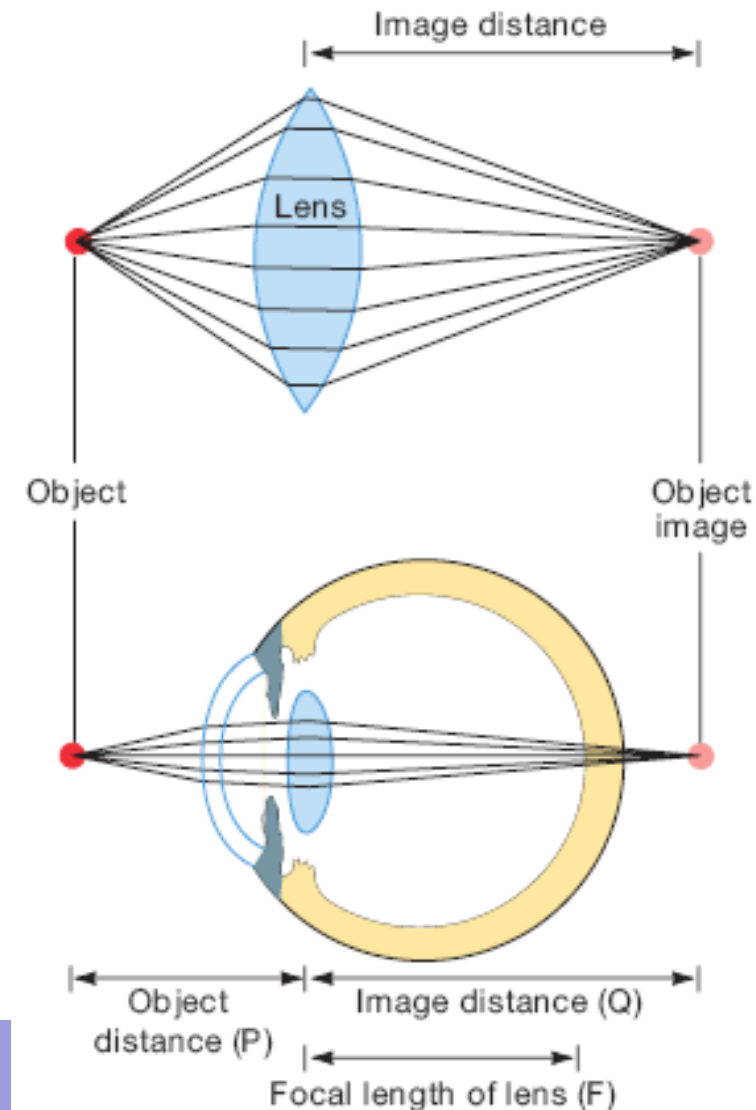
- Principles of Image formation
 - Convex (converging lenses) form real images of illuminated objects.
- The lens formula.

$$P = \frac{1}{o} + \frac{1}{i} = \frac{1}{f}$$

Where:

o, i, and f are the object, image and focal distances, respectively (in m)
P is the power of the lens in diopters.

Converging power (D) of the lens is the reciprocal of the focal distance (m)



Diverging lenses can not form an image.

-
- The cornea and lens provide the converging power of the eye (59.9 D)
 - The cornea is responsible for approximately *two-thirds* of the refractive power of the eye, and its refractive power can not be altered physiologically $\sim +53\text{D}$.
 - Anterior cornea = $+46\text{ D}$
 - Posterior cornea = $+7\text{D}$

-
- Cornea contribute 1/3 of the refractory power of the eye (20D)

because it is surrounded by a medium of similar RI

Anterior lens= - 6D

Posterior lens= +13D

Reduced eye: When all the refractive surfaces of the eyes are added up and considered as a single lens

Such a lens has focal point of 17 mm in front of the retina (~ 59 D)

However its convexity can change its power to 34D (Accommodation)

Refractive power - unlike the cornea the refractive power of the lens is under physiologic control.

Adolescent accommodate up to 14 D (7cm)

Old age only by 4 D (25 cm)

Normal reading distance 35 cm

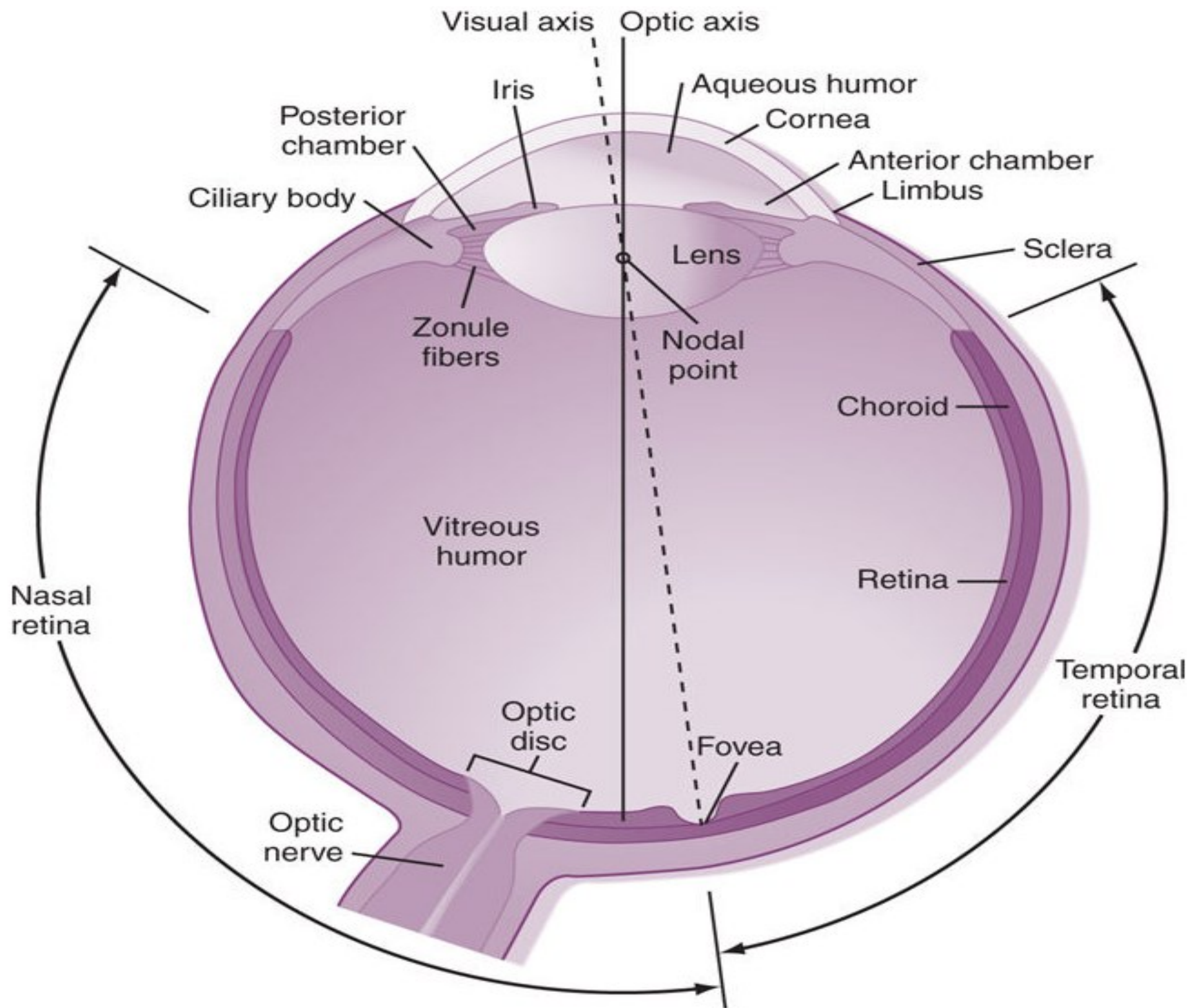
Visual axis:

- Axis along which the light is directed
- Is a line that passes through *fovea, nodal point, and fixation point*

Optical axis:

- Axis of optical symmetry
- Line the passes through the *center of cornea, pupil and lens*

Angle α :- angle between optical and visual axis



Transparency of cornea

- Regular and **uniform arrangement** of cells & fibers
- Absence of **blood vessels and myelinated nerve**
- Relative **corneal dehydration**
 - Water is imbibed by corneal tissue and expelled by **osmotic** and **metabolic pump** into the aqueous humor
- Maintenance of transparency
 - **Vitamin A** - maintains healthy epithelial cells
 - **Vitamin B2**- for formation of coenzymes for tissue respiration



Corneal reflex: Contact with cornea → **V** → **Superiour colliculi** → **VII**
→ **Orbiularis oculi**

Transparency/lens

- As an individual ages, the lens develops opacities called **cataracts** (*Proteins get denatured and coagulate* reducing transparency)
 - *Normal vision can be restored by surgically removing the opaque lens and replacing it with a plastic lens.*

Focusing of Image

- In a normal eye, an image of a distant object is focused on the retina mainly by the **cornea and lens**.

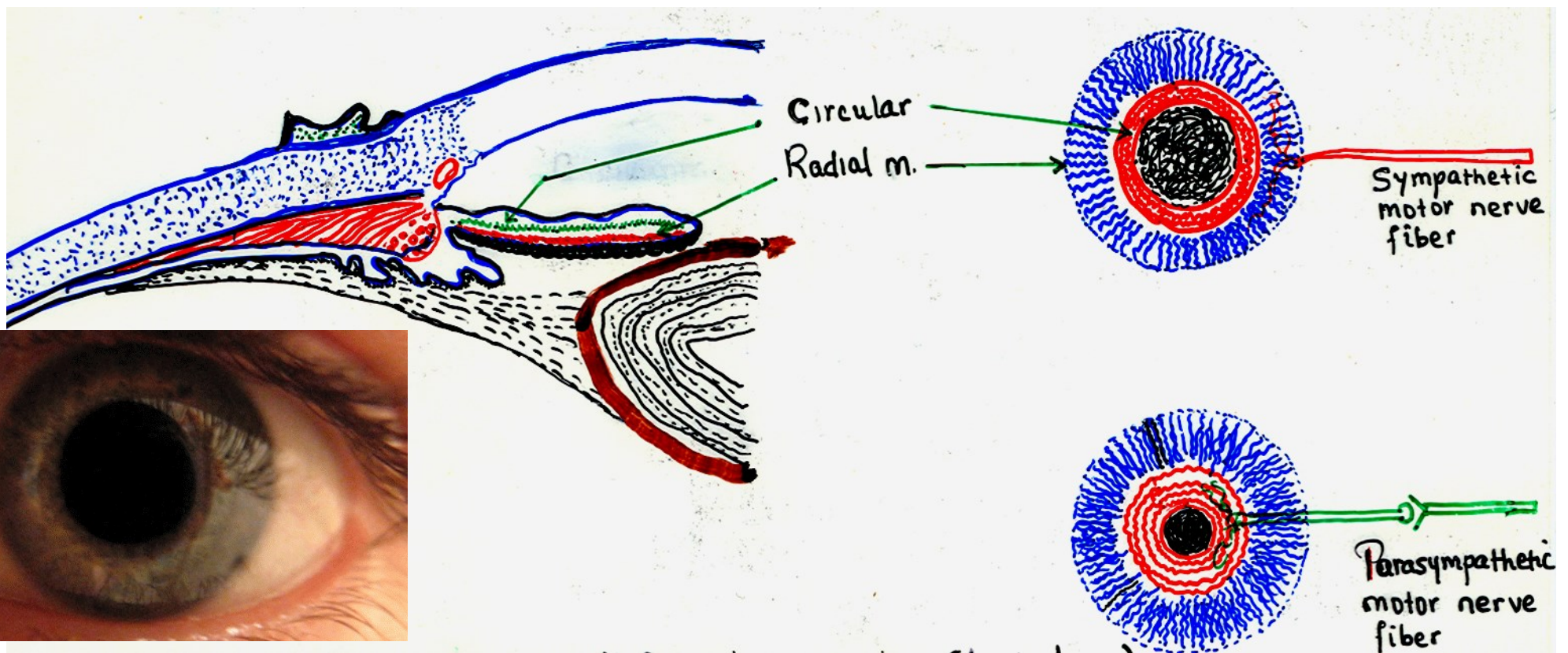
Iris and Pupil

Pupillary light reflex

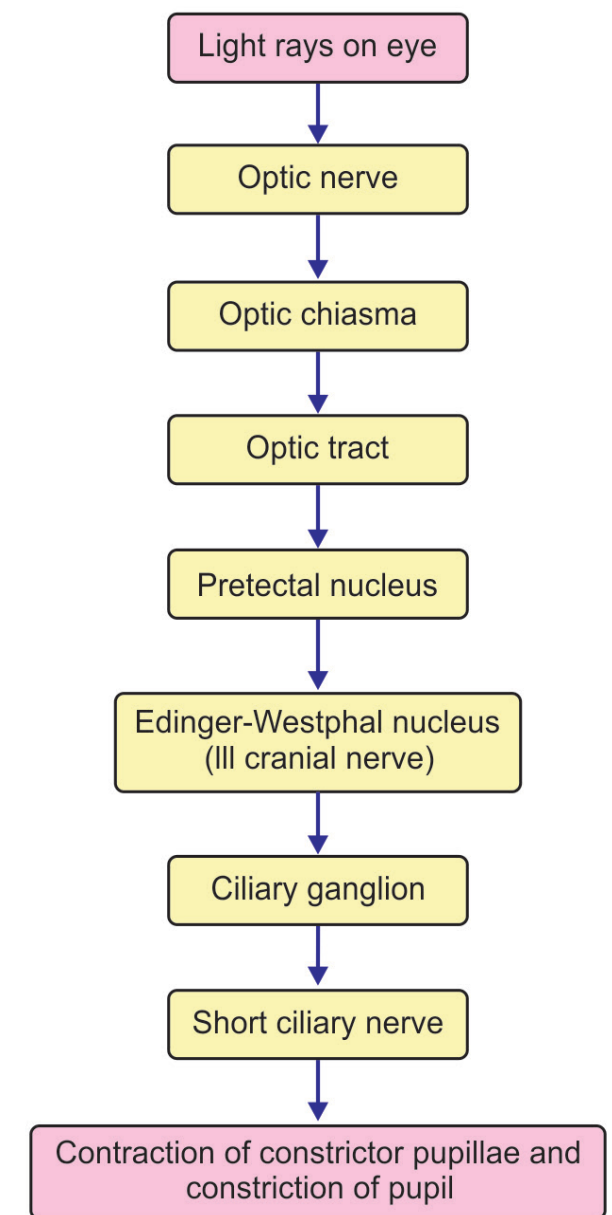
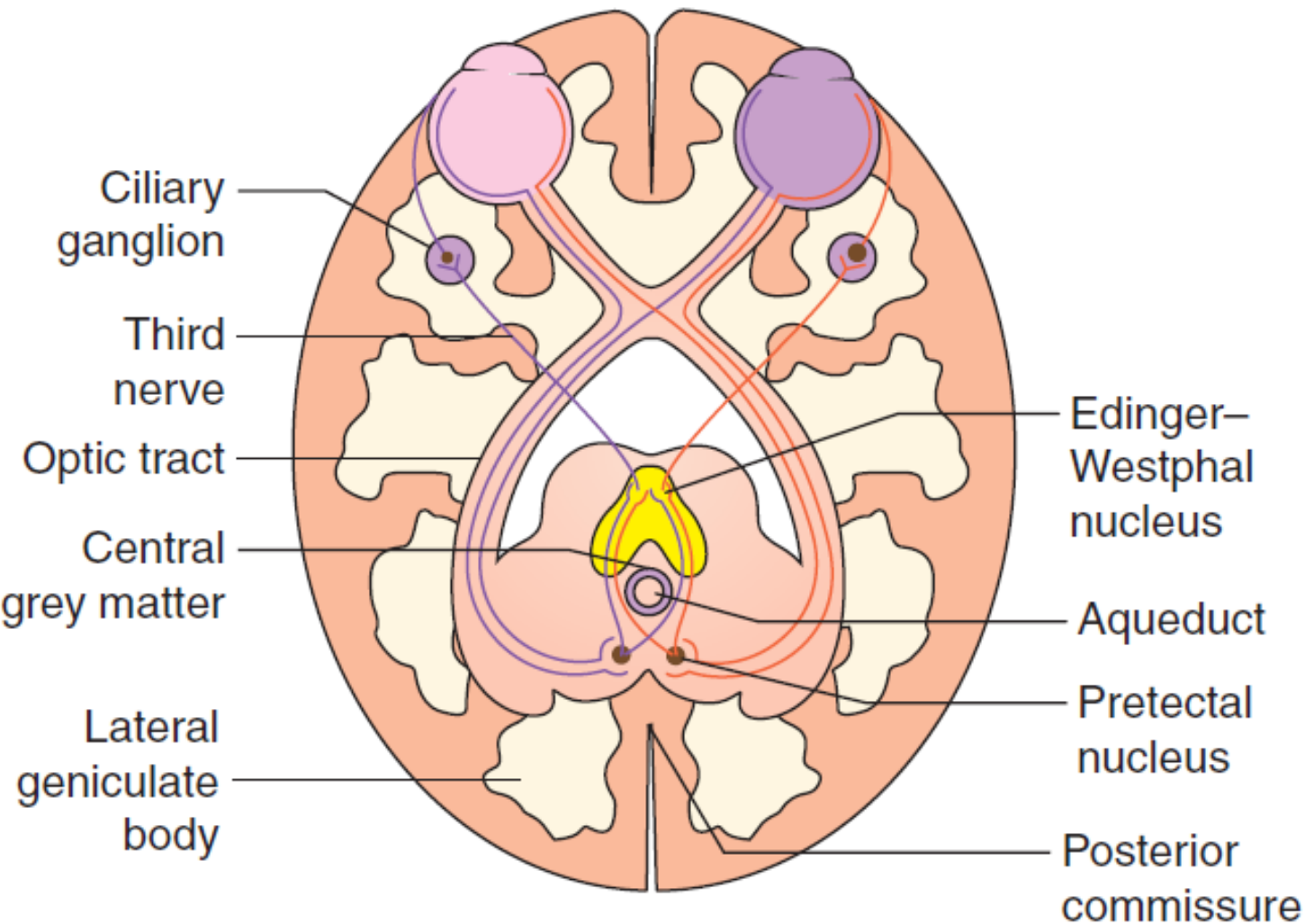
- Control amount of light entering the retina
- Restrict passage of light by the pigmented layer
- Increase depth of focus

Two types of pupillary light reflex

- Direct light reflex- When light fall on a retina its pupil constrict
- Consensual light reflex- when light falls on one retina the other pupil also constrict

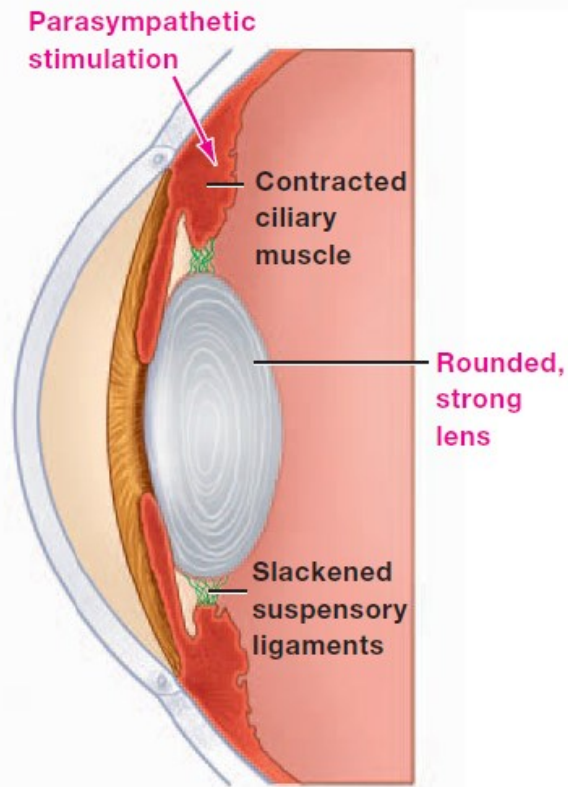


Significance of pupillary light reflex



- Lesions of tectonuclear pathway/pretectal n.: Loss of pupillary light reflex, small pupil and presence of accommodation reflex. *Argyll-Robertson Pupil* (syphilis, Alcoholism, encephalitis)

Bulging of the lens, pupillary constriction and convergence



Accommodation/Focusing

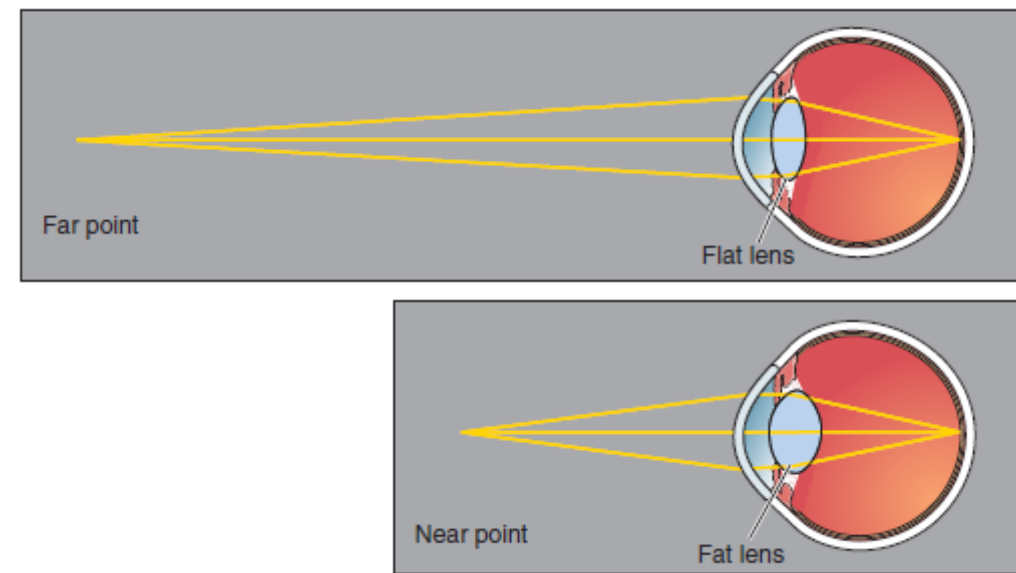
- A process whereby clear image is maintained as gaze is shifted from far to near object
- It is the relaxation of ciliary muscle that relieves the tension in
 - suspensory ligament= the lens becomes round
- Lens can increase the converging power of the eye in a young adult by about **12D**.

1. Accommodation Reflex

- Objects closer than 6m (20 ft) can be focused on the retina by the accommodation reflex, which has three components.

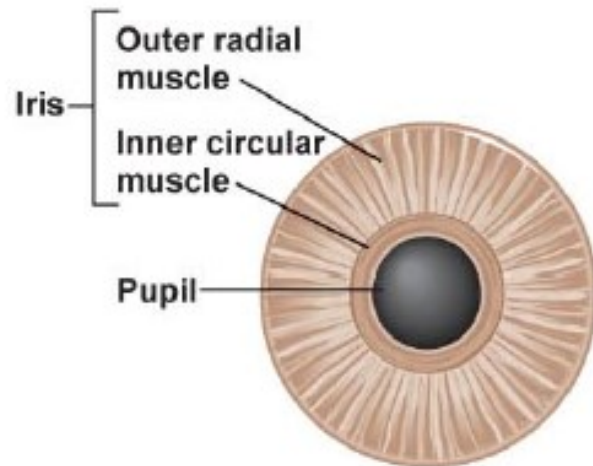
a. Bulging of the lens

- Lens changes shape to focus
 - Flatten for distance: Sympathetic
 - Thicken for near vision: PNS

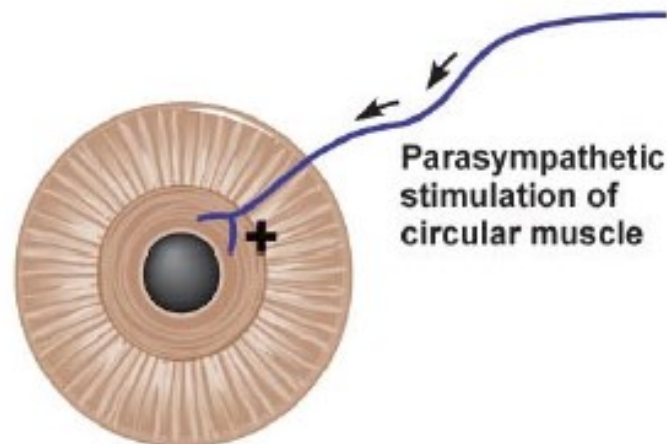


b. Pupillary constriction

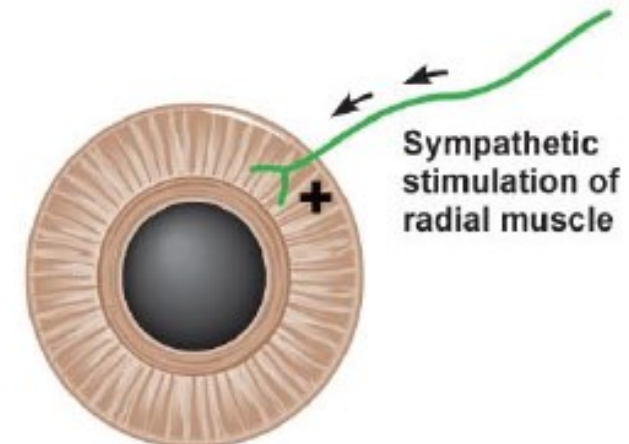
- Reducing the area through which light can enter the eye
- Reducing spherical aberration *increases the depth of focus*
 - Dim light stimulates sympathetic reflex to dilate pupils
 - Bright light stimulates parasympathetic reflex to constrict pupils



(a) Anatomy of iris and pupil



(b) Pupillary constriction



(c) Pupillary dilation

c. Convergence

- The gaze of the two eyes shifts toward the centre of the head in order to keep both eyes focused on the object.
- The near point and far point define the range of distances over which clear image can be formed by the eye.

The near point is the nearest point at which an object can be seen clearly.

A typical young adult can increase his or her converging power by 12D. Therefore, the near point is approximately 8.3 cm from the eye.

The far point is the distance from the eye that an object must be placed so that it can be seen clearly without accommodation for a normal person - 6 m.

Far point is at infinity and near point varies with age

Refractive Errors

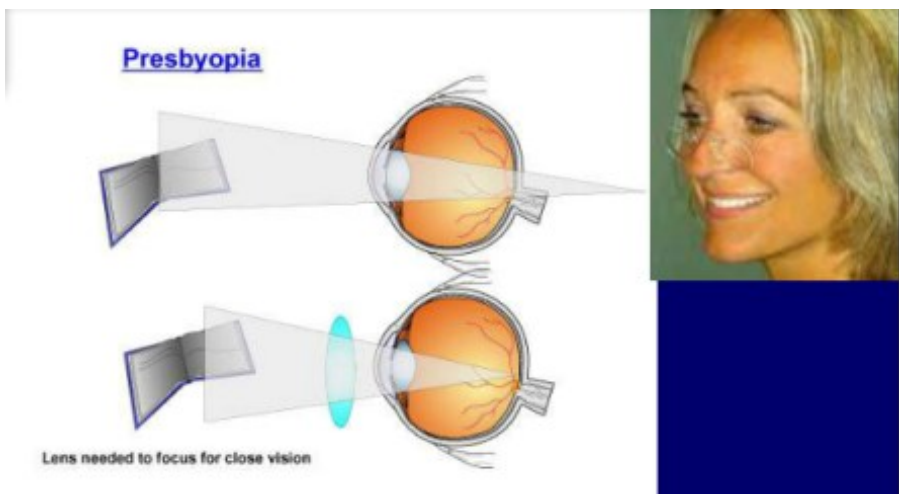
- Emmetropia: normal
- Myopia: nearsighted
- Hypertropia: farsighted
- Astigmatism: not uniformed curvature of lens
- Presbyopia: loss of the accommodation power of the lens

Presbysopia

- Occurs with age- his or her near point increases, making it difficult to see objects placed close to the eye
 - Each eye remains focused permanently at an almost constant distance
 - The eye can no longer accommodate for with near and far vision.
- This shifts near point from 7 cm at 8 yrs to 83 cm at 60 yrs

- *Reading glasses are converging lenses that compensate for the loss of accommodation.*

- *Bifocal lenses - upper segment far seeing
Lower segment near seeing*



- *As a person grows older, the lens grows **larger, harder and thicker**, and it also becomes less elastic, partly because of progressive denaturation of the lens protein.*
 - *decreases from $\sim 14D$ in young child to less than $2D$ at the age of 45 to 50 to about zero at the age of 70.*

Visual Acuity: Snellen's eye chart



- Test at 6m (parallel rays, min accommodation, illuminated, each eye separately).
- Numerator: distance of the patient from the letters
- Denominator: the smallest letters accurately read.
- Ratio of what a person can see compared to a person with normal vision
 - 6/6 is normal
 - 6/12 less visual acuity
 - What the subject sees at 6m, normal person could see at 12m
 - 6/5 better than normal visual acuity

- The letter of the top line of Snellen's chart should be read clearly at a distance of 60 m.
- letters in the subsequent lines should be read from a distance of 36, 24, 18, 12, 9, 6, 5 and 4 m.

The main mechanisms with vision:

- **Initiation of vision** (phototransduction): a function of photoreceptors (rods and cones)
- **Processing and transmission of visual sensation:** a function of the image processing cells of retina and visual pathway.
- **Visual perceptions:** a function of visual cortex and related areas of cerebral cortex.

Energy Transduction

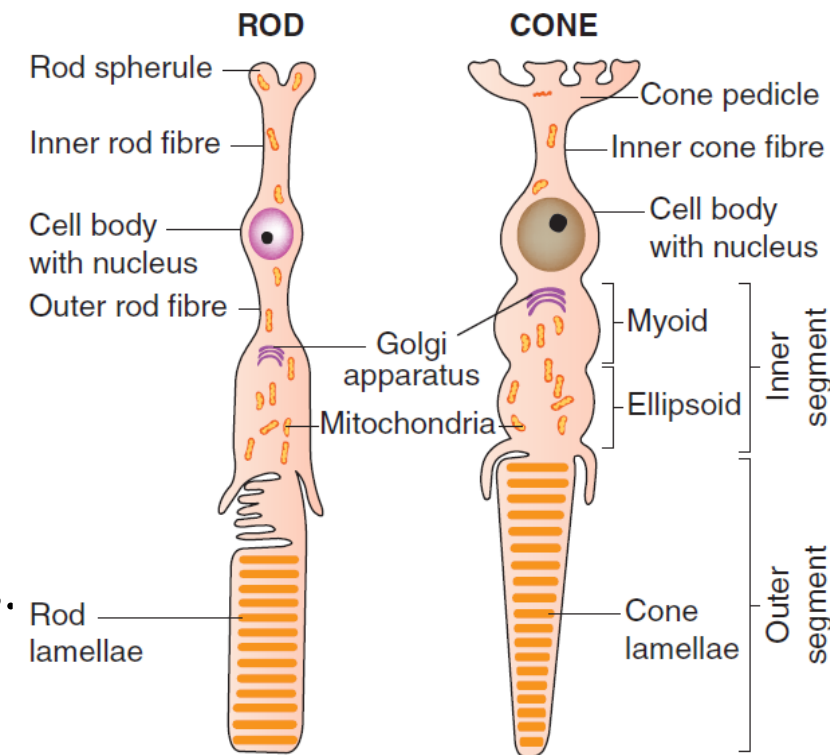
- *Rods and cones* are photoreceptors.

Morphology: both cell types consist of :

- **Inner segment:** the nucleus, mitochondria & synaptic vesicles.
- **Outer segment:** containing membranous disks.

1. The membranous disks are continuously formed segment and migrate toward the apex, where they are sloughed off.

2. The membranous disks contain a visual pigment, called **rhodopsin**, which absorbs light rays.



small number of photoreceptors contain no photopigment (rhodopsin or cone pigment) but they contain *melanopsin*

Visual pigments: rhodopsin & cone pigments

- Rhodopsin consists of a protein called **opsin** & a light-absorbing analogue of vitamin A(retinal) called **11-cis-retinal**.
- The **amino acid** composition of opsin determines the **wavelength of light** absorbed (493-505 nm) ~ yellow.
 - Rods contain a single type of opsin (scotopsin).
 - Cones contain three types of opsins (*blue, green, or red*) depending on the position of the visual spectrum they absorb best BUT the 11-cis-retinal is same as rods

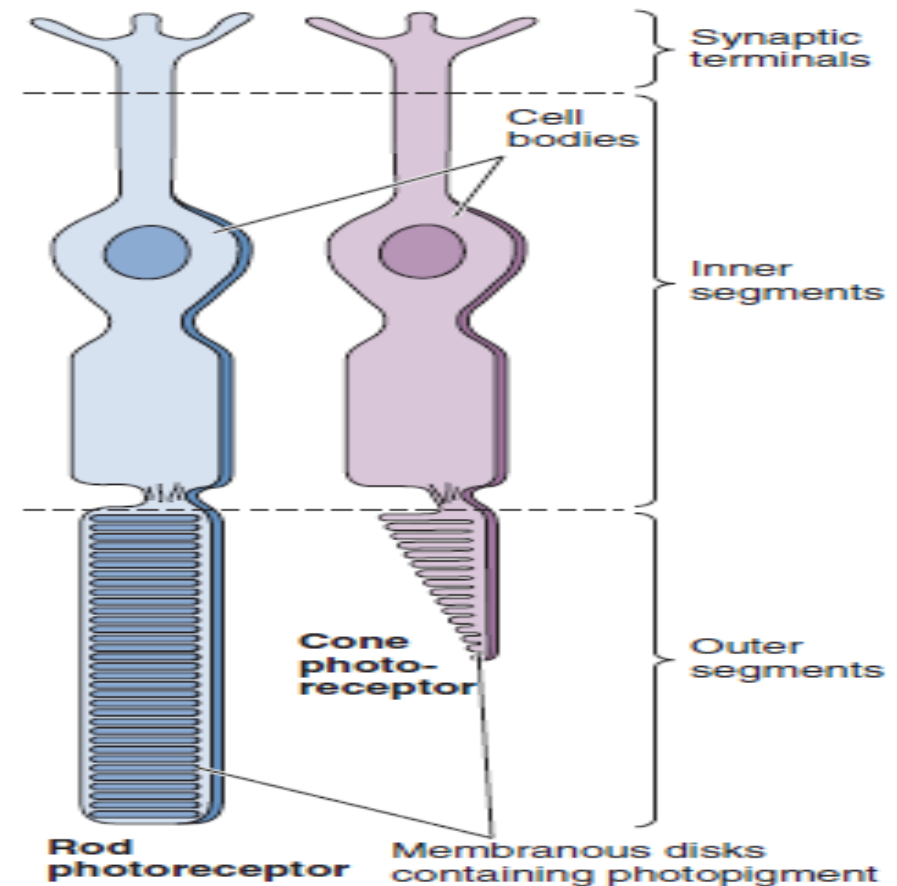


TABLE 2-7

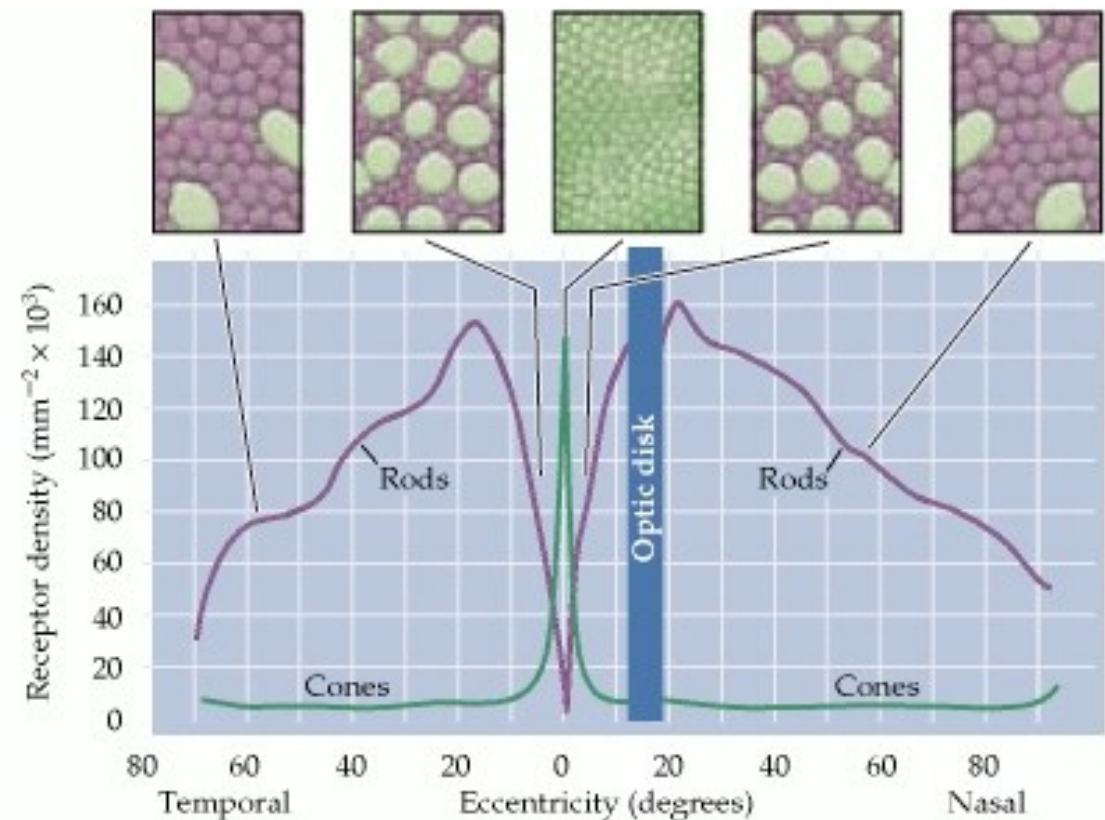
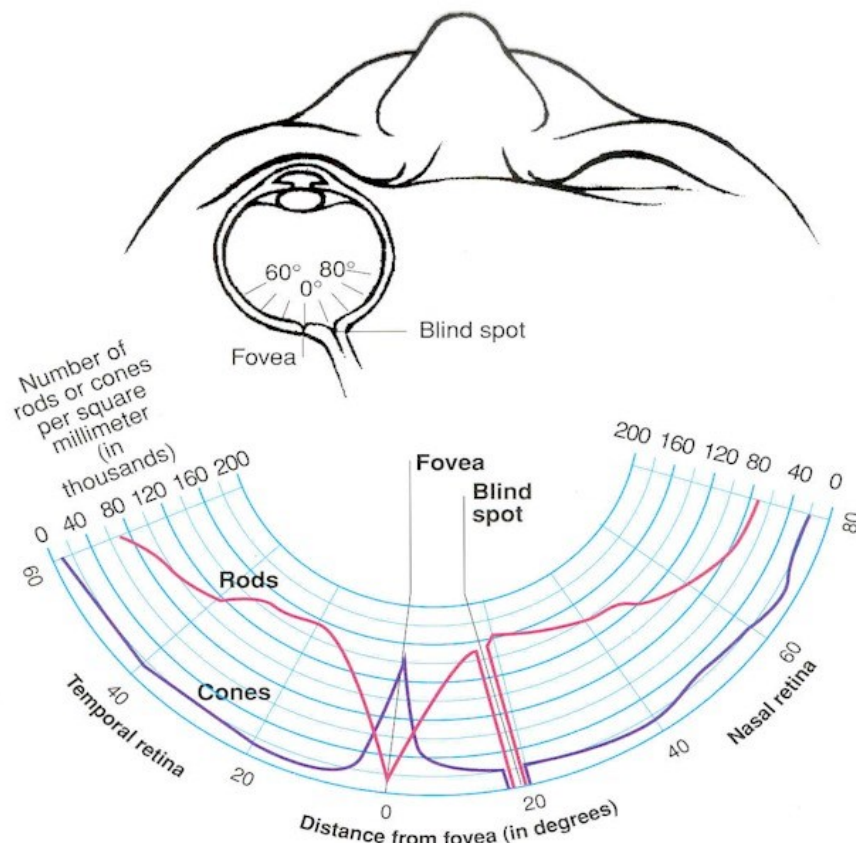
Functions of Rods and Cones

Function	Rods 120 million	Cones 6.5 million
Sensitivity to light	Sensitive to low-intensity light; night vision	Sensitive to high-intensity light; day vision
Acuity	Lower visual acuity Not present in fovea	Higher visual acuity Present in fovea
Dark adaptation	Rods adapt later	Cones adapt first
Color vision	No	Yes

Cones, which have three different photopigments are responsible for *day light, high acuity and color vision*.

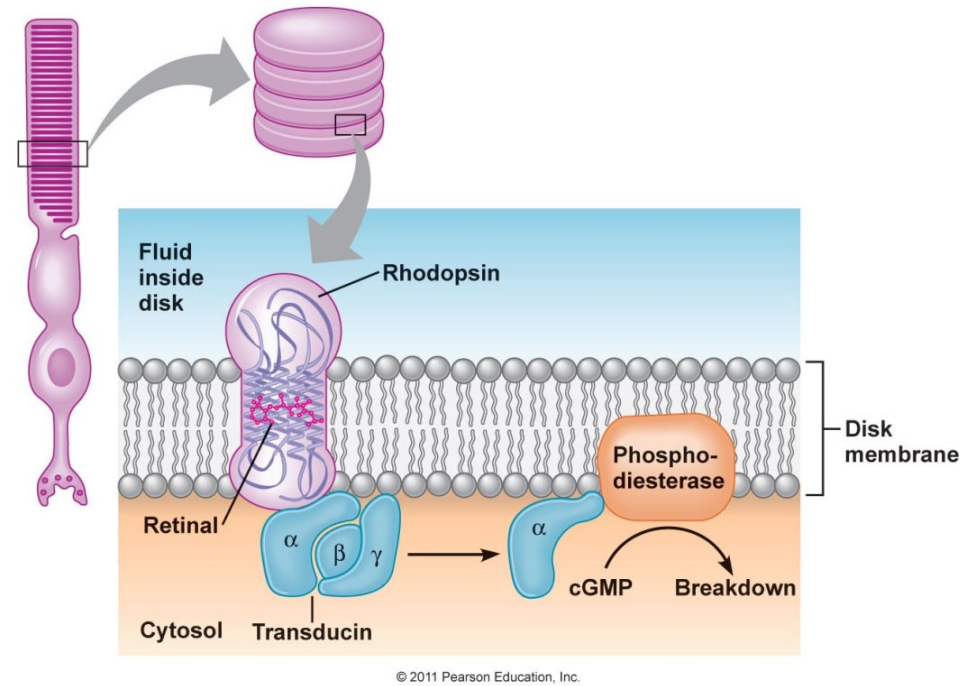
The rod cells are very sensitive to light and do not see color, that is why when we are in a darkened room we see only shades of gray

Rod and Cone Density on Retina



- Optic disc (blind spot/15 degrees lateral to the central point of vision): no rods and cones
- Fovea: sharpest vision (concentration of cones)
 - *High acuity and low sensitivity*
 - *Cones : bipolar cells = 1:1*

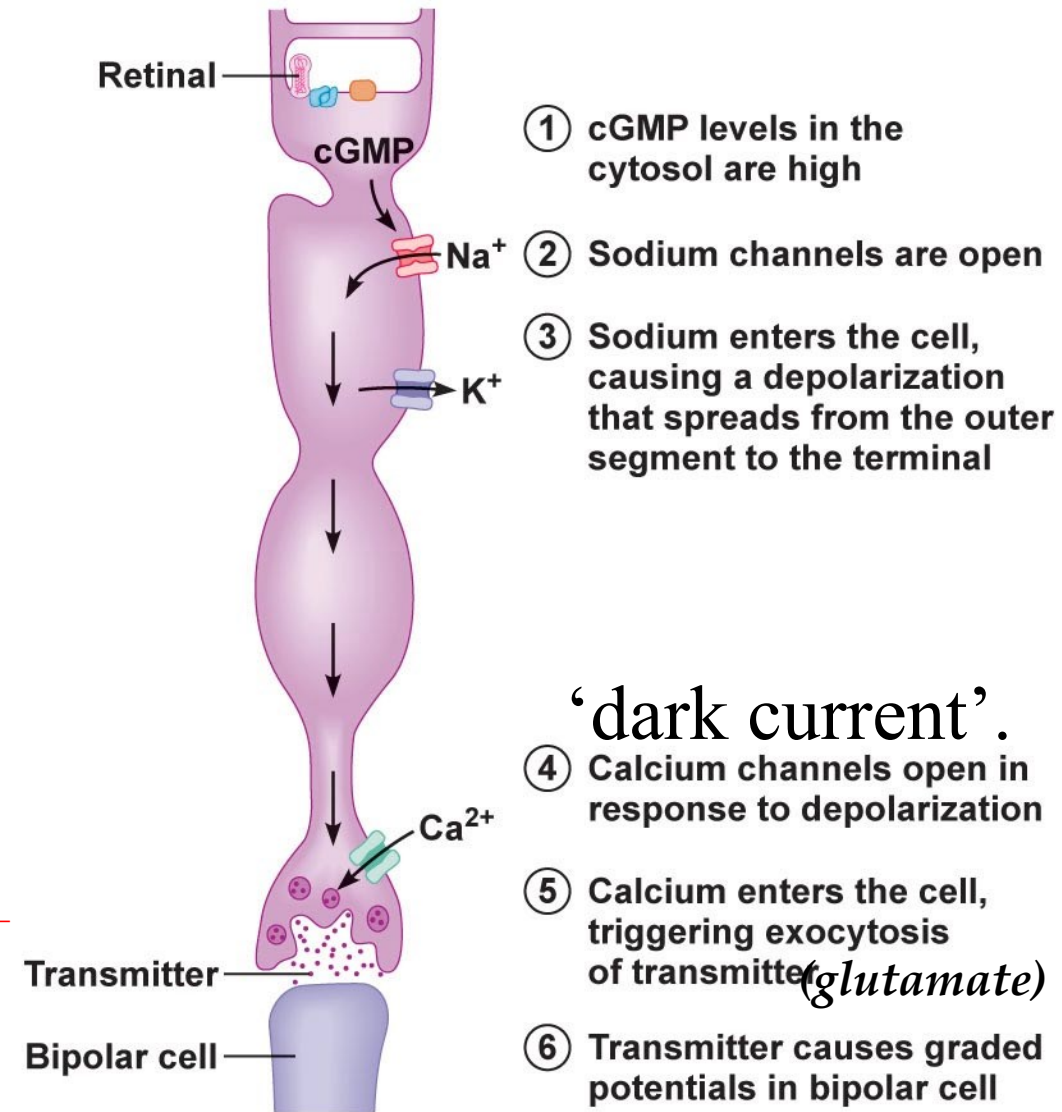
Phototransduction (in the dark)



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Darkness - rods and cones are **depolarized** in the dark.

- Low RMP (**- 40mV due to high Na^+ conductance of the outer segment.**)



(a) State of photoreceptor in the dark

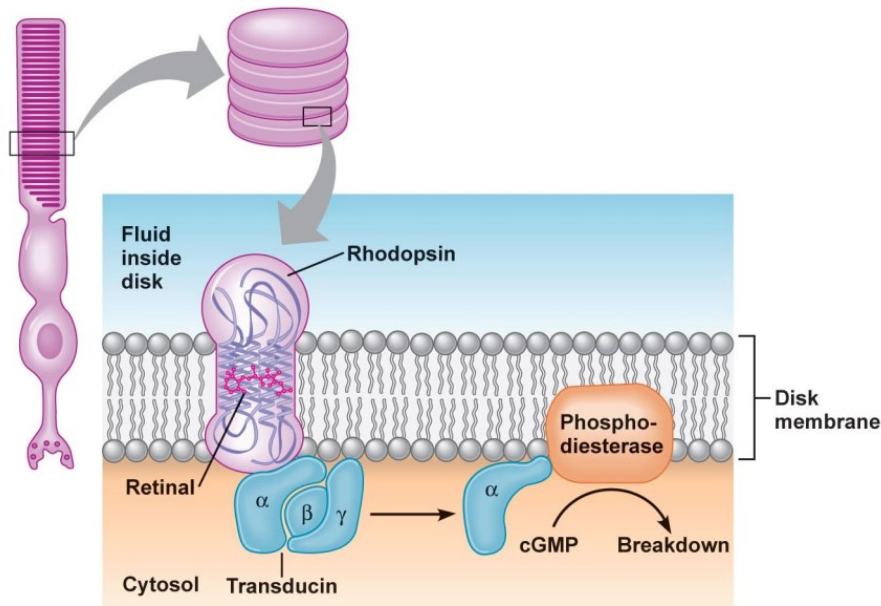
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Na^+ channels are maintained in the open state by cGMP

- When cGMP binds to the Na^+ channel, the channel opens. i.e. cGMP, acts by activating the channel directly
= MP changes from $\sim -70 \text{ mV}$ to -40 mV (depolarized)

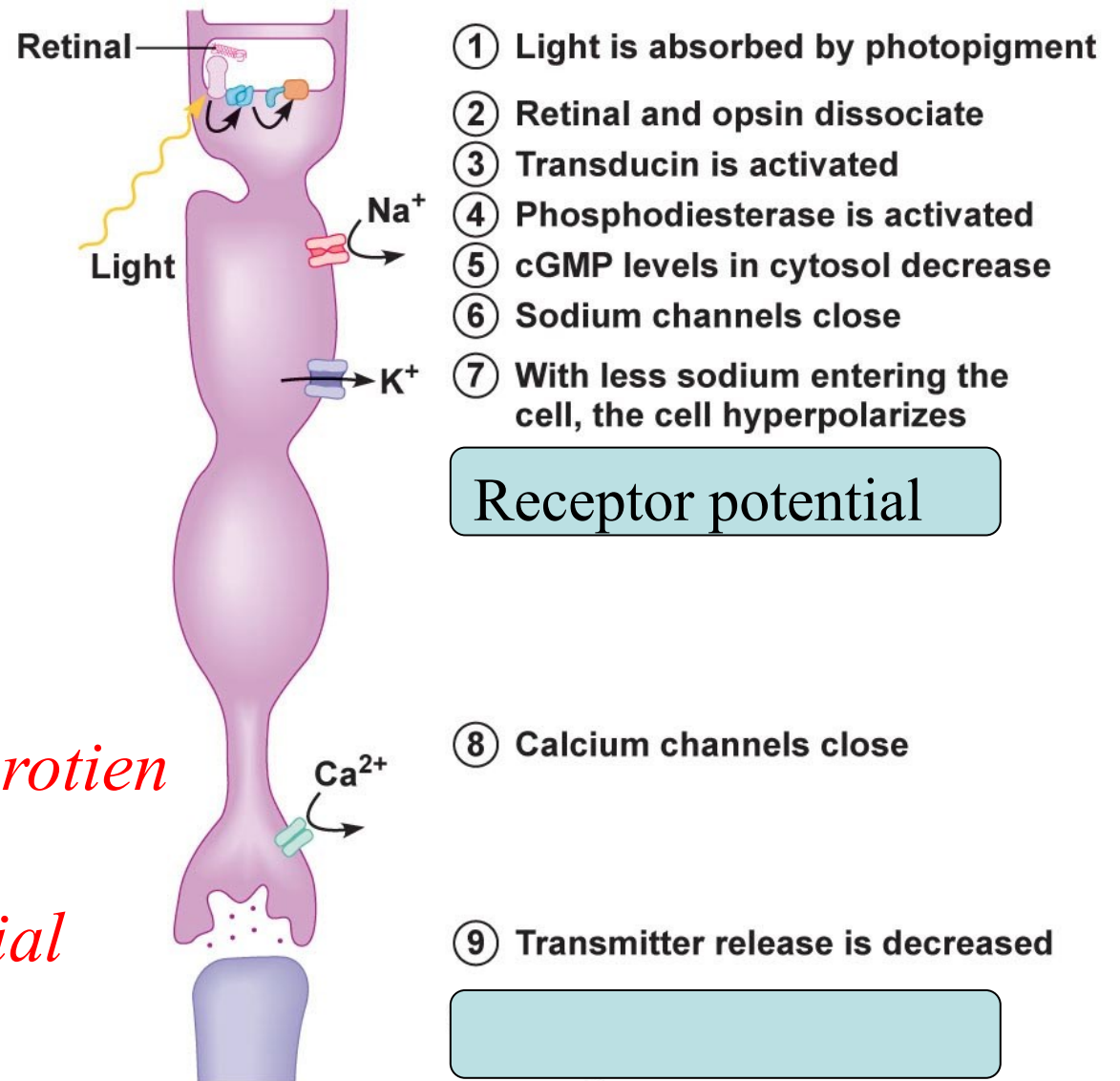
Phototransduction (in the light)

When light strikes the outer segment, rhodopsin (11-cis-retinal + opsin) dissociates into **all trans retinal** and **opsin** separates out = *photodecomposition*



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- ✓ *Activation of rhodopsin*
- ✓ *Activation of transducing/G-protein*
- ✓ *Conversion of cGMP to GMP*
- ✓ *Production of receptor potential*



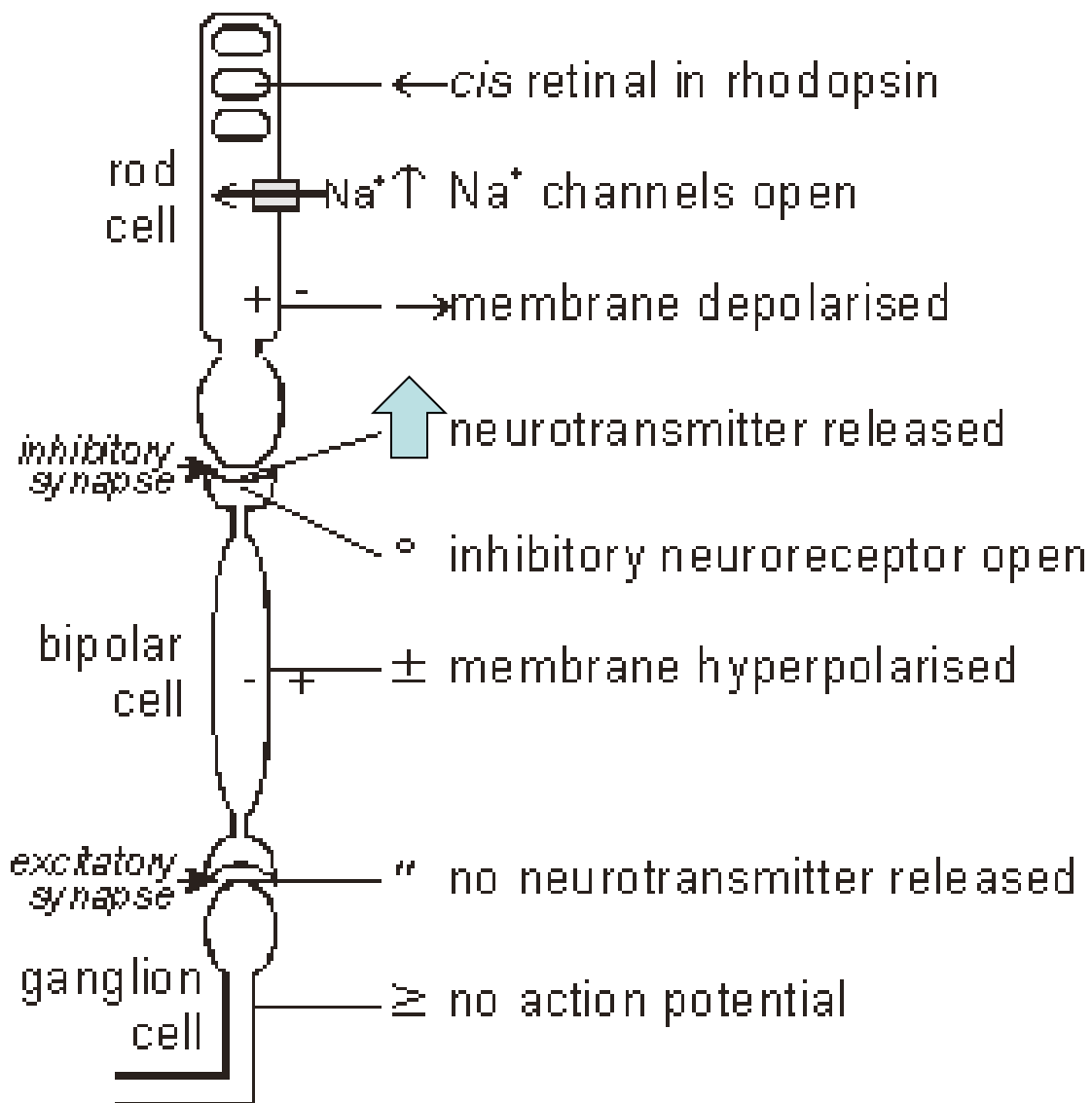
- ① Light is absorbed by photopigment
- ② Retinal and opsin dissociate
- ③ Transducin is activated
- ④ Phosphodiesterase is activated
- ⑤ cGMP levels in cytosol decrease
- ⑥ Sodium channels close
- ⑦ With less sodium entering the cell, the cell hyperpolarizes
- ⑧ Calcium channels close
- ⑨ Transmitter release is decreased

Receptor potential

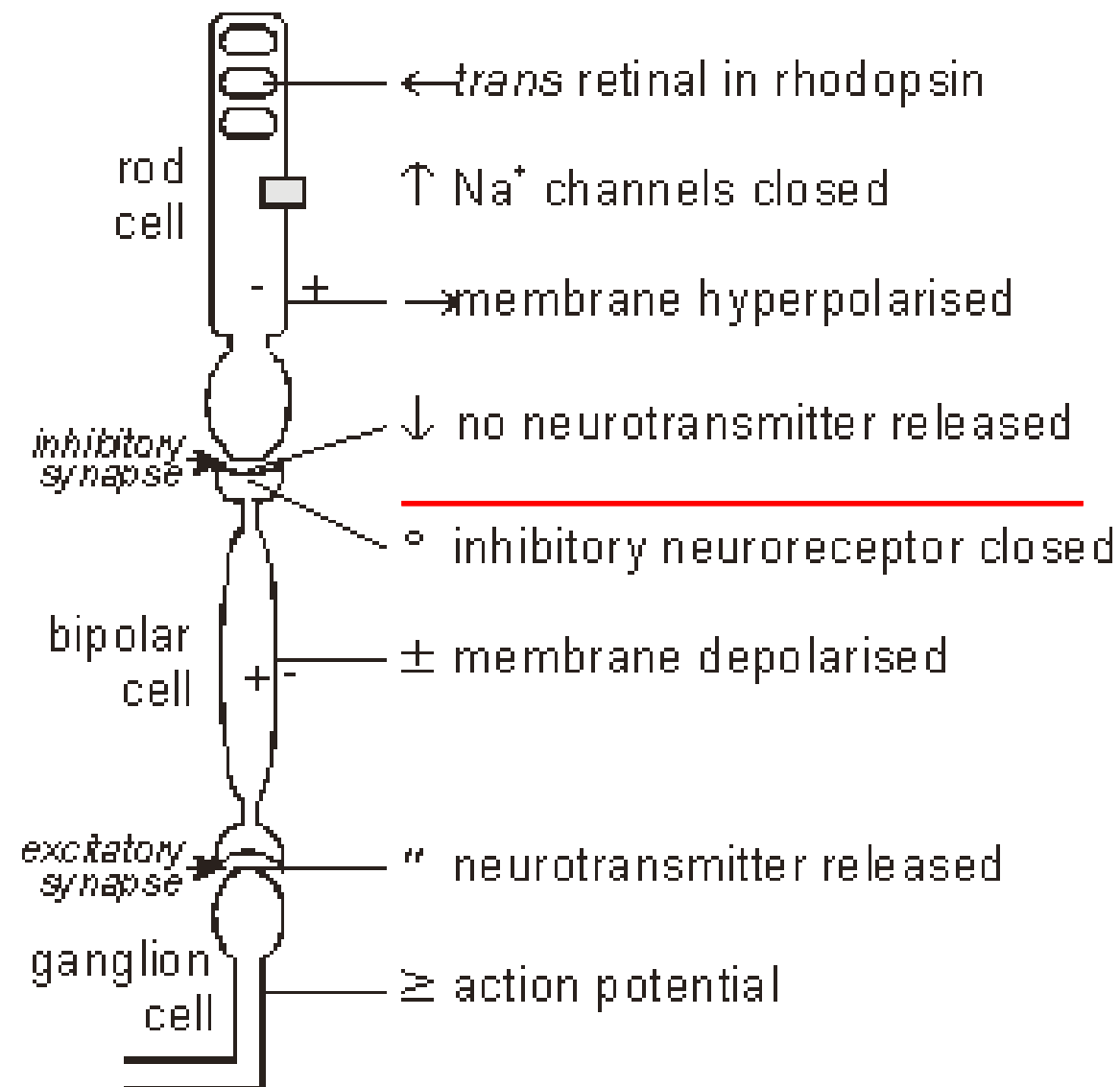
(b) Response of photoreceptor to light

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In The Dark



In The Light



RODS

- More sensitive to light than cones - responsible for the night vision.

1. Rods can absorb more light than cones.

Rods contain more rhodopsin in their outer segments.

Rods can detect light entering the eye from any direction, whereas cones respond only to light directly along their axis.

2. Rods *remain polarized for a longer time* than cones

The visual cycle: equilibrium between the photodecomposition and regeneration of visual pigments.

Light adaptation: decreased sensitivity to continued light exposure

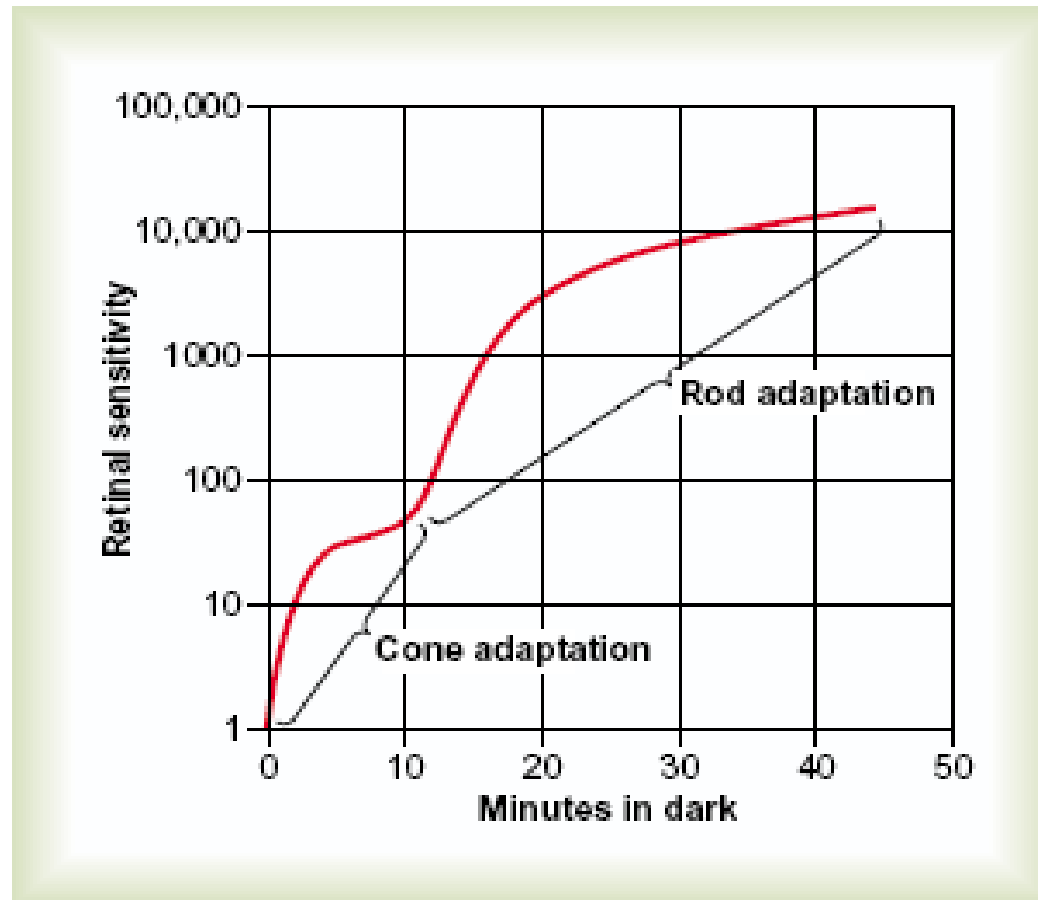
- Bright light induce **bleaching**
 - breaking down of most photo pigments into **retinal and opsin**.
 - Retinal conversion into Vit-A (cis to trans/retinol)
- Concentration of photopigments is considerably *reduced*
- This *reduces sensitivity* of cones and rods to light (~ 5 min)

Dark adaptation: *increased sensitivity/to dim illumination*

- Requires ~20 min
 - Pupil dilates
 - Regeneration of rhodopsin/ ***unbleaching***
 - ✓ Vitamin A (retinol) goes back to retinal
 - ✓ Opsin unite with retinal to form more photopigments
 - *convergence 100 rods on one ganglionic cells (sensitivity inc 6000 – 24,000x)*

The course of dark adaptation

cones can not achieve the same degree of sensitivity as the rods



In minute

*10 fold increase
in sensitivity*

After 20 minute

6000 fold

After 40 min

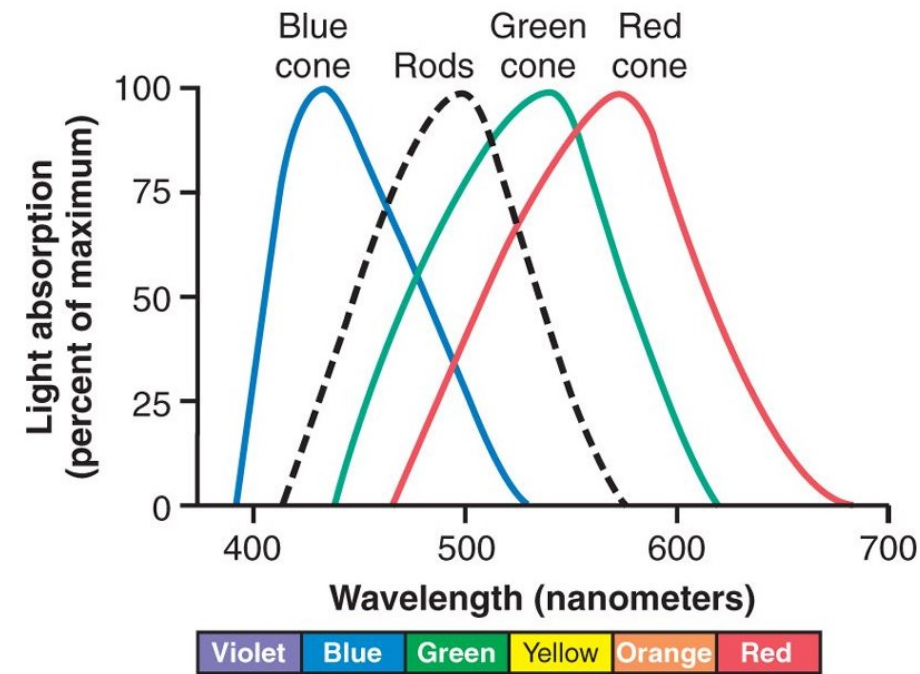
25,000 fold

Severe deficiency of vitamin A elevates the threshold for dark adaptation curve due to depletion of the photosensitive pigment/Nyctalopia, i.e. night blindness

CONES

- *Achieve high visual acuity because they are concentrated in the FOVEA (where the clearest images are formed), they do not respond to scattered light.*
- Color vision is achieved by combining the information in cones, - *in the red, green or blue* range of the visual spectrum.
- Information about color is relayed from the *lateral geniculate nucleus* to cortical cells which are organized into oblong regions called **Blobs**.
 - The blobs decode *contrast and depth*.

Photochemistry of color vision



- **Basis: Cone Sensitivity**

- Opsin of rods = scotopsin
- Opsin of cones = photopsin

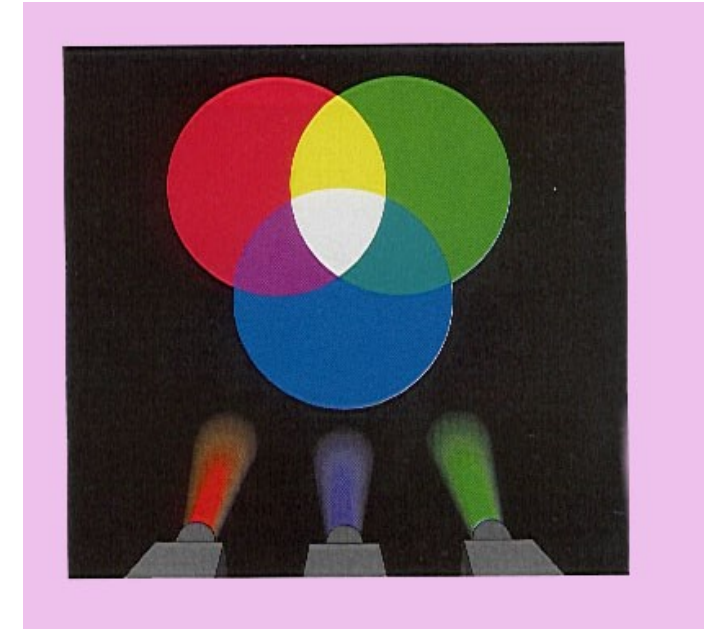
- **Trichromatic theory:** Three types of cones (based on opsin types/spectral sensitivity)

- Blue cones/S - 430 nm (435)
- Green cones/M - 530 nm (535)
- Red cones/L – 560 nm (570)

Rods have peak sensitivity to light at **500 nm** - perceived as blue-green by **photopic vision**

Young Holmerz Trichromacy Theory

- The ratio of stimulation of the three types of cones is interpreted as a given color
 - Stimulus value ratios of **red**: **green**: **blue**
 - ✓ $99:42:0 = \text{Orange color}$
 - ✓ $0:0:97 = \text{Blue color}$
 - ✓ $83:83:0 = \text{Yellow color}$
 - ✓ $31:67:36 = \text{Green color}$
 - ✓ *Equal activation of the three pigments = white*



Ganglion cells - Opponent process theory

- Relaying information about color to the lateral geniculate nucleus respond oppositely to **red and green-light**; therefore they are called **color-opponent** cells
 1. **Red centre, green surround cells** - are stimulated by red cones in the centre and inhibited by green cones in the surround.
 2. **Green centre, red surround cells** - are stimulated by green cones in the centre are inhibited by red cones in the surround.

Red vs green, blue vs yellow, black vs white.

Activation of one member of this pair inhibits the other.

Cortical cells

- Interpret the information received from the LGB to assign color to each region of the image formed on the retina.
- For example., if green centre and red centre cortical cells are equally excited, the color assigned that region of the cortex is **yellow**.⁴¹

Analysis of color signals in the visual cortex

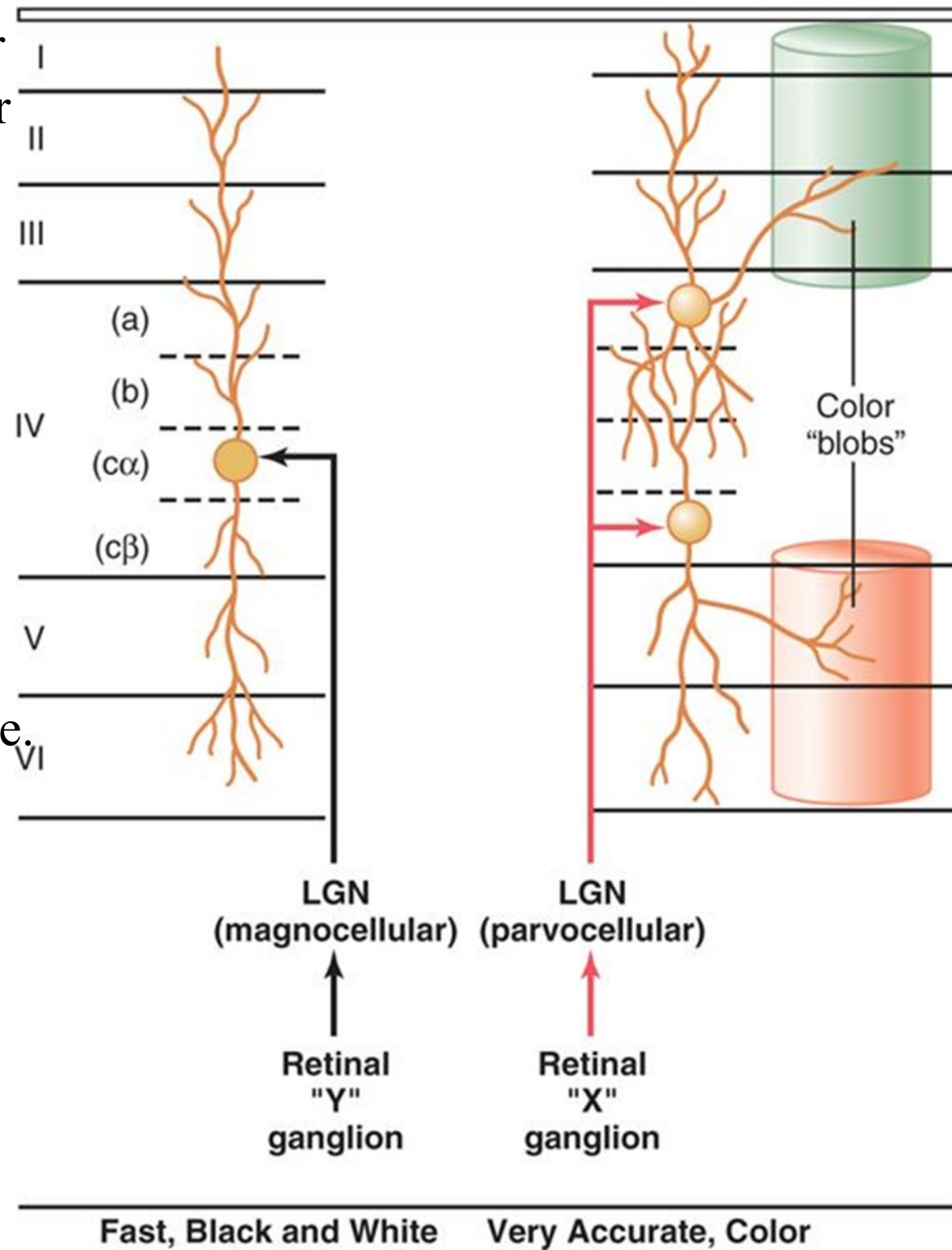
Color information from the parvocellular portion of the LGB is relayed to the layer IVc of the *striate cortex* (area 17).

then to the blobs in layers II and III. (the blobs respond to colours)

visual association area

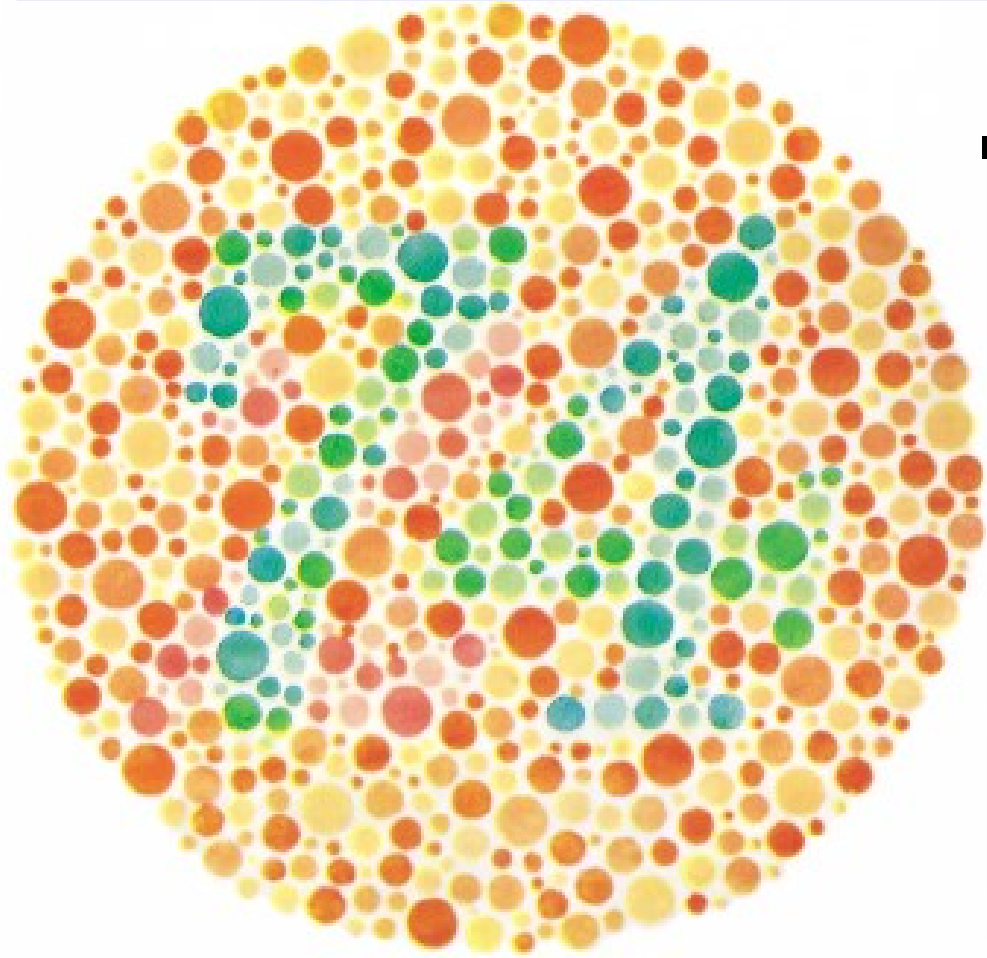
Specialized color area

lingual and fusiform gyri of occipital lobe.



Color blindness

- Results from the absence of the gene that codes for either the *red, green or blue opsins*.
- *Trichromats: Normal person with three cones is trichromat. BUT weakness of one of the cones*
 - *Red cones weak: Protanomaly*
 - *Green cones weak: Deuteranomaly*
 - *Blue cones weak: Tritanomaly*
- *Dichromats: Have only two functional types of cones.*
 - Absent of:*
 - *Red cones: Protanopia*
 - *Green cones: Deuteranopia*
 - *Blue cones: Tritanopia*



- Absence of both red and blue cones will lead to total color blindness (**monochromatic vision**)

Ishihara chart

- Normal - 74
- Red - green blind = 21

Encoding

The retina has three major functional layer

1. Photoreceptors (outer nuclear layer)

Rods

Cones

2. Interneurons (inner nuclear layer)

Bipolar cells

Amacrine cells

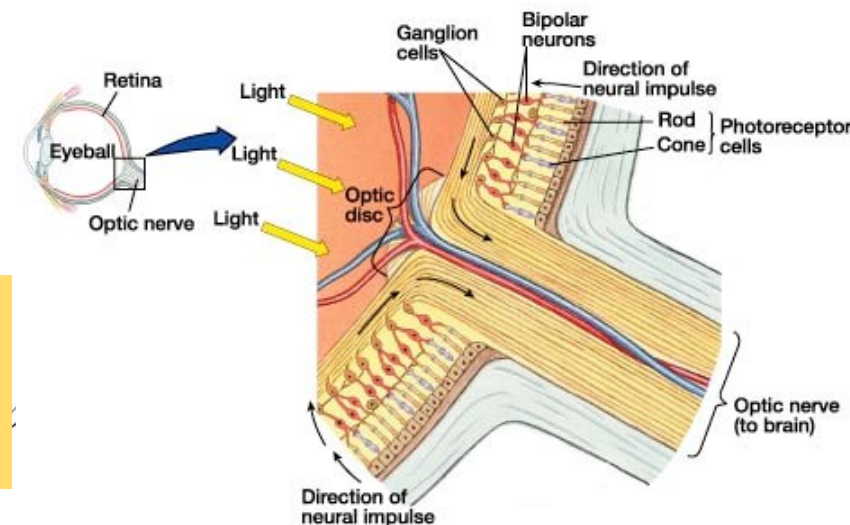
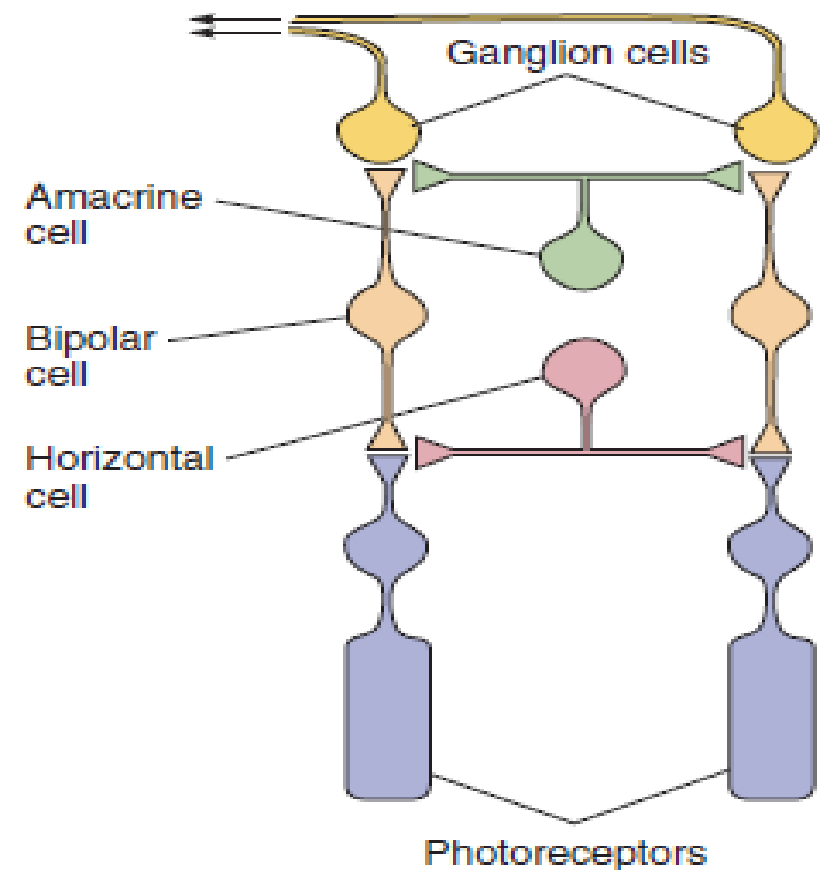
Horizontal cells

3. Ganglionic cells layers
(only source of retinal output)

Midget (M) ganglion cells

Parsol (P) cells

Ganglion cell axons
projecting to forebrain



Retina built “backwards”

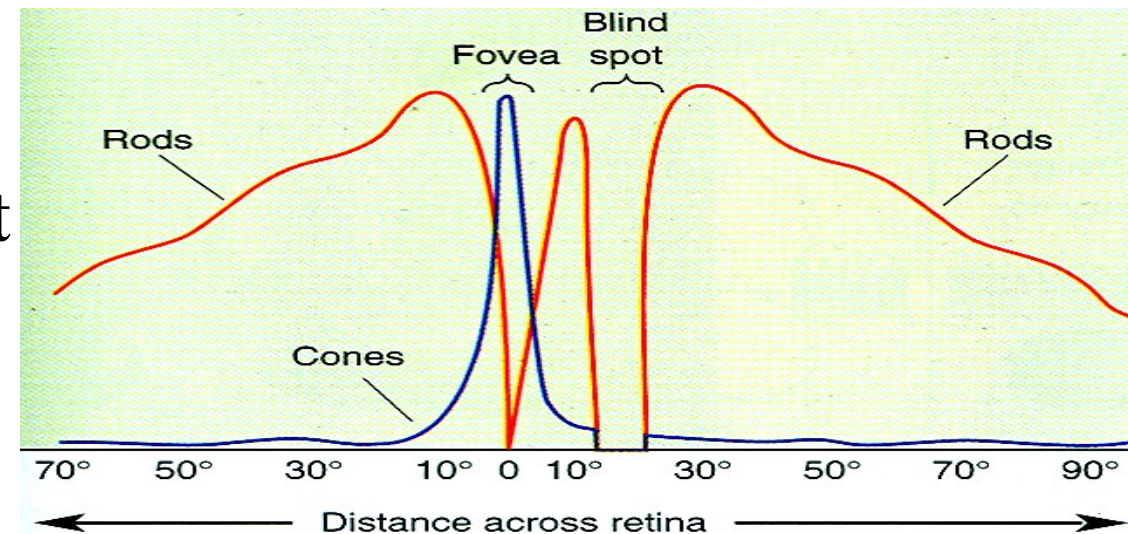
- Light passes through 3 layers of cells before reaching photoreceptors -**NEXT SLIDE**

Central retina / Fovea

Only cones are found
= **high resolution vision**

Fovea is thinning of the retina at center of macula

Ganglionic cells are laterally displaced

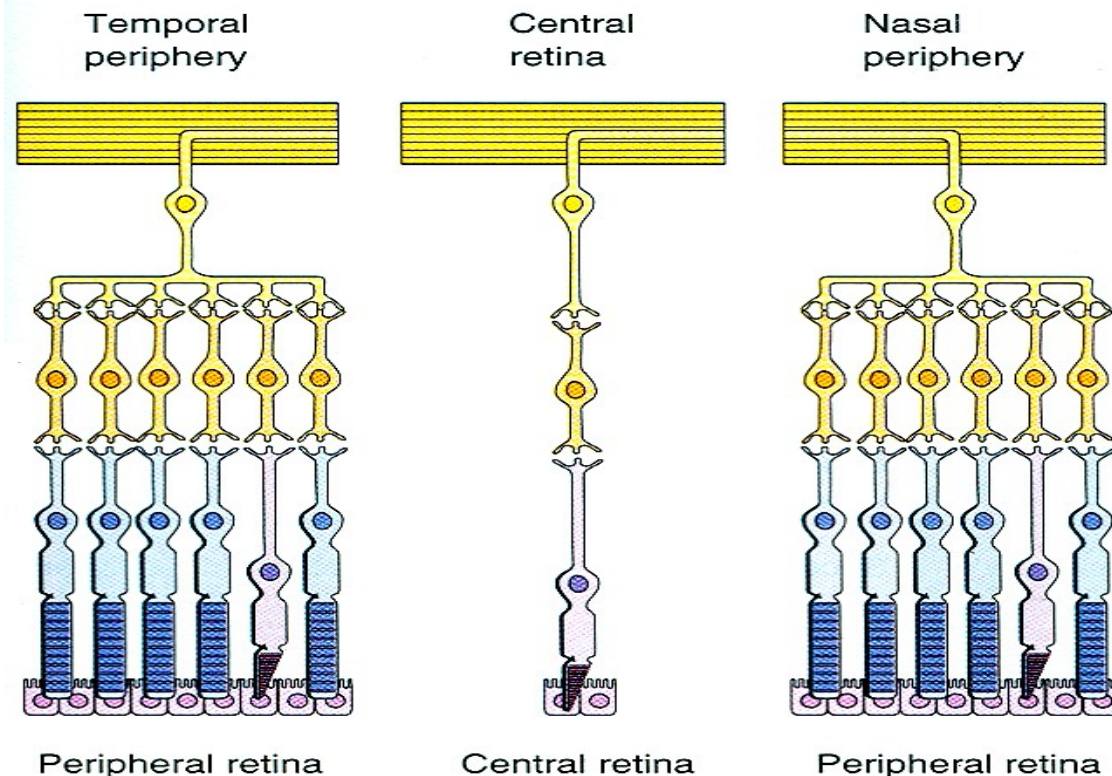


Lateral retina

Many rods and few cones

Many photoreceptors provide input

Better at detecting dim light



Retinal layers

A. The Pigment epithelium

- A single sheet of melanin containing epithelial cells.
- Functions include
 - a. Light Absorption*
 - the light reaching the back of the eye is absorbed by the pigment epithelium so that light scattering does not degrade visual acuity.
 - b. Phagocytosis*
 - Membranous disk & other debris sloughed from the photoreceptors
 - c. Vitamin A (Retinal) storage*
 - d. Blood-retinal-barrier*

Clinical significance :

Retinal detachment

- *Detachment of the rest of the retina from the CHOROID/pigment*
---- blindness/ reattachment can be accomplished by *laser surgery*.

Retinitis pigmentosa

- *Disruption of phagocytosis* of discarded membranous disc
- Progressive degeneration of photoreceptors resulting in macular damage with *central vision* loss or *peripheral* (night) vision loss

B. Cellular layers

1. Outer Nuclear Layer -

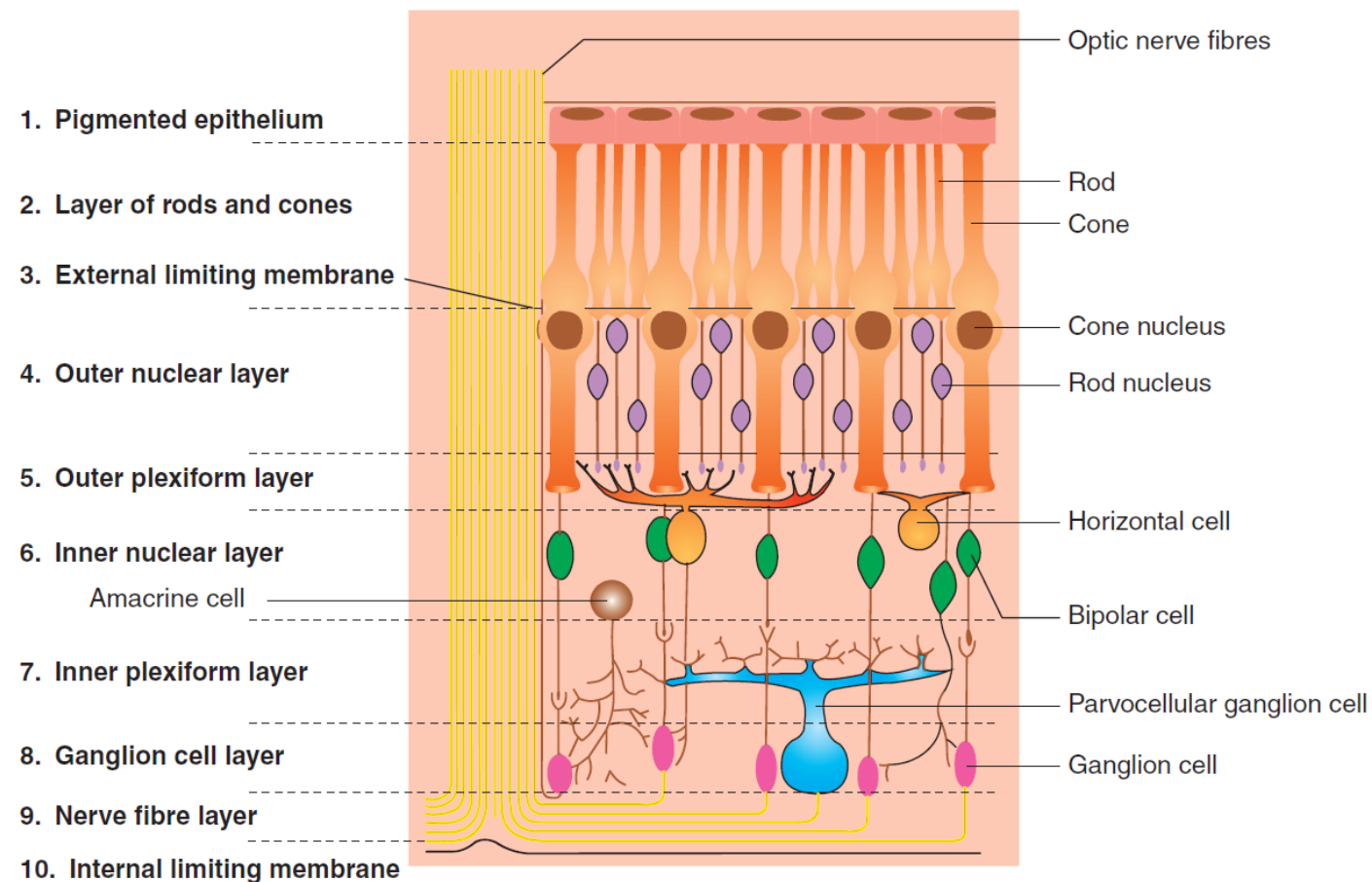
The nuclei of rods and cones.

2. Inner Nuclear Layer -

The nuclei from neurons (*bipolar, horizontal, and amacrine cells*)

4. Ganglion cell layer

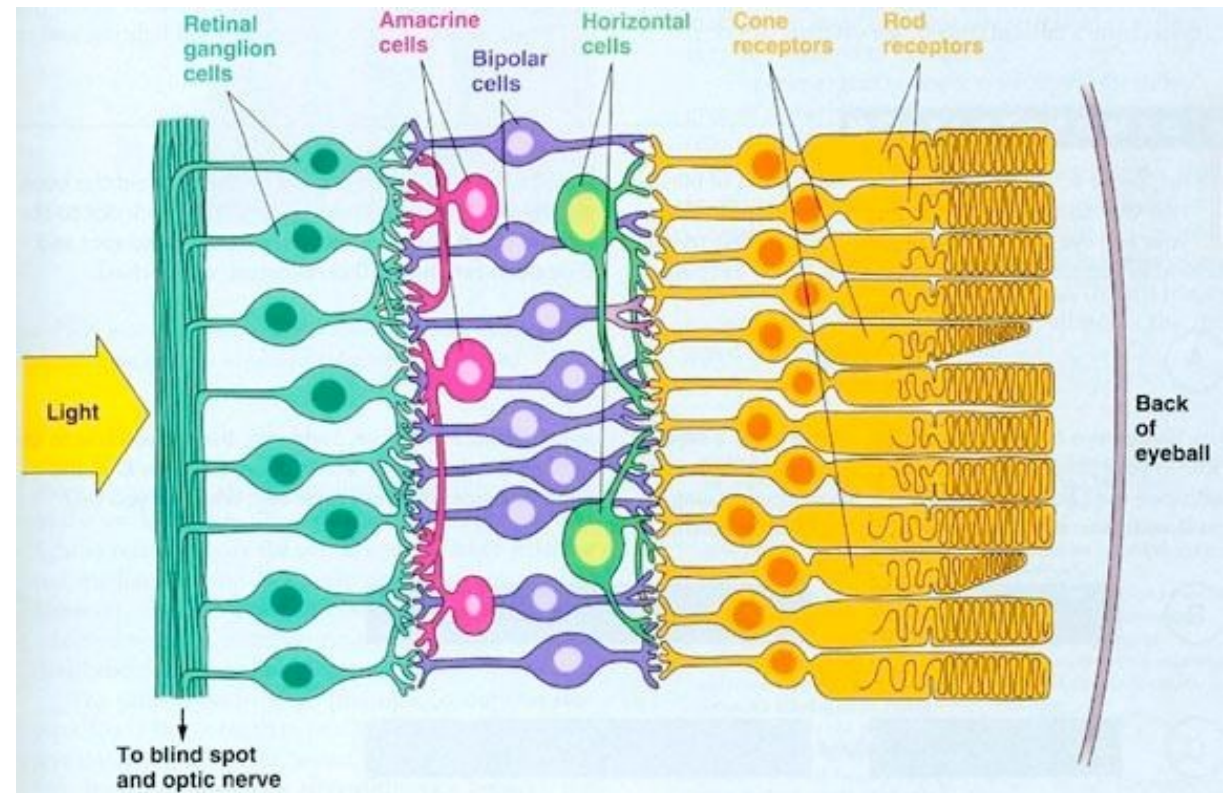
The third cellular layer of the retina. Axons of the ganglion cells form the *optic nerve*.



C. Synaptic layers

1. Outer Plexiform Layer

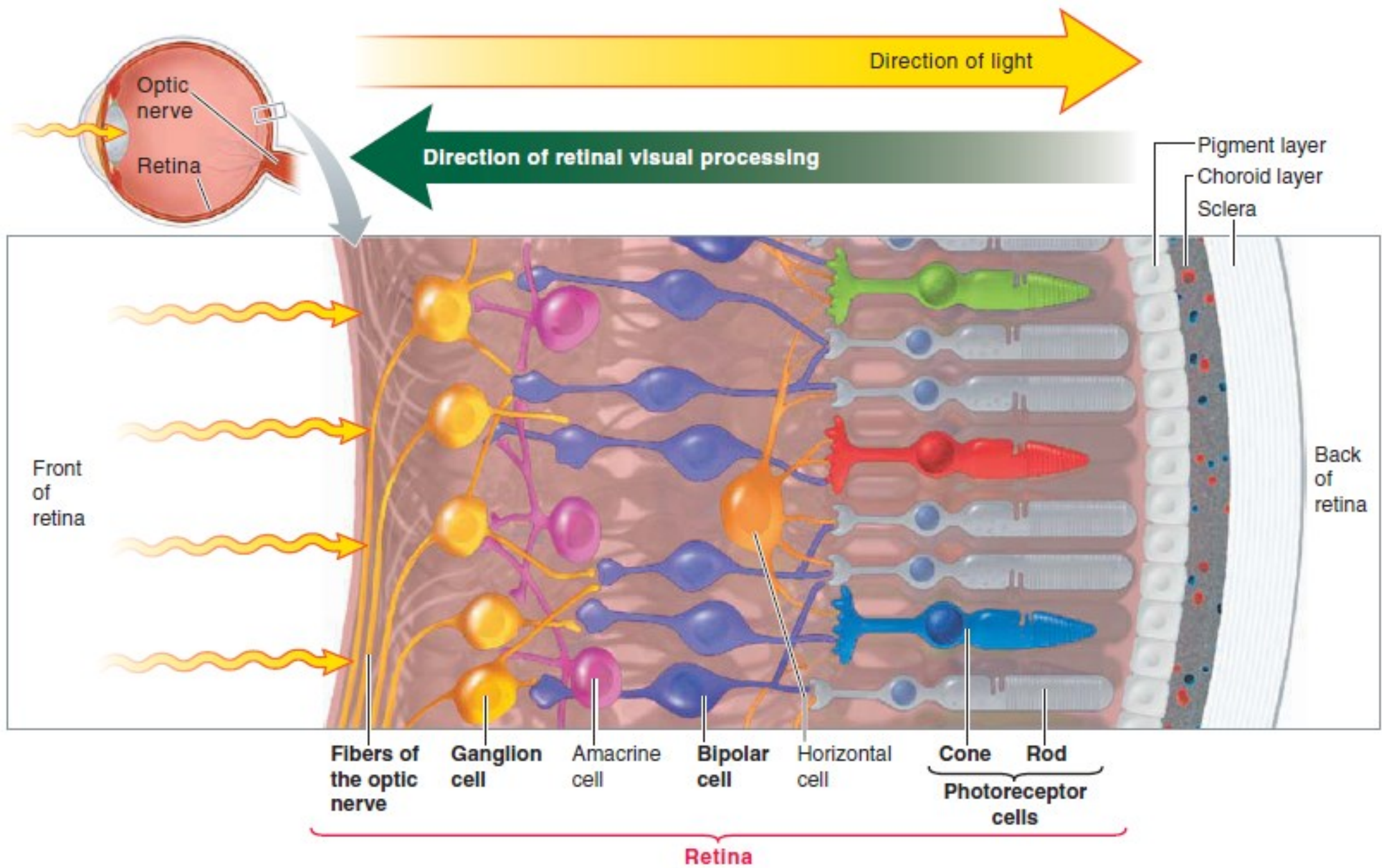
- Synaptic connection between the *photoreceptors* and *horizontal and bipolar cells*



2. Inner Plexiform Layer-

synaptic connections between the *ganglion cells and bipolar and amacrine cells*.

---- control visual contrast by inhibiting spread of visual signal from bipolar cells in outer plexiform layer



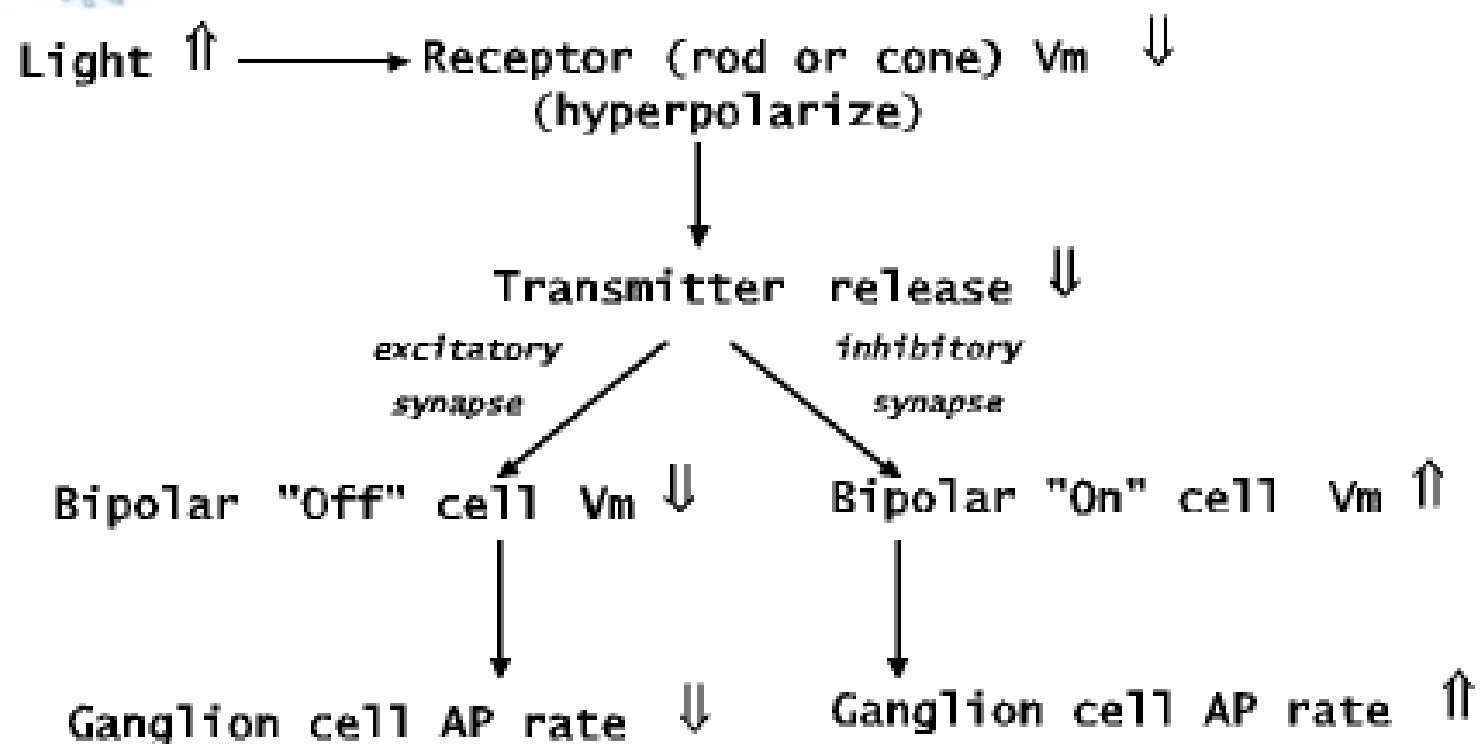
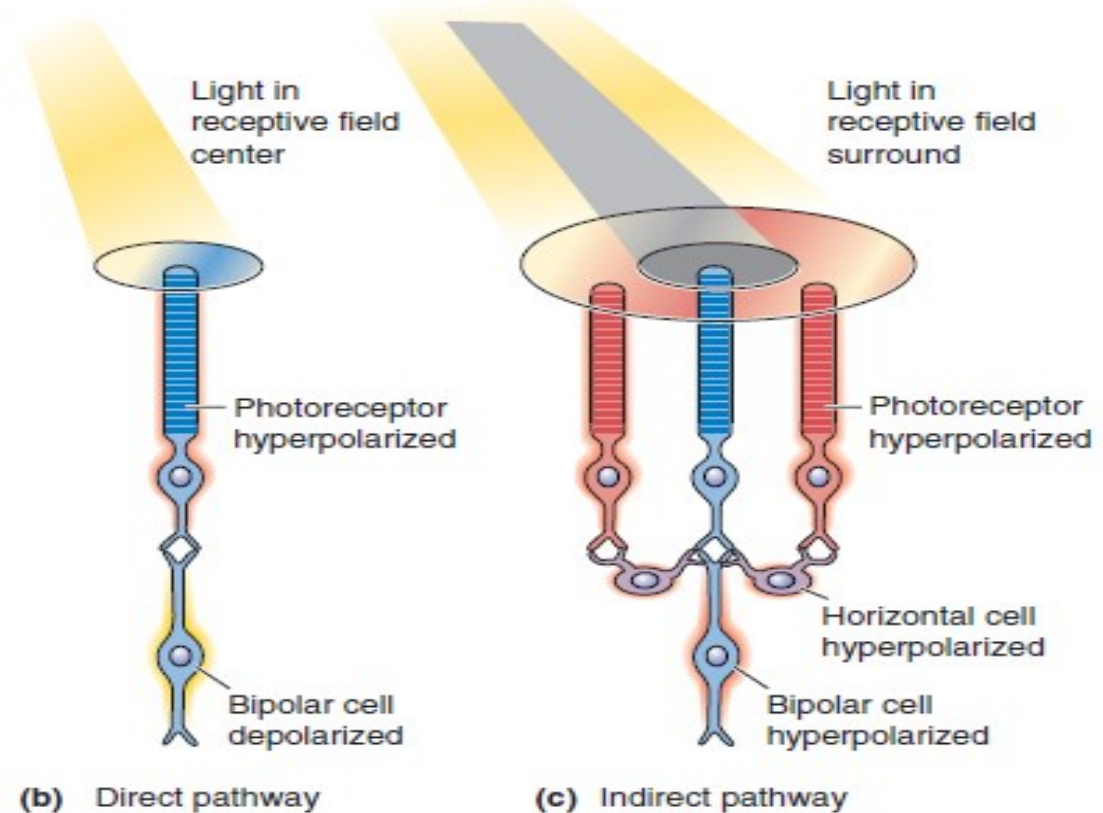
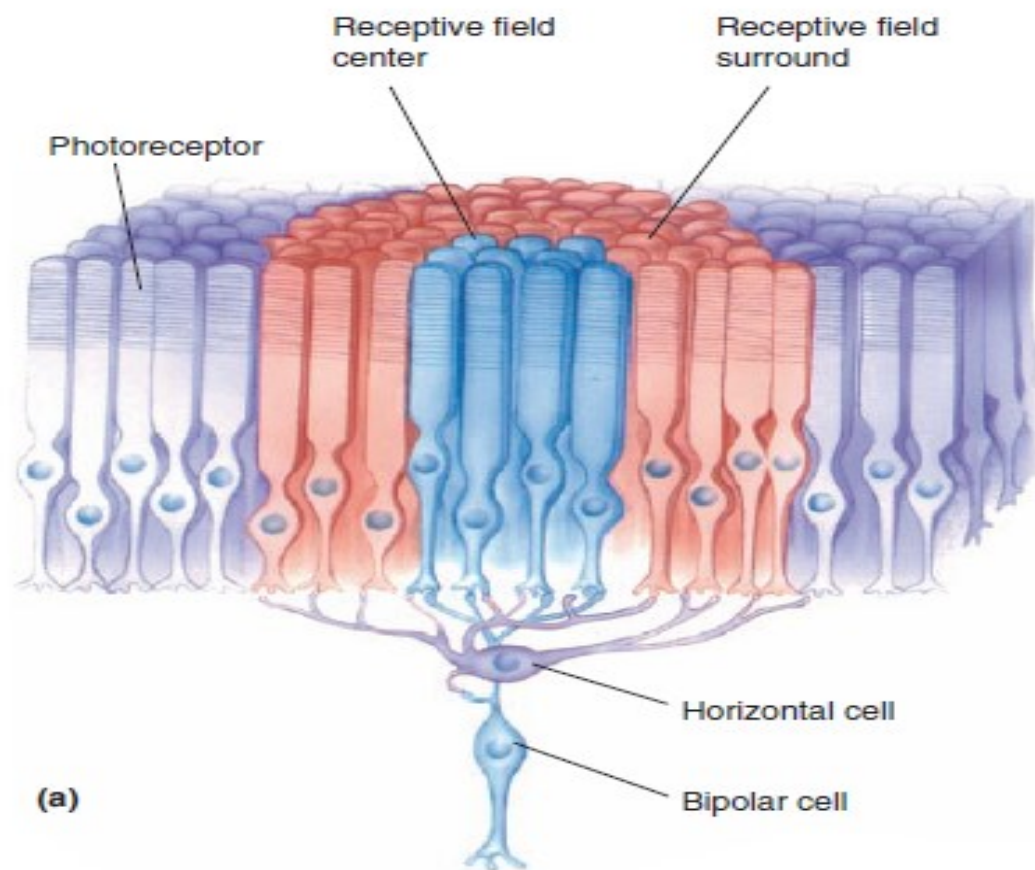
Receptive field of each ganglion cell occupies a circular region of retina.

- In the fovea where visual acuity is greatest- the receptive field may be only 10 μ m in diameter.
- Near the periphery of the retina, receptive fields are larger (~1 mm).

*1. The cones in the centre of the receptive fields - the field **centre-** convey information **directly** to the ganglion cell by synapsing with **bipolar** cells.*

*2. The cones at the periphery of the receptive field- the field **surround** reach the ganglion cell **indirectly** through **horizontal cells**.*

Reduced release of rod or cone synaptic transmitter may cause depolarization or hyperpolarization of bipolar cell, depending upon type of synapse with bipolar cell

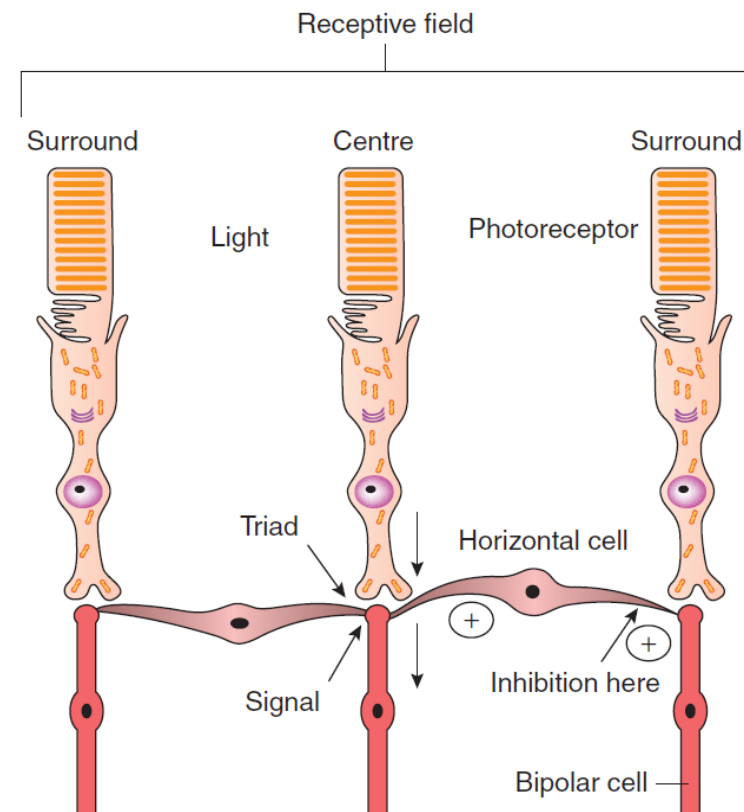


Neurotransmitter within the retina

1. Cones release glutamate

- Glutamate's inhibitory effect on bipolar cells (in **on-centre** receptive fields) is produced by opening K^+ -selective ion channels or by closing Na^+ -selective ion channels --- G- protein (second messenger system).

2. Horizontal cells release GABA: generally inhibitory



Intensity

- Light intensity is encoded by the *frequency of ganglion cell firing rate*
 - *At moderate light levels, both cones and rods are involved* in the stimulation of ganglion cells.

Rods activate cones through gap junctions, therefore, they communicate with ganglion cells through cone bipolar cells.

- *At very low light levels, only rods are active.*

The gap junctions between rods and cones are closed, therefore, the rods communicate with ganglion cells through rod bipolar cells.

*Because rods are not organized into **centre-surround** receptive fields, the ability to see contrast and form is diminished.*

Ganglionic cells and optic nerves

Ganglion cells have Two channels

1. M-cells (Magno ~ large, 40 %, Rods vision)

- Slow conducting
- Have broad receptive field in the peripheral retina
- Project to **kinio cellular layer of LGB**, superiors colliculus and hypothalamus.
- Responsible for
 - Crude rod vision at dim light
 - Subconscious activity of visual processing,
 - Directional sensitivity (movement) and circadian rhythm

2. X or P-cells (Parvo ~ small, cone vision)

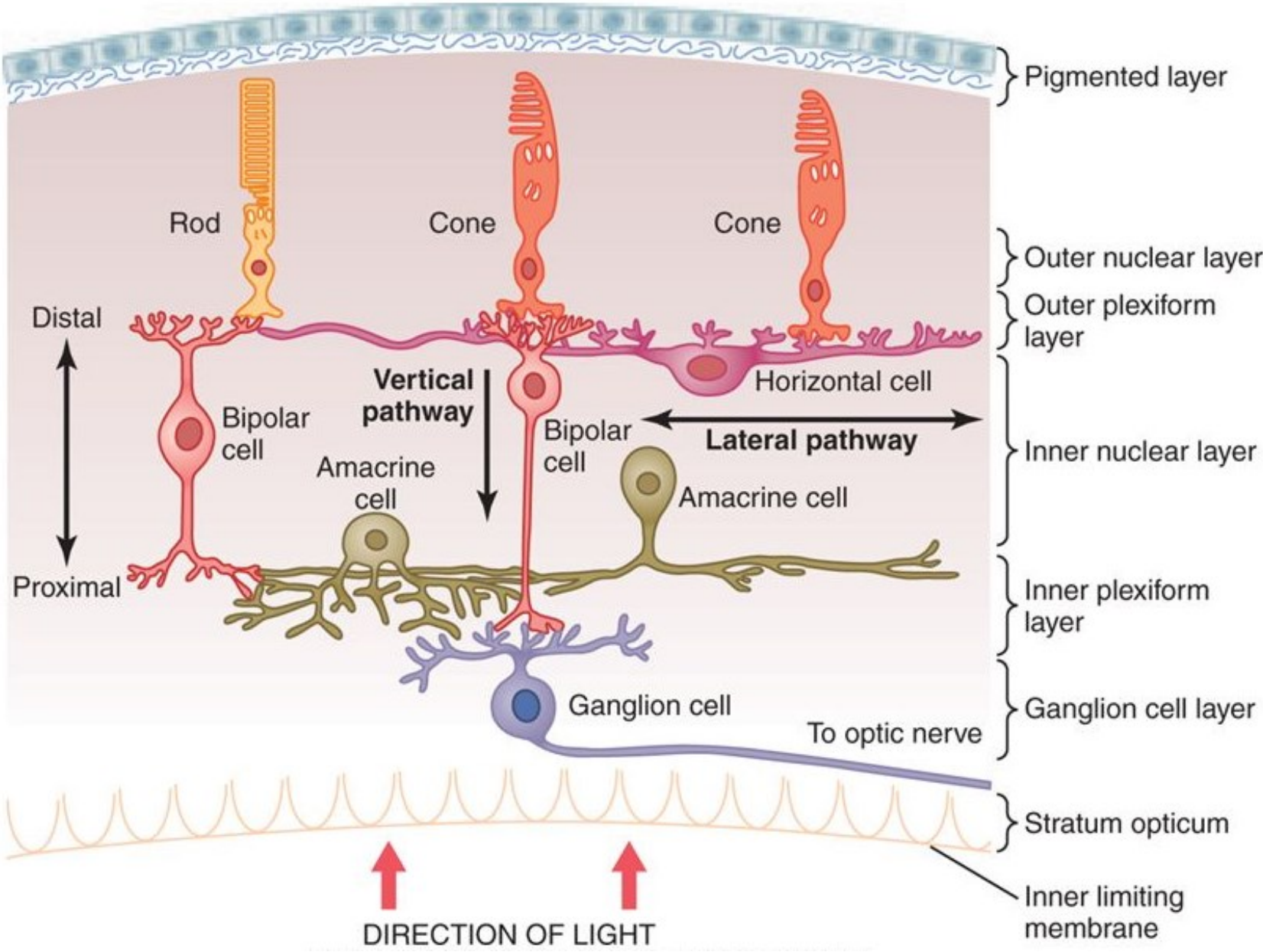
- 55% or more of ganglionic cells
- Have less dendritic arborization and **small receptive field**
- Medium sized fibers and better conduction velocity
- Transmit signal to **parvocellular layer of LGB**
 - **Color vision, Acuity and Detailed visual image**

Optic Pathways: Neural circuitry of the retina

1. **Photoreceptors** – Transmit signals/**electrotonic conduction**/ to outer plexiform layer (bipolar & horizontal cells)
2. **Horizontal cells** – Transmit from photoreceptors horizontally in outer plexiform layer to bipolar cells- enhance contrast/lat inhibition
3. **Bipolar cells** – transmit vertically in inner plexiform layer to ganglionic cells and amacrine cells (-/+ by glutamate)
4. **Amacrine cells**- transmit vertically to ganglionic cells and horizontally to bipolar cells (Analyze signal before leaving retina/**temporal processing**)
5. **Ganglionic cells**- output from the retina
 - are the first cells in the chain that generate APs

- 100 million rods/ retina (60)
- 2 million cones/retina (2)
- 1.6 million ganglionic cells (1)

Ratio= 60:2:1

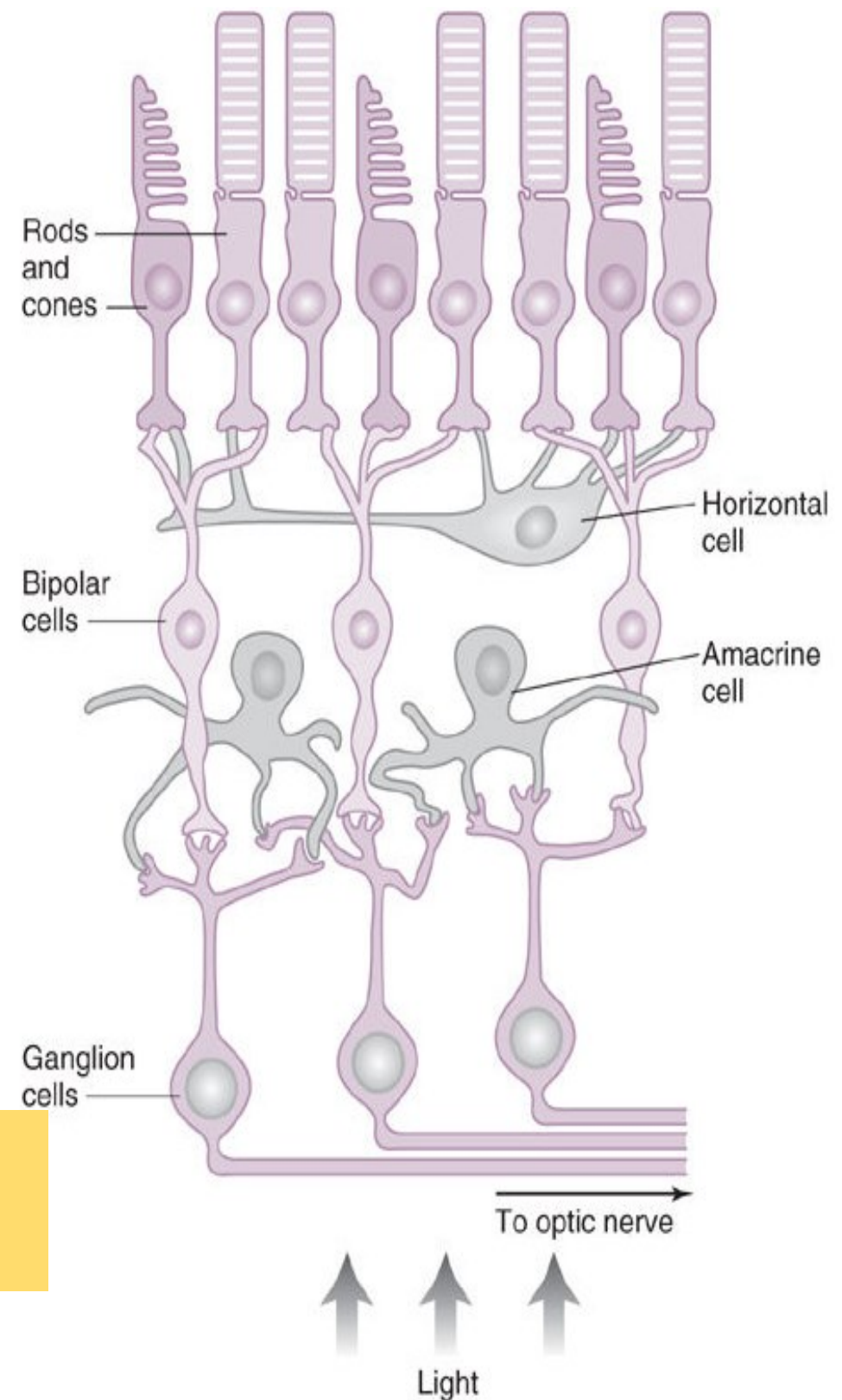


Information about light flows from

- Photoreceptors to
- Bipolar cells to
- Ganglionic cells

Horizontal and **amacrine** cells modify the response of bipolar cells and ganglionic cells via lateral inhibition

Provide second mechanism of lateral inhibition and contrast adjustment



Information Relay : Optic nerve -- Optic Tract -- Thalamus -- Visual Cortex

- visual information is relayed from the ganglion cells to the cortex by cells within the lateral geniculate nucleus/LGN.

Retinofugal fibers

Optic nerve

Exit at optic disk

Optic chiasma

Union from both eyes

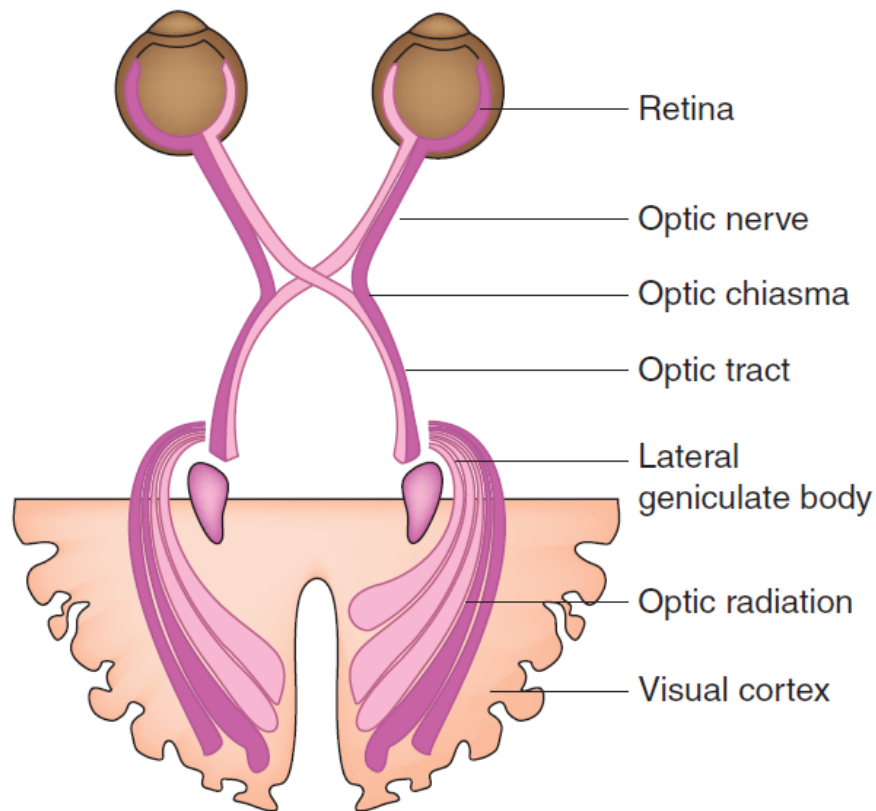
Anterior to pituitary gland

Decussation of axons from nasal retina

All fibers from temporal half pass through the lateral portion of the chiasm without crossing

Optic tract

Convey signal from opposite visual field



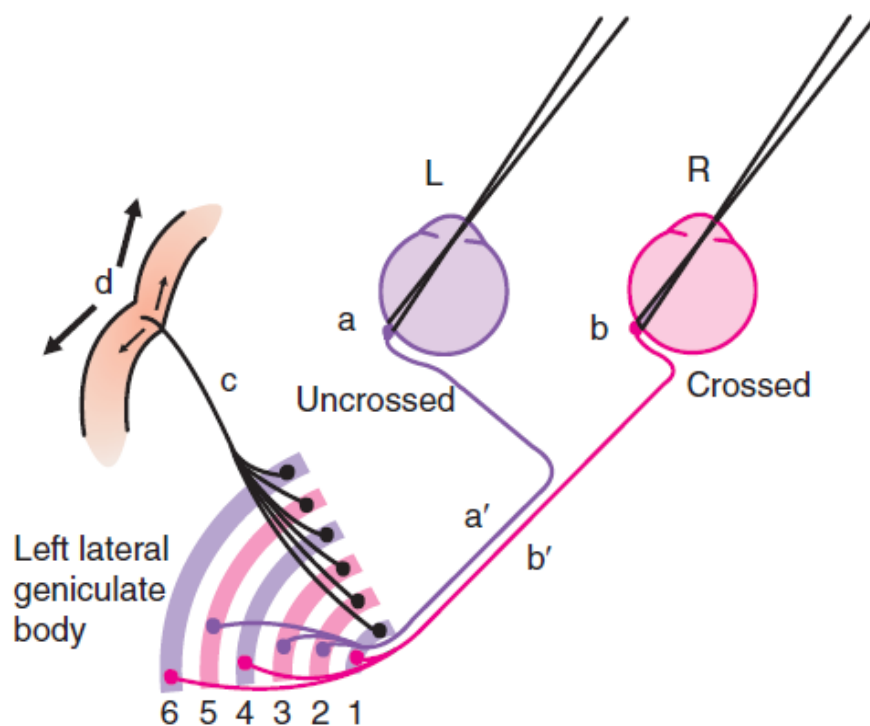
right visual field is conveyed in the left optic tract

Optic tract

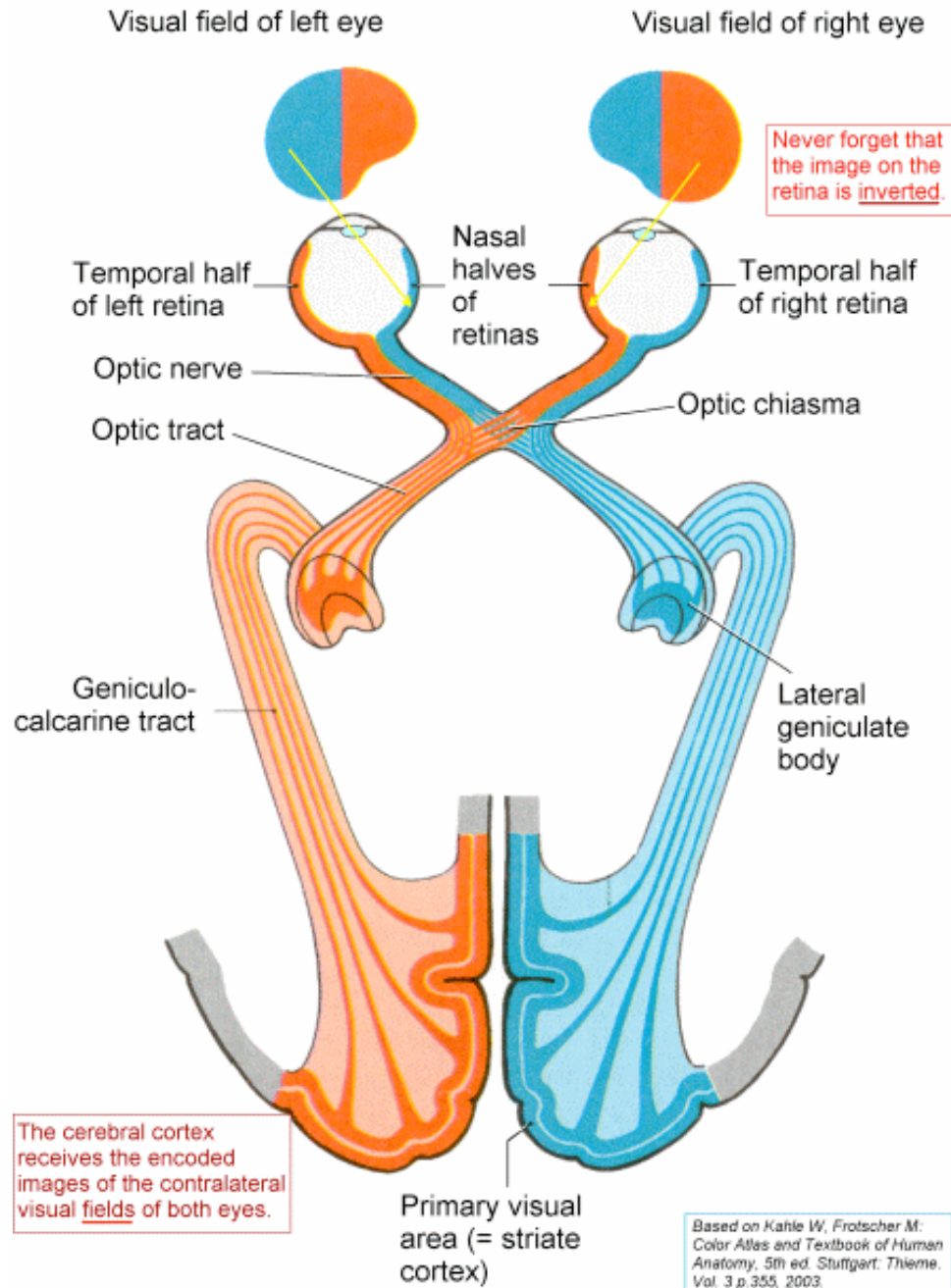
- Axons from the right half of each retina project to the right LGB and vice versa.
- *This means that fibers of the temporal retina on each side pass through the chiasm without crossing, while fibers of the nasal half of each retina decussate at the chiasm.*

Neurons in the lateral geniculate also retain on centre/off-surround (or the off-centre/on surround) organization of the ganglion cells.

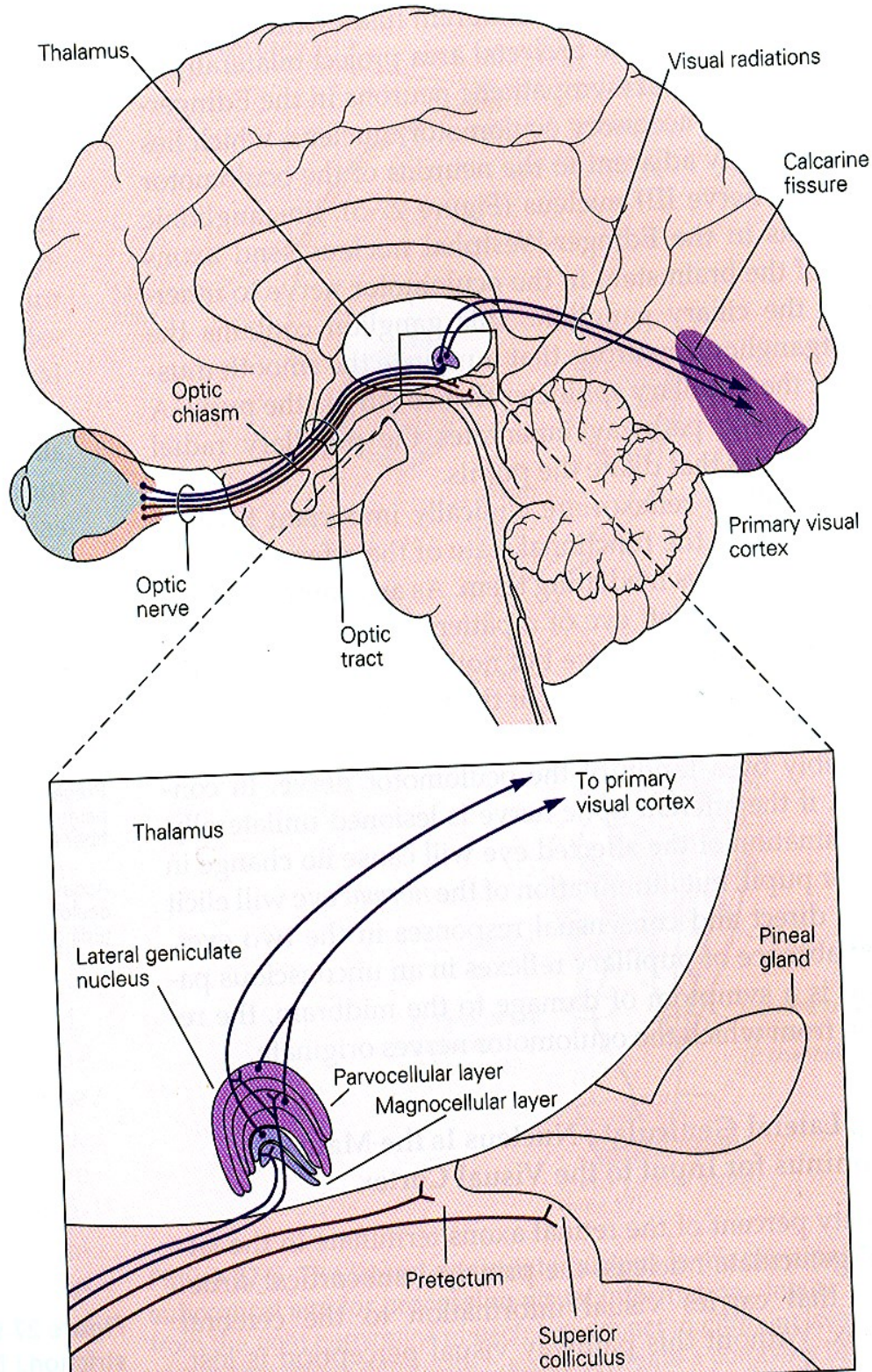
Fibers arising in a corresponding point of the retinas are located near each other (*~ precise point-to-point localization*).



On each side, layers 1,4 and 6 receive input from the nasal half of the contralateral eye, while layers 2, 3 and 5 receive input from the temporal half of the ipsilateral eye

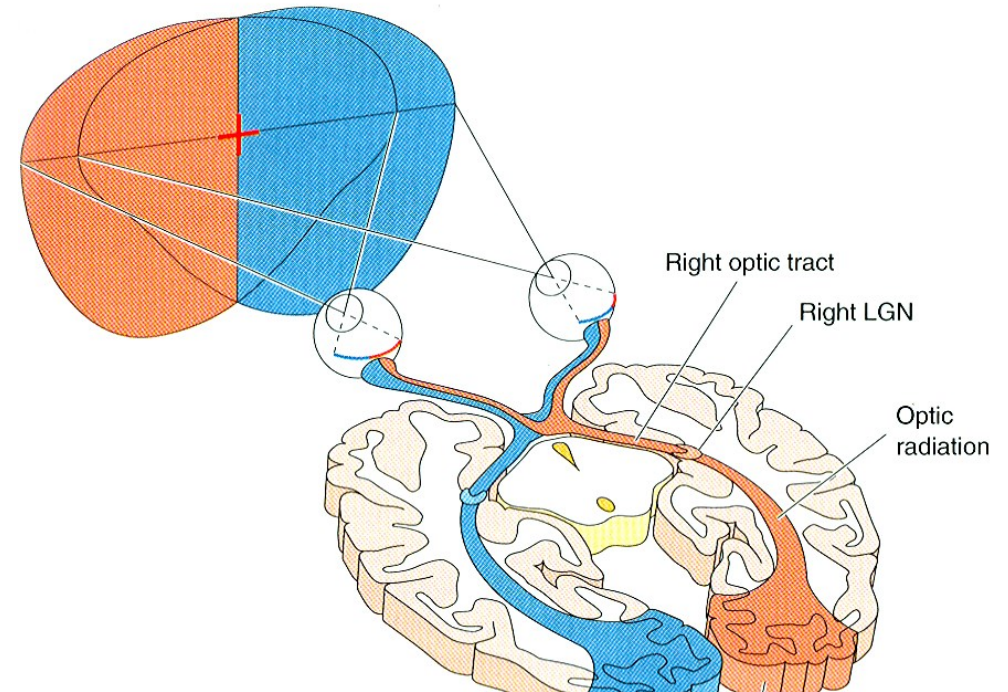
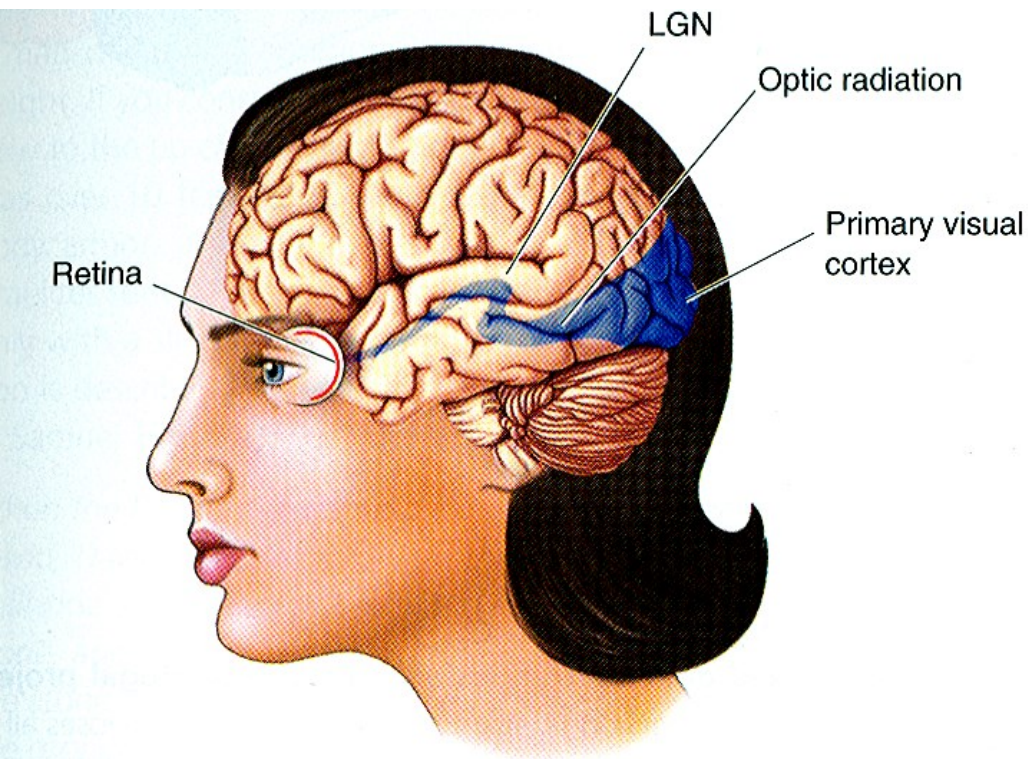


Targets of optic tracts



- Hypothalamus-
 - Small fraction (*diurnal, circadian cycle*)
- Midbrain- 10%
 - Pretectum (pupillary size)
 - Sup. colliculus
- Lateral geniculate body

Optic radiations



Projections from the LGB to visual cortex along retrolenticular part of internal capsule = *geniculocalcarine tract*

The optic radiations maintain a retinotopic organization in their passage to visual cortex (~ serial analysis of visual image)

- *impulses from corresponding points of two retinae meet.*
- *right visual cortex ~ to the left of the visual fields,*

Visual Perception

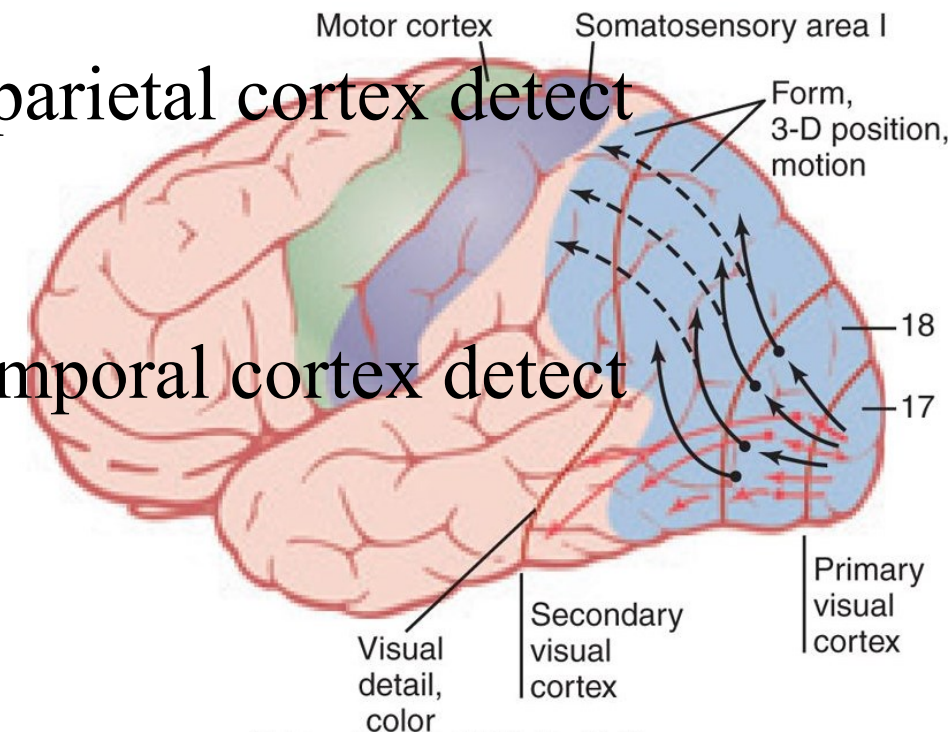
- The primary visual cortex (17) decodes about *contrast/color/depth*.
- Visual association area (18 is Visual area-II).
 - Dissecting, analyzing and meaning of visual image

- Superior occipital cortex and posterior parietal cortex detect

- **Form, 3-Dimension, Motion**

- Anteroventral occipital and posterior temporal cortex detect

- **Visual detail, color**

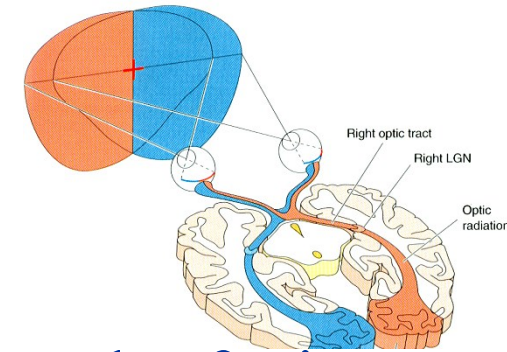


Parallel processing:

The various aspects of the visual image are decomposed and projected to a number of areas of the cortex for separate analysis

Stereopsis/depth perception/Binocular single vision

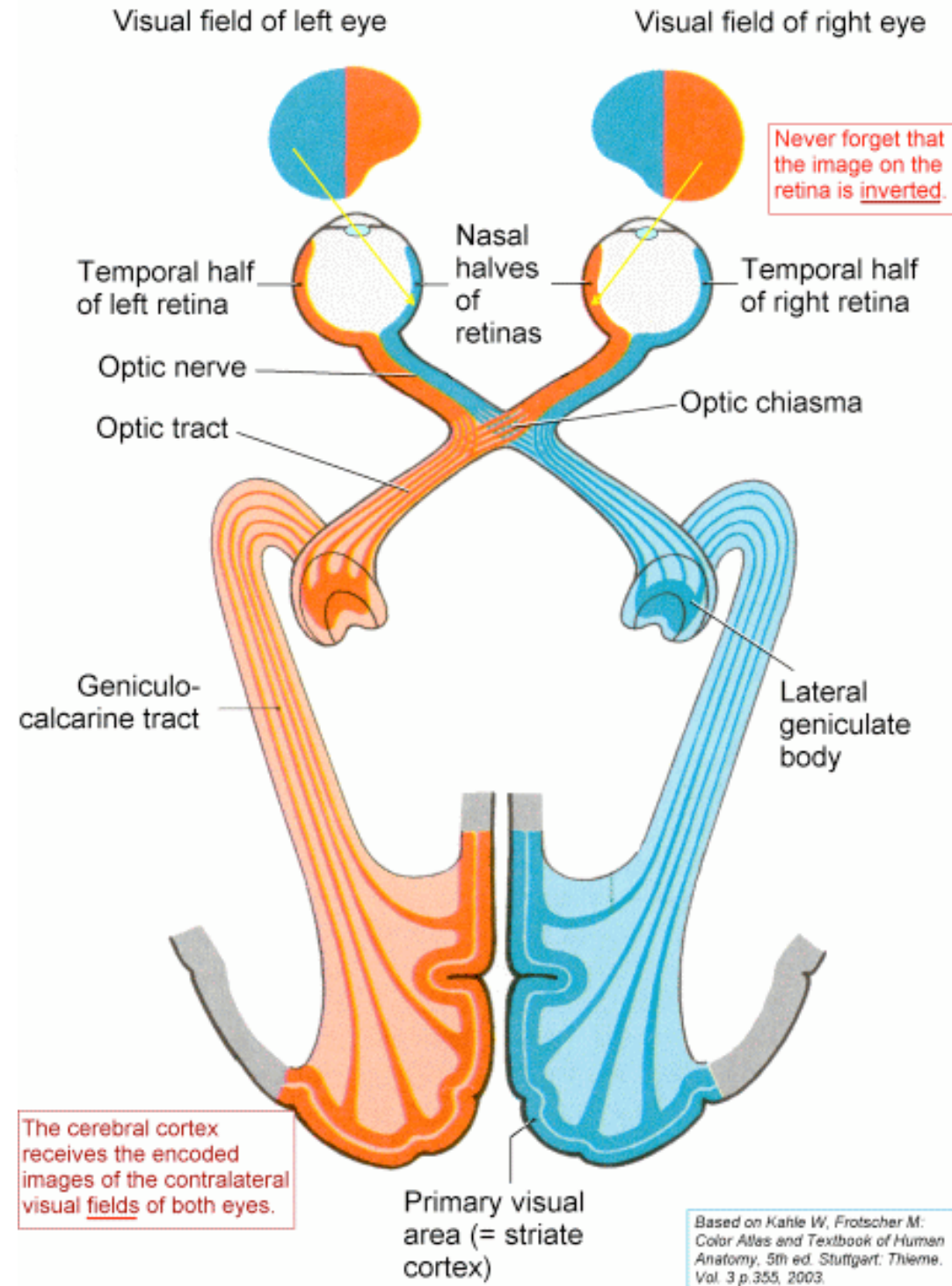
- Fusion of visual images from the two eyes fused at cortical level as one.
 - Makes visual perception more meaningful
 - When corresponding points of two retinal are not in register in a corresponding layer of cortex (un fused) excitation result in a corrective oculomotor induced eye movement
- The image on the two retinas are not exactly the same
- Right eye sees a little more of the right hand side
- The nearer the object, the greater the disparity (the lesser the fusion or register)



Depth perception (ie., stereopsis) is possible because each eye perceives a slightly different image.

Visual field defect:

- A. If the optic nerve of the left eye is cut, the eye loses all vision of same side (loss of light reflex but near/accommodation reflex is present).
- B. If the optic tract on the left eye is cut the right visual field of the eye loses vision (*homonymous hemianopia*)
- C. If the optic chiasm is split down in the middle, only the crossing fibers are damaged, and both eyes lose peripheral vision (*bitemporal hemianopia*)

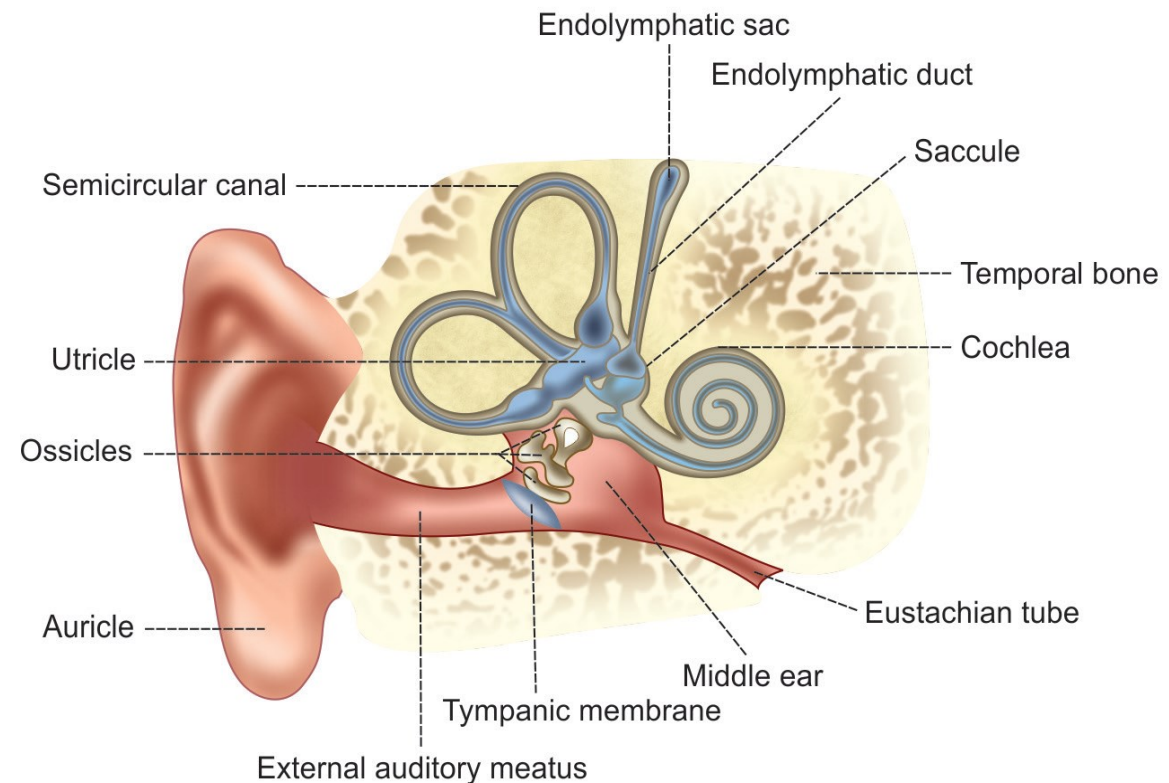


Special Senses

1. Vision
2. Hearing
3. Equilibrium
4. Olfaction
5. Taste

The Ear: Hearing and Equilibrium

- Three parts - the **inner**, **outer**, and **middle** ear
 - Outer & middle ear - involved with **hearing**
 - Inner ear - both **hearing and equilibrium**
 - *Respond to separate stimuli*
 - *Are activated **independently***

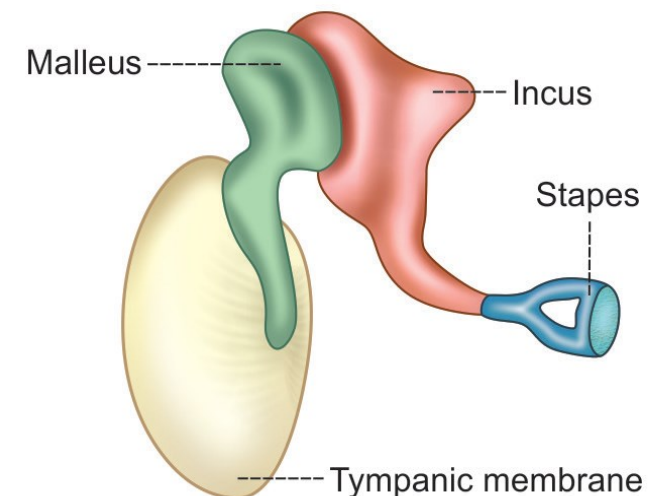


The Outer Ear

- A funnel for conducting vibrations to the tympanic membrane (eardrum)
- **Auricle (pinna)**
 - directs sound down the auditory canal
 - Sound localization
- **External auditory canal**
 - tube directs sound to tympanic membrane
 - filled with **cerumen** (ear wax)

Middle Ear (Tympanic Cavity)

- Air-filled, mucosa-lined cavity in the mastoid region of the temporal bone
 - *Volume = 2 ml*
 - *Flanked laterally by the **eardrum***
 - *Flanked medially by the **oval and round windows***
- **Contain**
 - *Tympanic membrane*
 - *Pharyngotympanic tube/Eustachian tube*
 - *auditory ossicles*
 - *stapedius (smallest muscle) and tensor tympani muscles*



Middle Ear (Tympanic Cavity)

Tympanic membrane (eardrum) – 1cm

- vibrates in response to sound waves from the auditory canal
- provides the *first mechanical step*
 - *Transfers sound energy to the middle ear ossicles*
 - *Amplifies sound (10 – 15 db) = resonator*

Pharyngotympanic tube/Eustachian tube

- Connects the middle ear to the nasopharynx
- Equalizes pressure in the middle ear with the external pressure
- If pressure is not equal → pain + impaired hearing
 - Ex. rapid pressure change during elevator riding, airplane takeoff
 - Fluid in middle ear due to infection/otitis media

Ear Ossicles - three small bones: *Malleus, incus and stapes*

–Transmit vibratory motion of the eardrum to the oval window

Sound amplification by ossicles and TM– *impedance matching*

Sound energy into mechanical vibration in fluid by matching the impedance/perilymph (60% transferred)

1. Ossicles act as levers

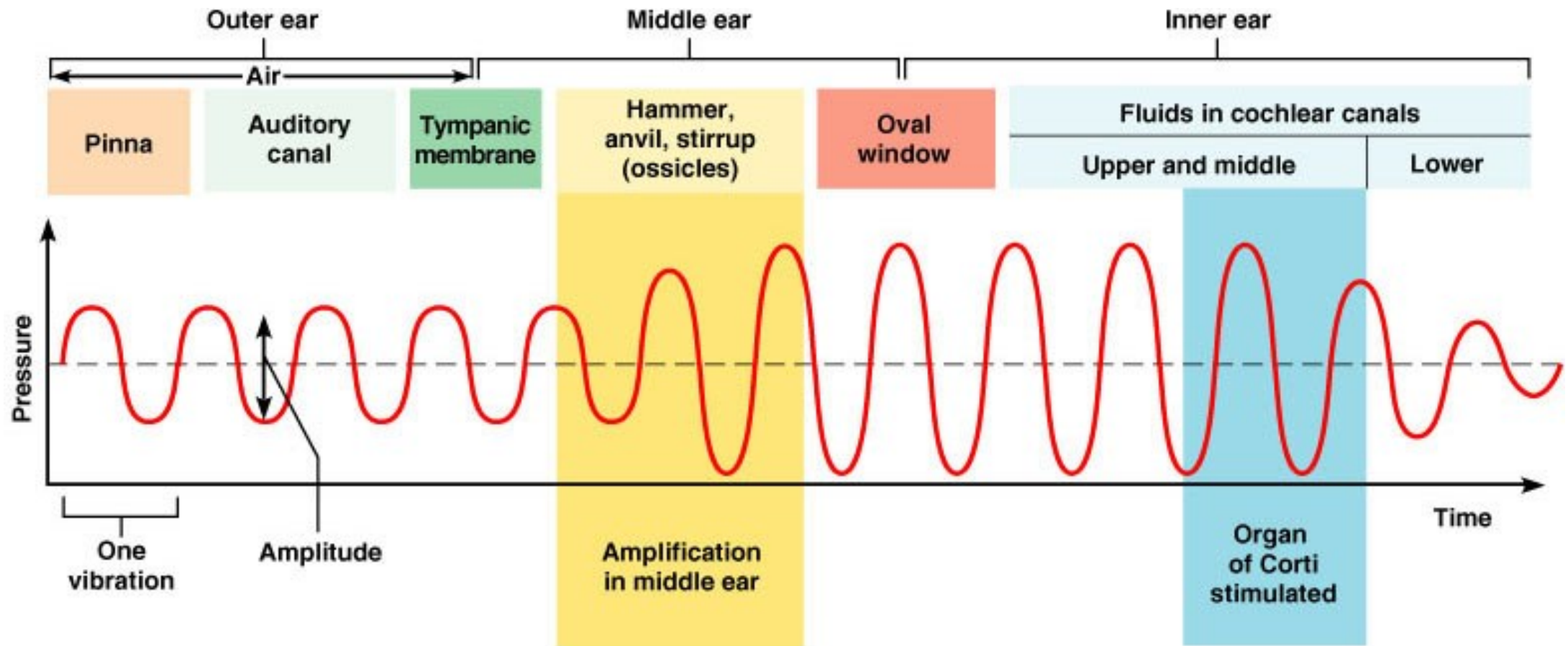
- lever ratio (M vs. I) $\sim 1.3:1 \rightarrow$ a slight gain in force (1.3x)

2. The surface area of the oval window smaller than TM

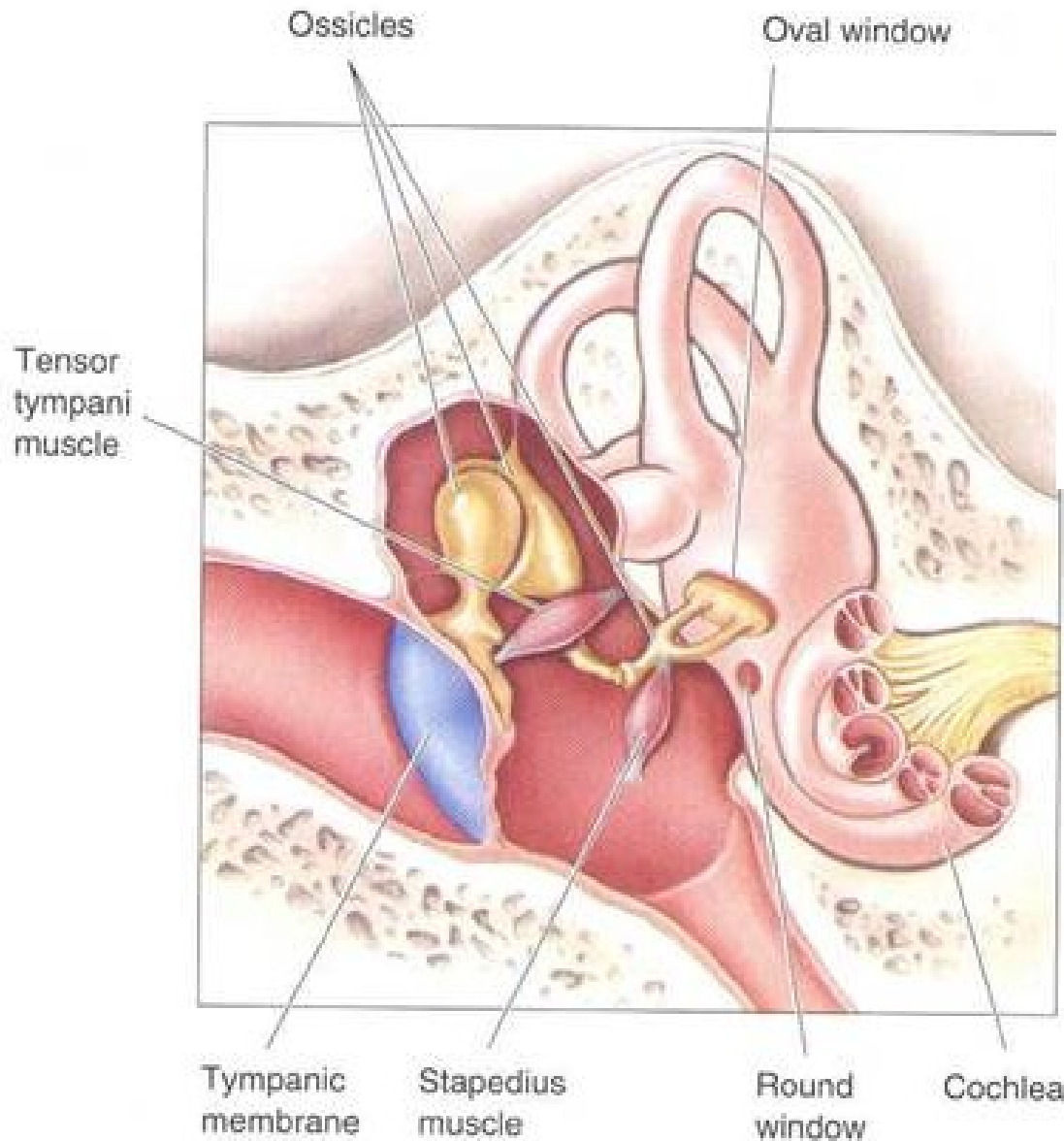
- 17:1 ratio
- the sound pressure at the oval window about 20 times greater than the pressure at the tympanic membrane = sufficient to move the fluid in the inner ear

Total an amplification of $1.3 \times 17 = 22$ db

Transmission of Sound to the Inner Ear



The middle ear muscles - *The Attenuation Reflex*

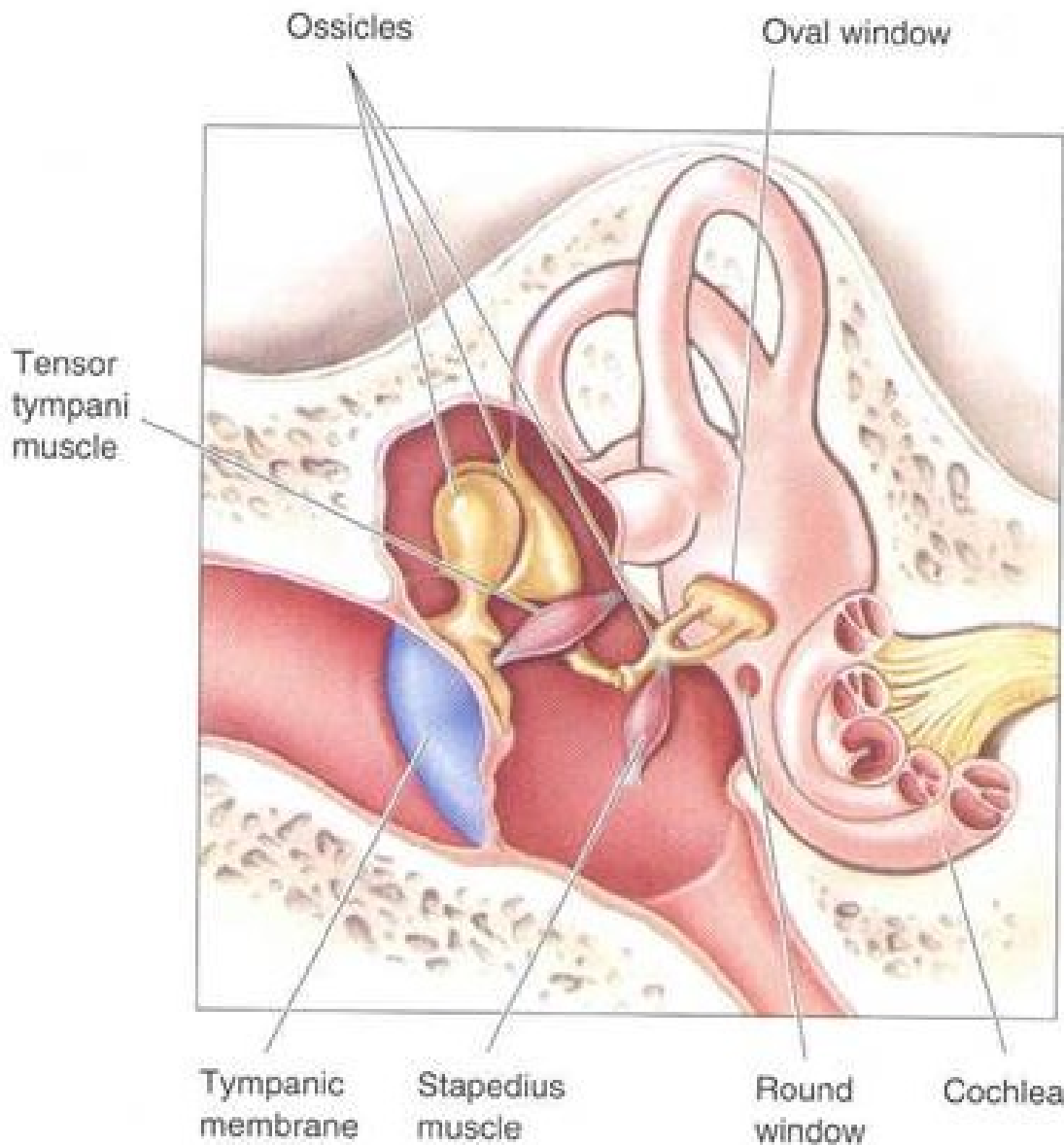


The middle and inner ear. Both the stapedius muscle and the tensor tympani muscle are attached to the wall of the middle ear at one end and to the ossicles at the other end.

Tensor tympani muscle is attached to the *malleus* and is innervated by the *trigeminal nerve (V)*.

The stapedius muscle is attached to the *stapes* and is supplied by the *facial nerve (VII)*.

The middle ear muscles - The Attenuation Reflex



- When the two muscles contract in response to *a moderate or loud sound*, the chain of ossicles becomes much more ridged and *sound conduction to the inner ear is greatly diminished*
- **Reduce the transmission of sound to the inner ear and → protect its delicate structures**
- Cannot provide protection from **sharp or sudden bursts of sound.**
- Adaptable (cant work > 10 min)

Bone conduction

- Sound transmission can bypass the ossicular chain .

Ex. If a vibrating object, (tuning fork) is placed against a bone of the skull (the mastoid), the vibrations are transmitted mechanically to the fluid of the inner ear – normal hearing process occurs.

- Used as a means of diagnosing hearing disorders.

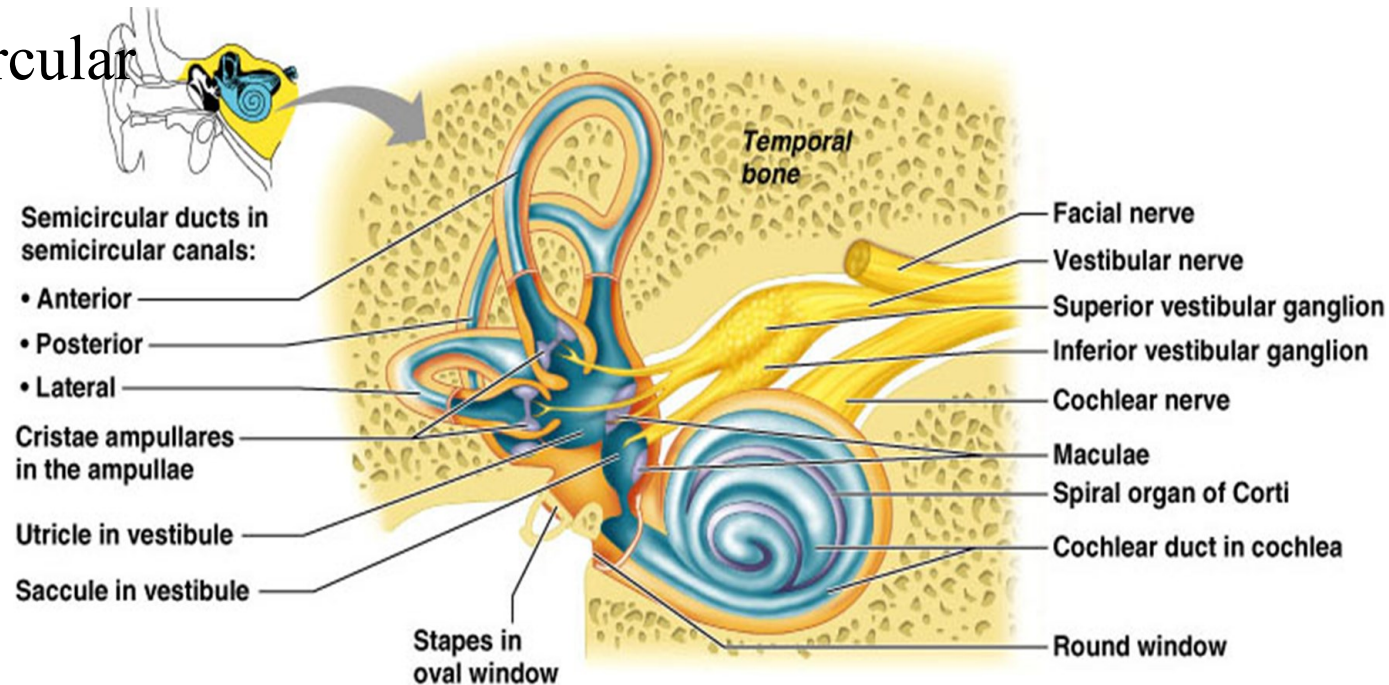
ex . Due to lesions in the ossicular chain.

Inner Ear = labyrinth

- Fluid-filled cavity within the temporal bone
- *Energy transduction by the mechanoreceptors takes place here*

Bony labyrinth: vestibule, semicircular canals and the cochlea.

Membranous labyrinth: filled with endolymph surrounded by perilymph



■ Components :

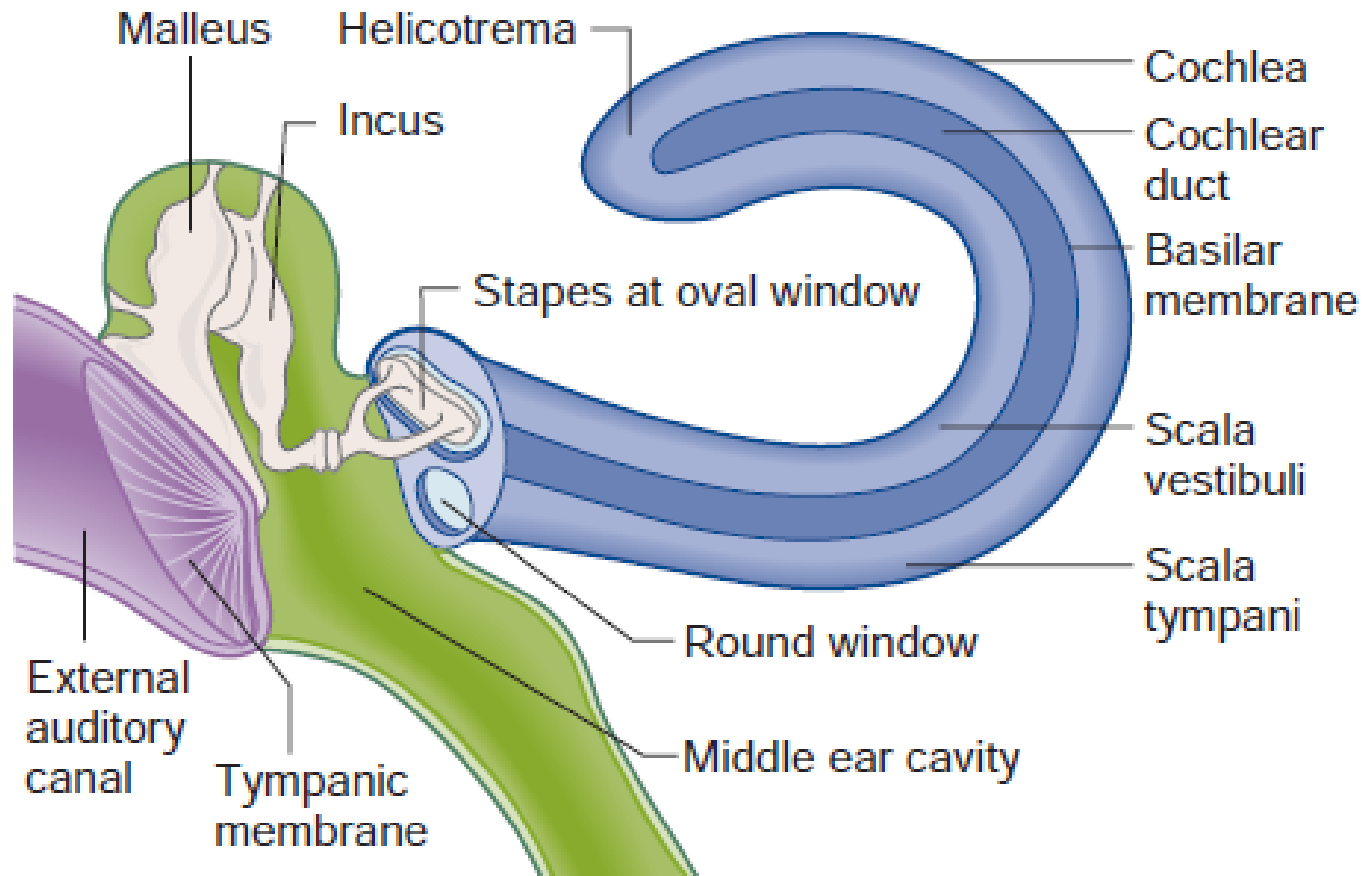
- **Vestibular apparatus**

- Contain vestibular sensory receptors (SCC, U, S)

- **Auditory apparatus**

- Contain sound sensory receptors/cochlea

The Cochlea



- A spiral, conical, bony chamber
- 35 mm long and 2 mm in diameter.
- Contains the hearing receptors - **organ of Corti**

- *At the base are two membrane-covered holes*
(oval & round windows)

The cochlea is divided into three chambers:

Scala vestibuli

Scala media (Cochlear Duct)

Scala tympani

The Cochlea

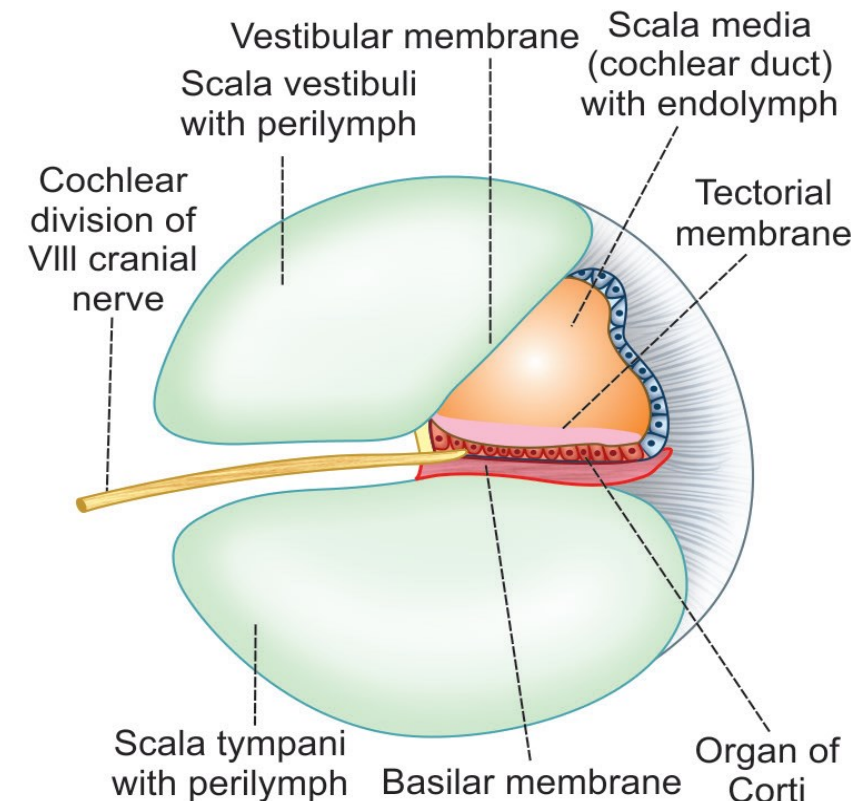
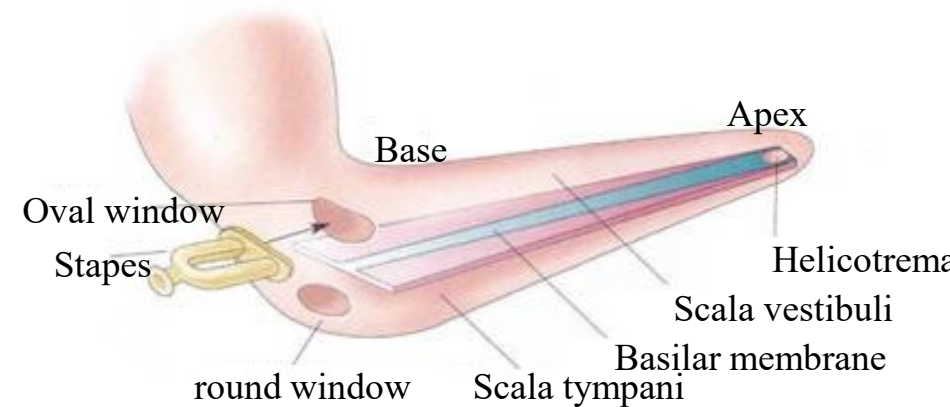
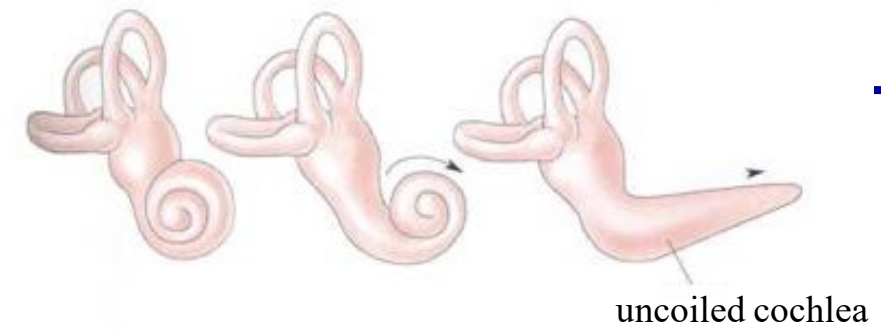
- **Scala tympani**
 - Terminates at the round window
- **Scala vestibuli**
 - The chamber that meets the oval window At the base of the cochlea
- **Scala media** is closed off at the apex of the cochlea (= cochlear duct)

The scalas tympani and vestibuli:

Are continuous with each other via the **helicotrema**

Are filled with *perilymph w/c bathes bottom of hair cells*

Scala media is filled with **endolymph** and bathes the tops of hair cells/**organ of corti/BM**



The Cochlea - membranes separate chamber

■ Basilar membrane

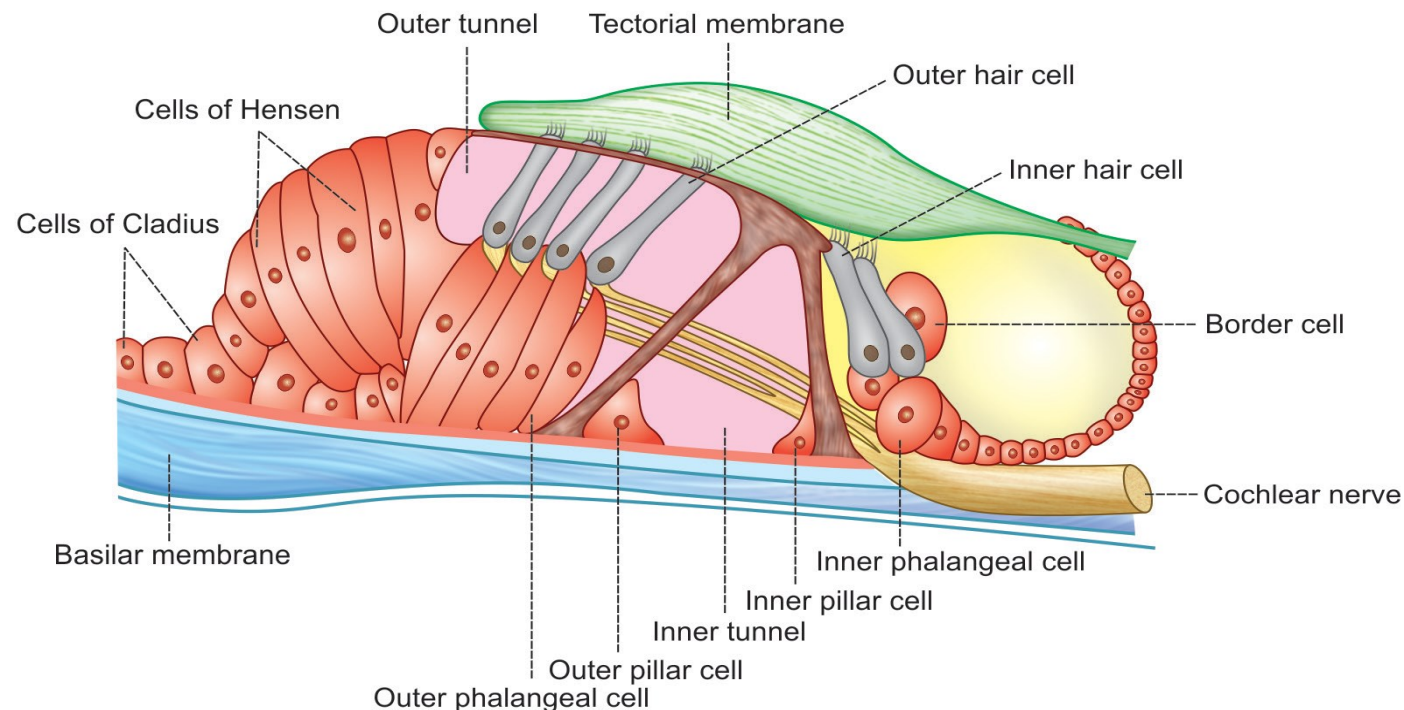
- contain the **organ of Corti**/ auditory receptor neurons = **hair cells**

■ Tectorial membrane

- hanging over the organ of corti

■ Stria vascularis

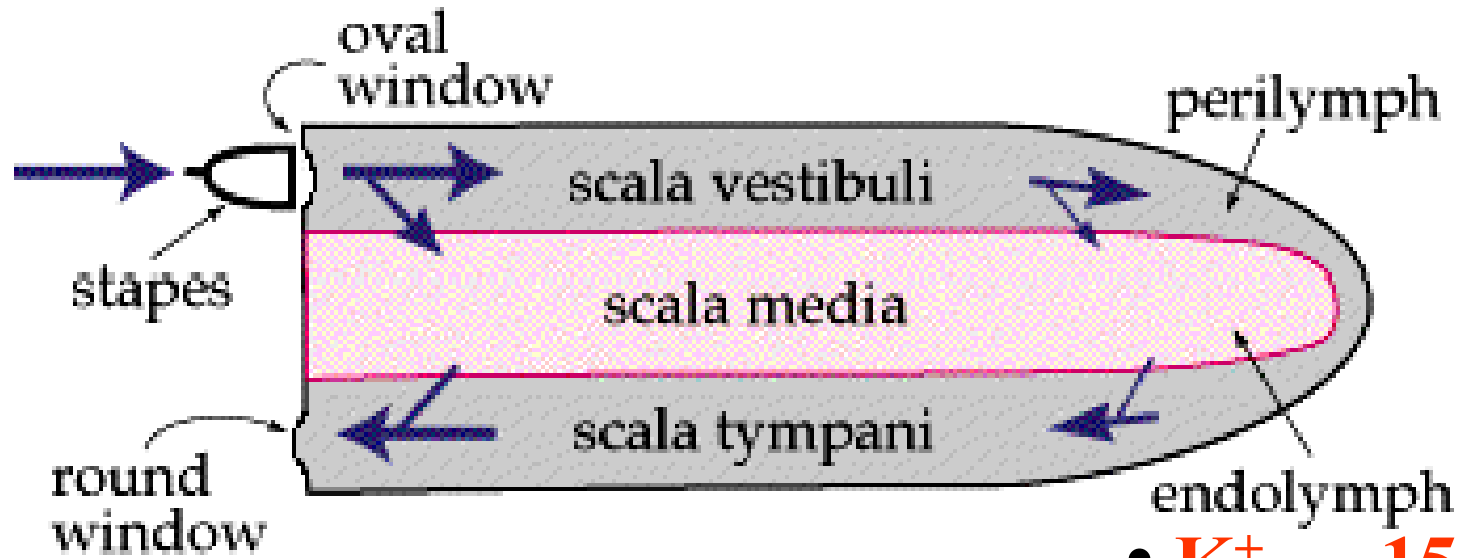
- vascular tissue /along the outer wall of the scala media
- *Secretes the endolymph (high K^+)*



Function of Cochlea

- Change *mechanical vibrations in fluid into action potentials* in the CN VIII.
- Sound vibrations created in the fluid cause movement of the basilar membrane
- Increased displacement
 - increased neuronal firing
 - *some hair cells only activated at high intensity*

Cochlear Fluid



- **K⁺** 7mM
- **Na⁺** 140mM
- ~~~ CSF

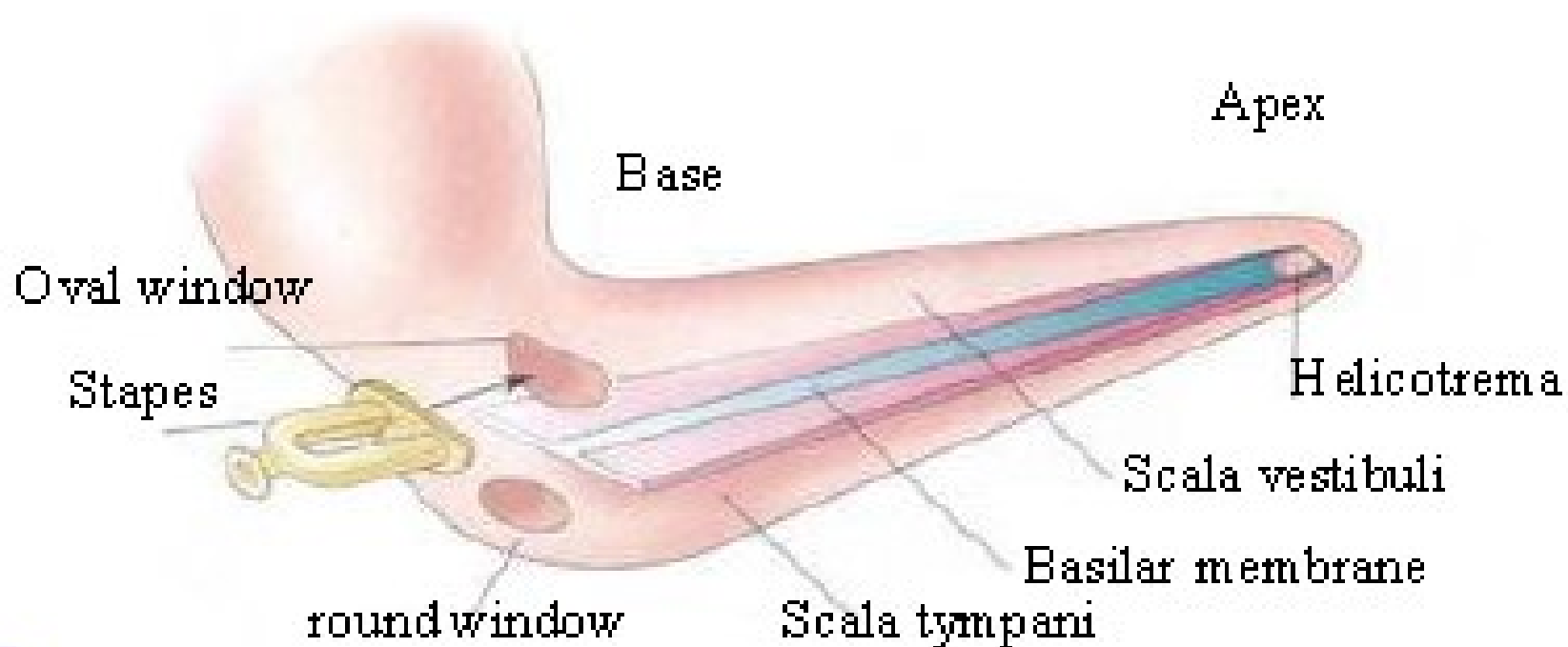
- **K⁺** 150mM
- **Na⁺** 1mM

Endocochlear/Cochlear microphonic potential

- the endolymph - about 80mV more positive than the perilymph due to *active pump of K⁺ into endolymph (~ ICF) by striae vascularis*
- enhances auditory transduction and sensitivity

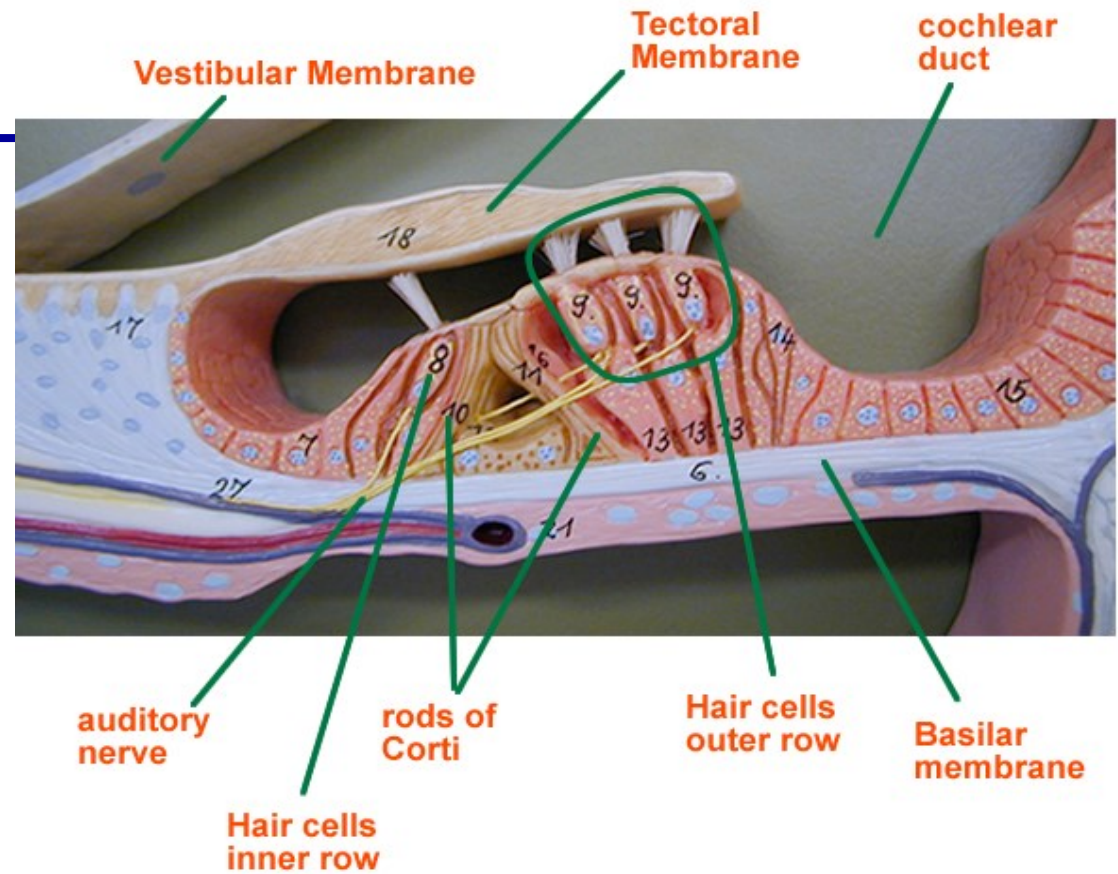
Basilar membrane

- The *stiffness & width* of the membrane vary with its length
 - Wider at the apex than at the base
 - The stiffness decreases from the base to the apex
 - The base is stiffer than the apex



The Organ of Corti

- Is composed of
 - *supporting cells*
 - *outer hair cells*
 - *inner hair cells*
- Contains the cell bodies CN VIII
- **The stereocilia (hairs):**
 - Rod-like structures At the apical end of each inner hair cell
 - Protrude into the endolymph and Touch the tectorial membrane
 - strands link the free ends of the stereocilia together → the bundle tends to move as a unit



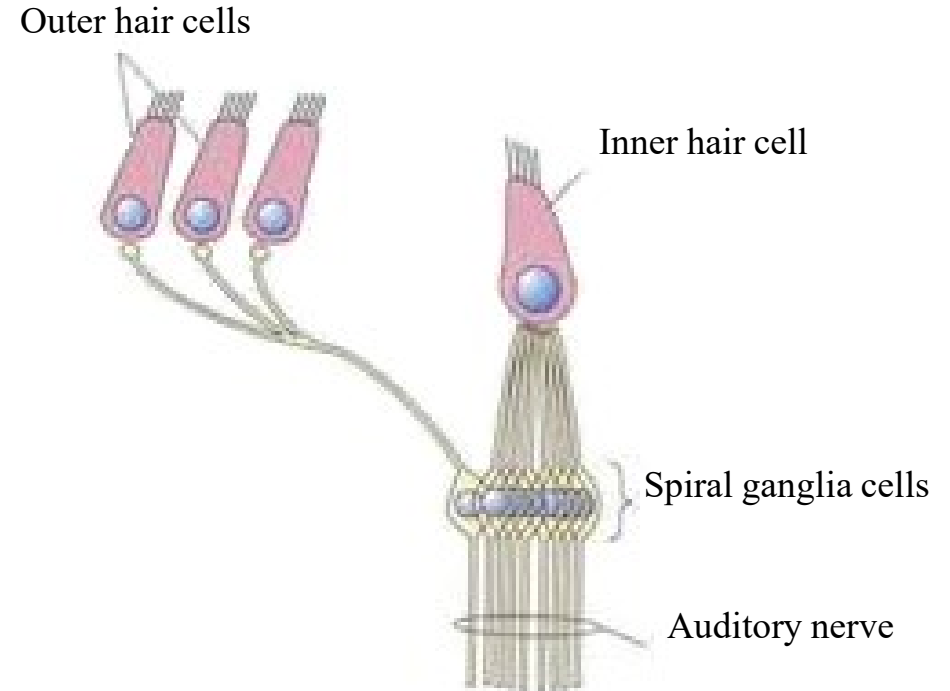
Hair cells

■ Inner hair cells

- one row/about 3,000
- receive *95% of the afferent fibers*
- Site of *fine auditory transduction*

■ Outer hair cells

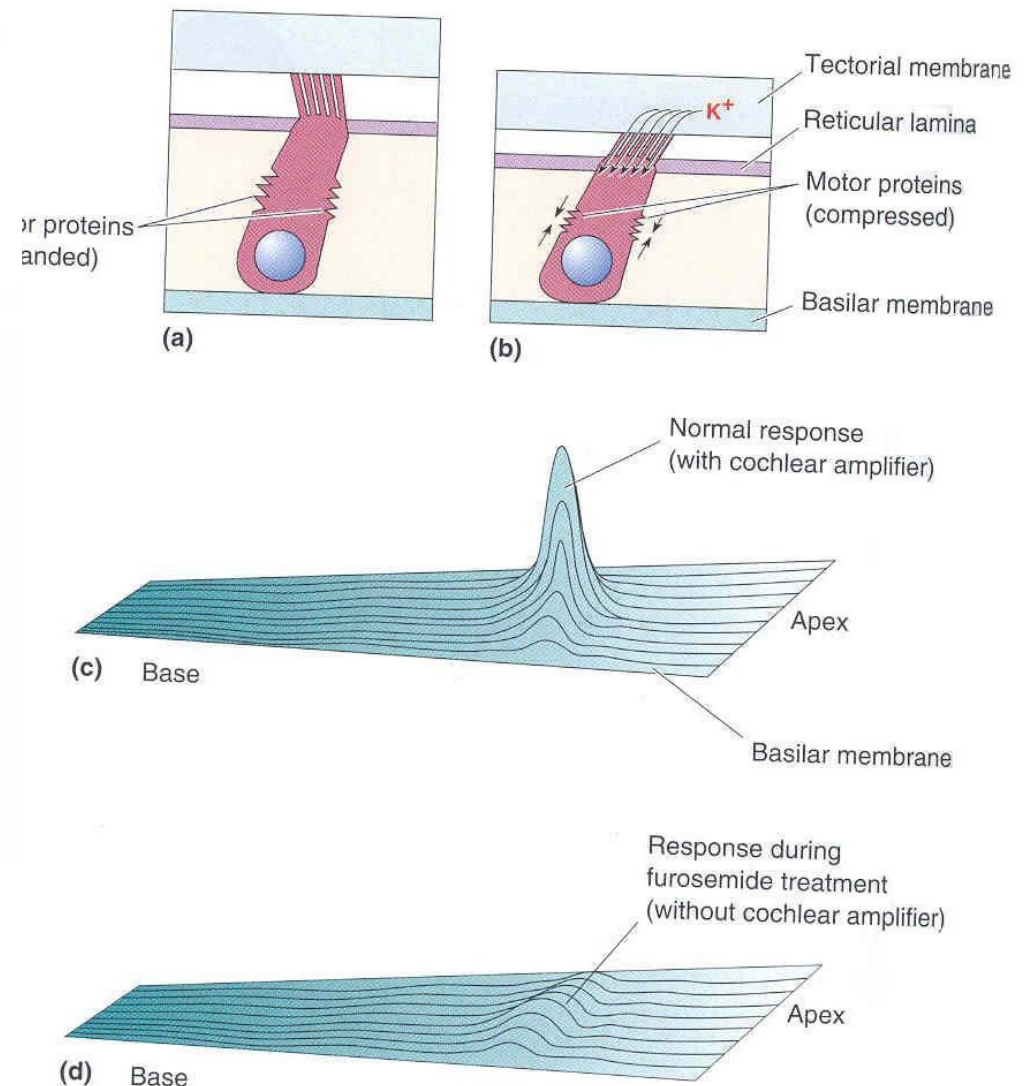
- three or four rows
- about 15,000 – 20,000
- detecting the presence of sound.
- Receives 5% of the afferent fibers
- mainly receive efferent innervation from the olivary complex and are concerned with modulating the function of inner hair cells.



The majority of the information leaving the cochlea comes from the inner hair cells.

Efferent innervations of the cochlea

Amplification by outer hair cells. (a) Motor proteins in the membranes of outer hair cells. **(b)** When potassium enters the hair cell, motor proteins are activated and shorten the hair cell. **(c)** The conformational change of the hair cell increases the bending of the basilar membrane and is therefore called cochlear amplification. **(d)** Furosemide decreases hair cell transduction, consequently reducing the bending of the basilar membrane.



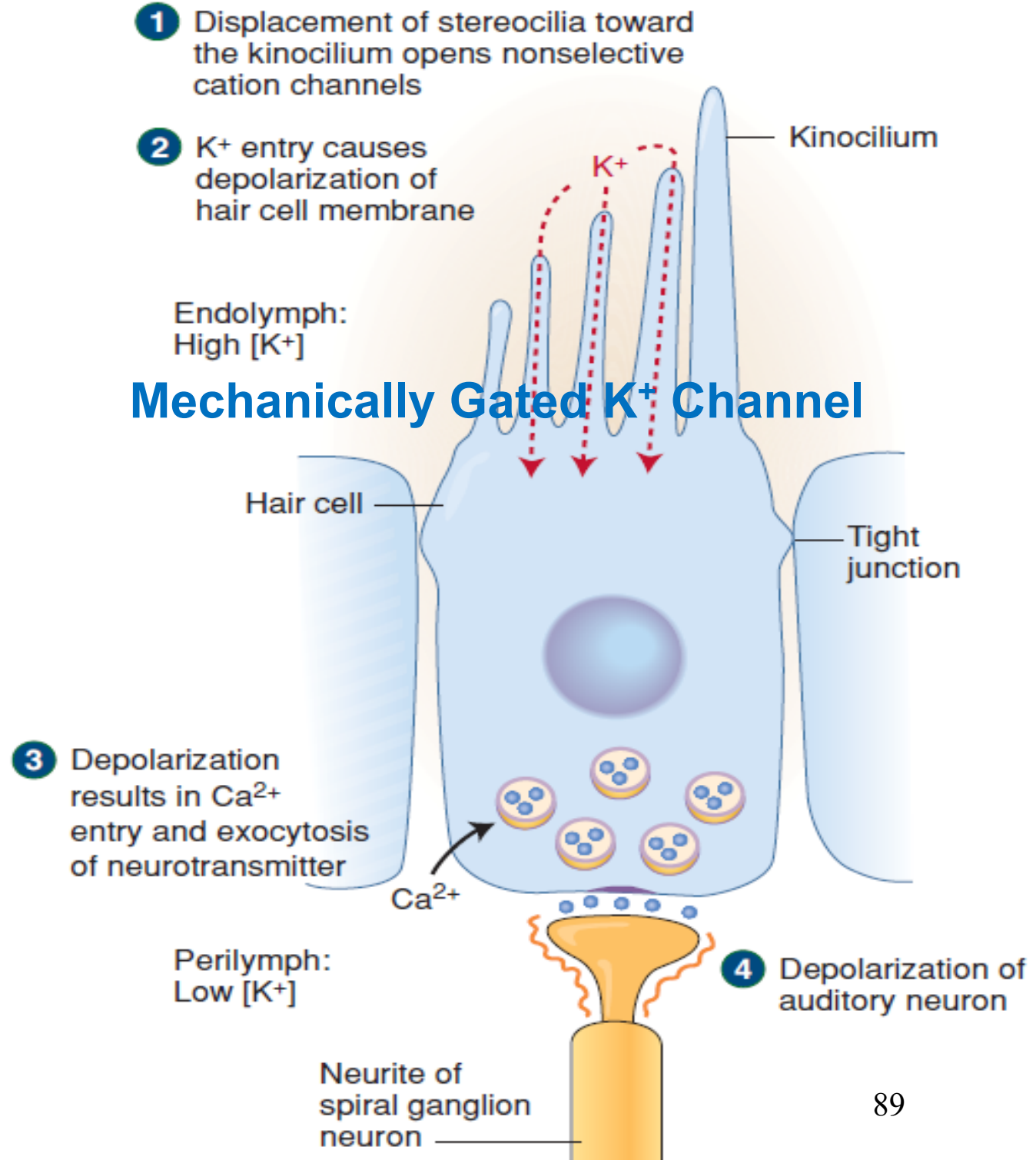
Cochlear Tuning (Cochlear amplification)

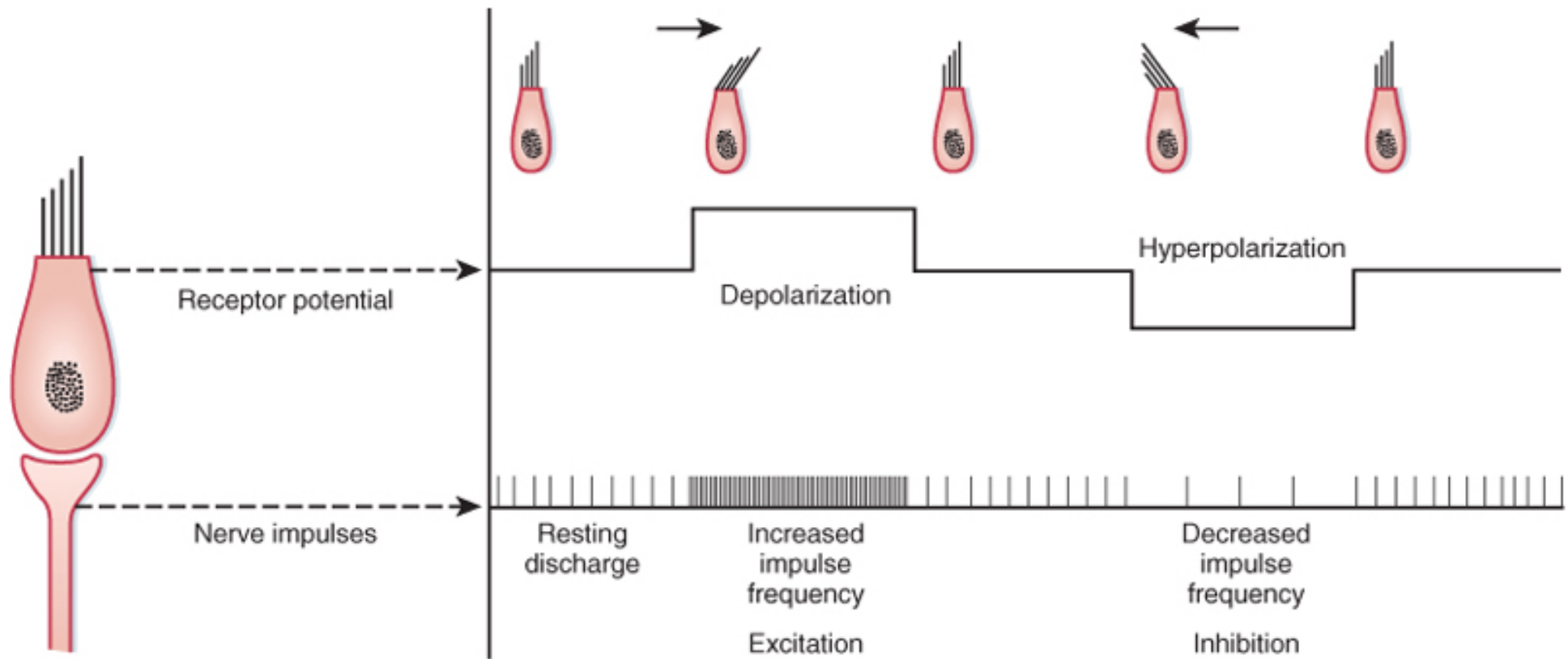
Increases ability of cochlea to receive some sound frequencies

Mechanical Transduction in the hair cells of the ear

- Sound causes hair cells to *bounce and touch tectorial membrane* causing *bending of the stereocilia*
- Bending cilia:
 - Opens **mechanically gated ion channels**
 - Causes a **graded potential** and the release of a neurotransmitter (**glutamate**)
- The neurotransmitter causes cochlear fibers to transmit impulses to the brain, where sound is perceived

Mechanical Transduction in the hair cells of the ear





- When stereocilia bend in the **direction of the tallest** one, the increased *tension on stereocilia* open the K^+ channel more, more Ca^{2+} in, more neurotransmitter release, increase the *frequency of AP*.
 - *The greater the intensity of sound, the greater the bending of the stereocilia.*
- When stereocilia bend away from the tallest one, K^+ channels close and the results are opposite.

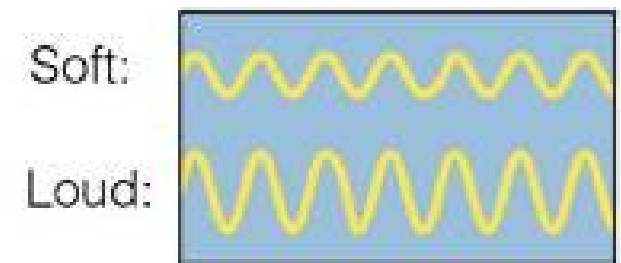
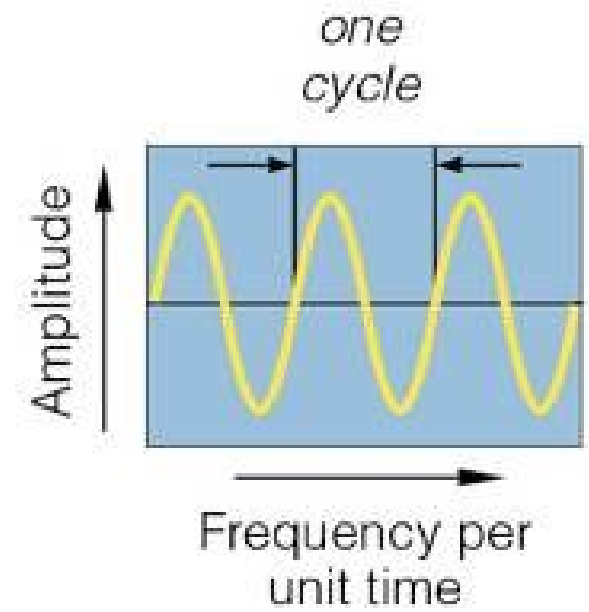
Properties of Sound

Pitch

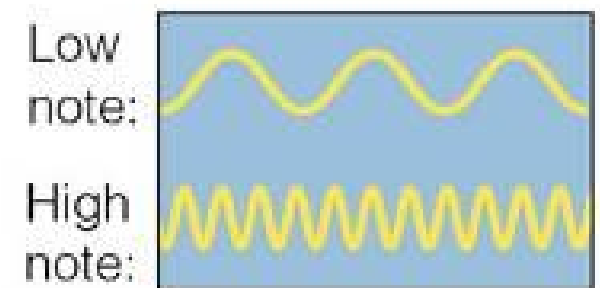
- Depend on perception of different *frequencies/Hz*
- Sense of a sound as “*high*” or “*low*”
- We hear from 20–20,000 Hz (↓ with age)

Loudness

- Amplitude
- Subjective interpretation of sound intensity (*amplitude* in decible/dB)

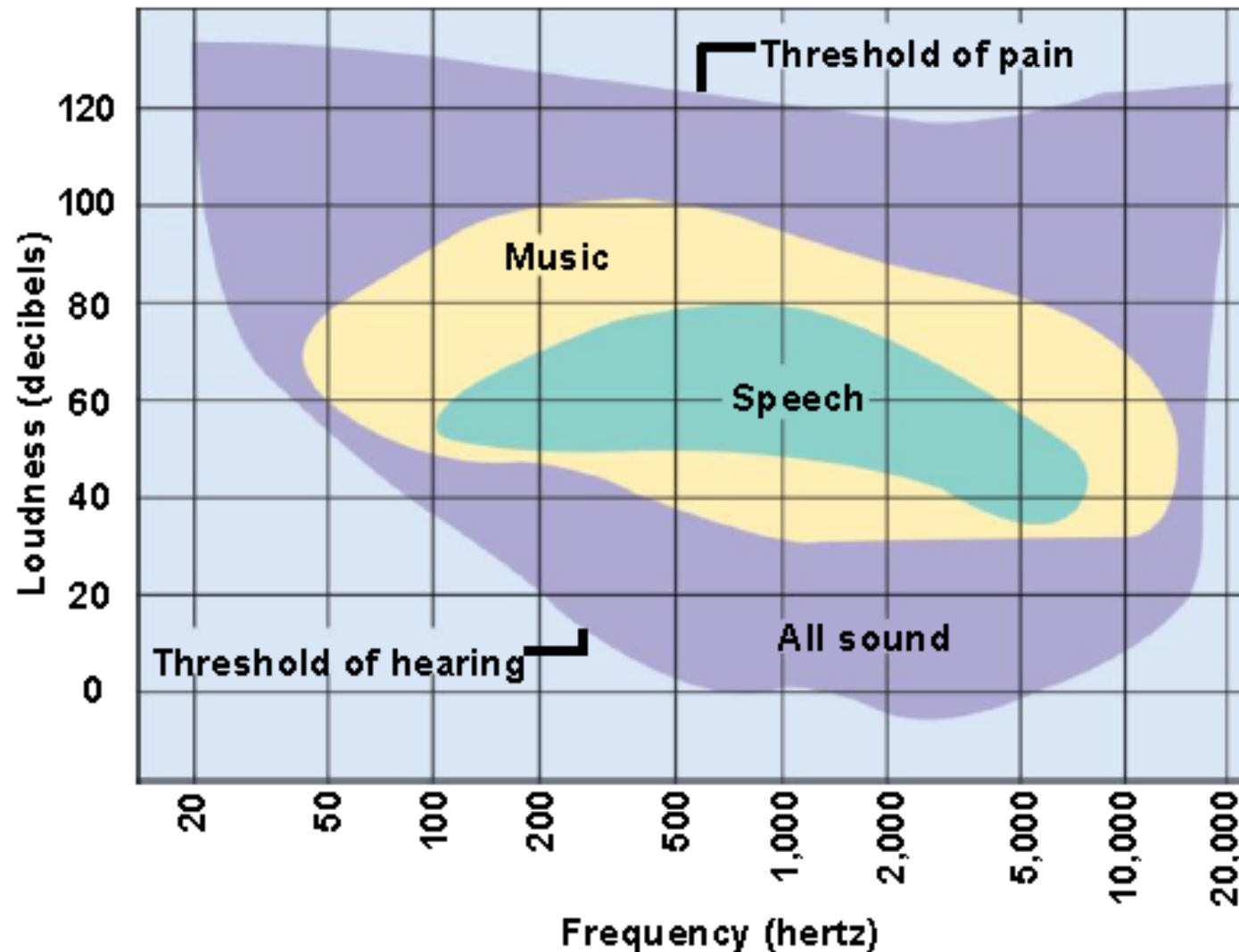


Same frequency,
different amplitude



Same loudness,
different pitch

Properties of Sound



- 60 dB- normal conversation
- 130 dB- threshold of discomfort
- 140 -160 dB- threshold of pain

Sound and Mechanisms of Hearing

Sound wave moves the tympanic membrane



Tympanic membrane moves the ossicles



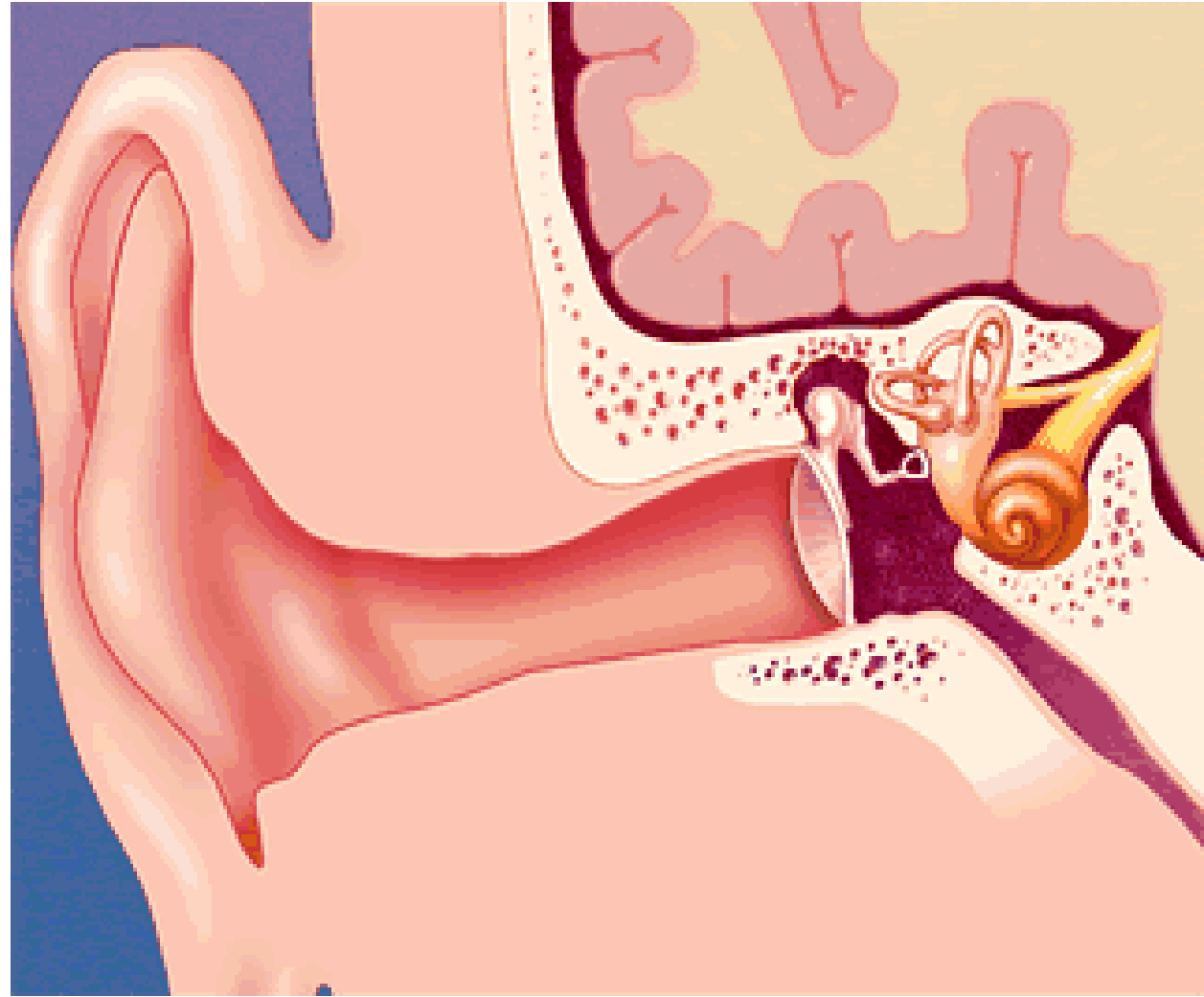
Ossicles move the membrane at the oval window

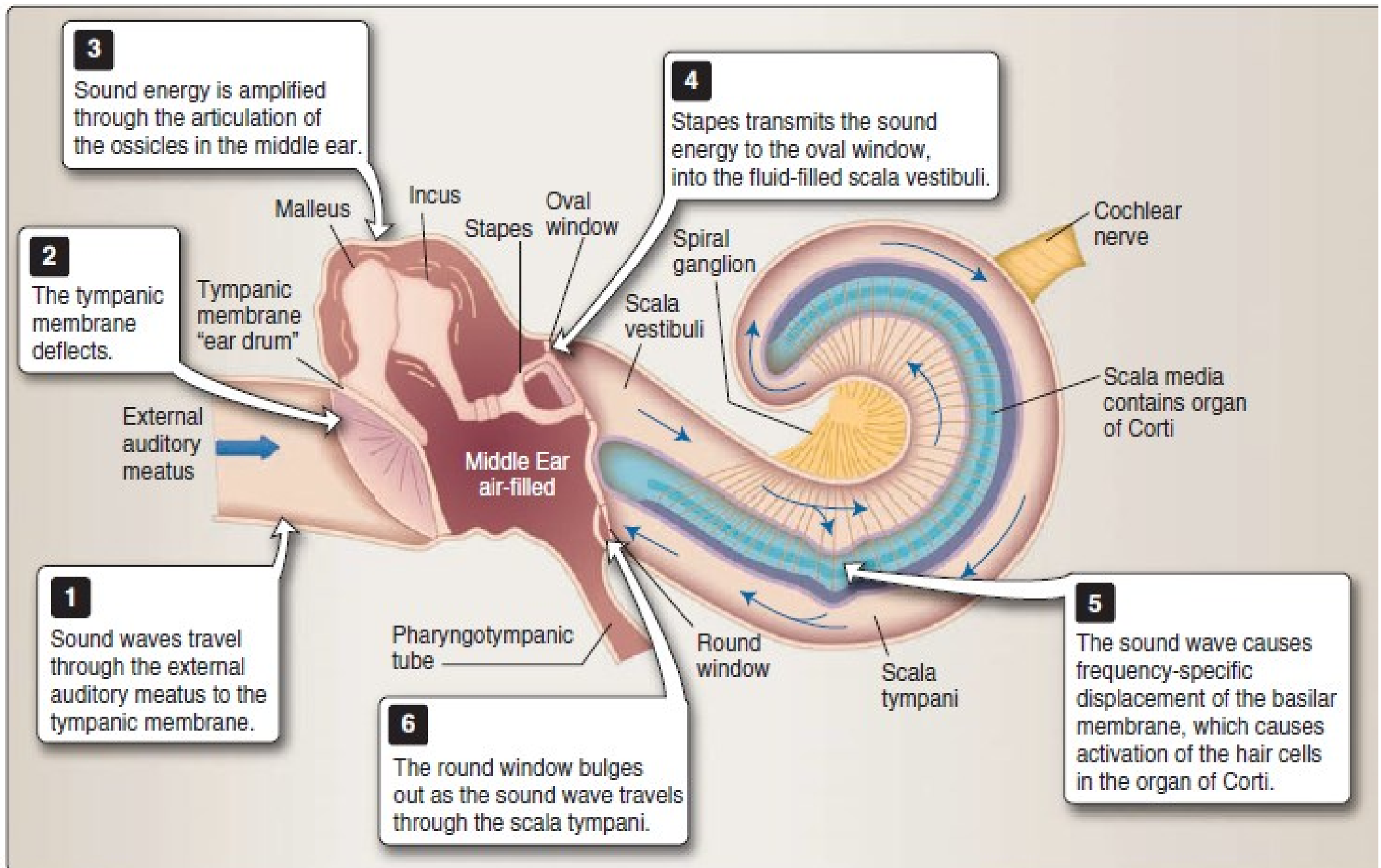


Motion at the oval window moves fluid in the cochlea

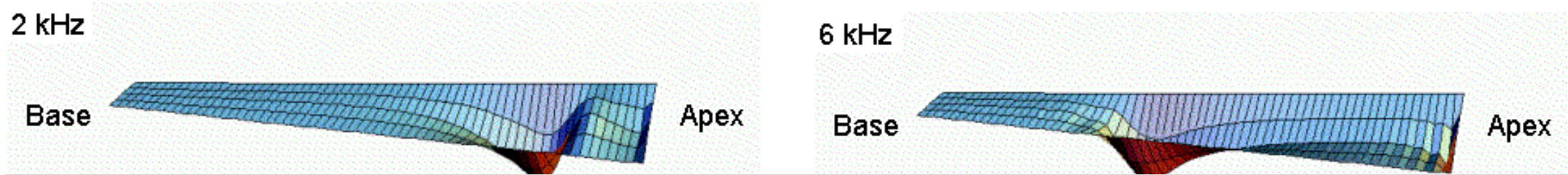


Movements of fluid in the cochlea causes a response in *organ of corti* and then the *sensory neurons*





Tonotopic organization of Basilar membrane



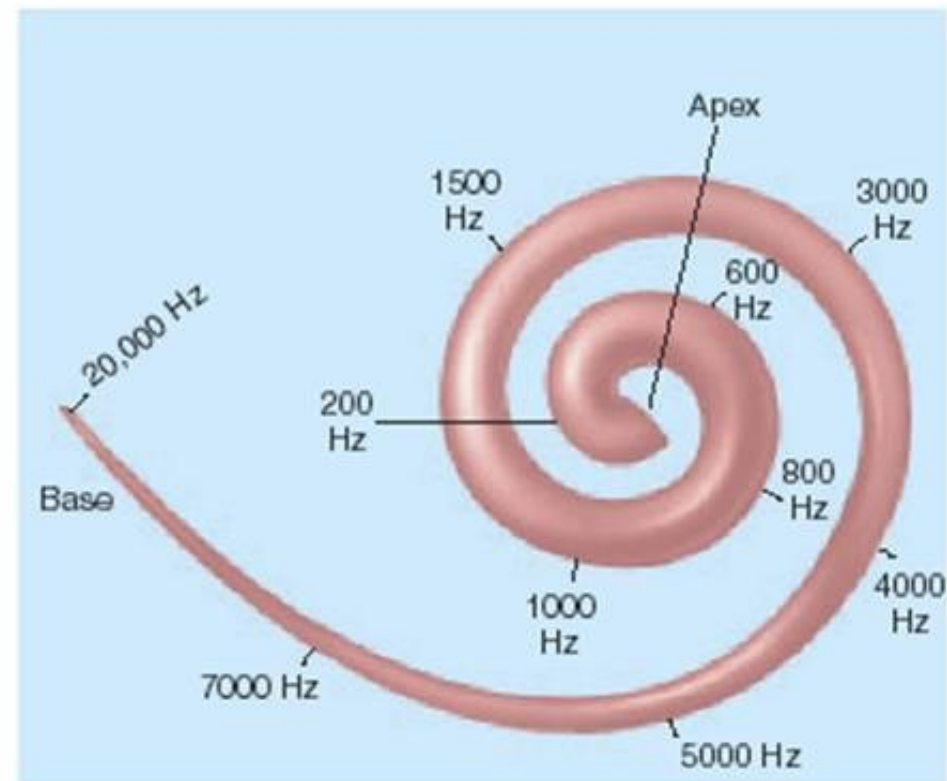
- High frequency sounds cause maximal stimulation of basilar membrane near base of cochlea/OW

Tonotopic organization of Basilar membrane

- mapping of specific tones (pitches) to specific areas
- The response of the basilar membrane establishes a **place code**
 - different locations of the membrane are maximally deformed at different sound frequencies
 - responsible for the **neural coding of pitch**

Place Principle

- Different sound frequencies displace different areas of the basilar membrane
- Hair cells near oval window (base)
 - short and thick- respond best to higher frequencies
- Hair cells near helicotrema (apex)
 - long and slender - respond best to lower frequencies



Determination of loudness of sounds

Three ways

- Amplitude of vibrations of BM increases & hair cells produce **impulses at rapid rate.**
- Rise of quantity of excited hair cells. **Loud sound stimulates also special high-threshold hair cells.**
- Outer hair cells become recruited at a significant rate.

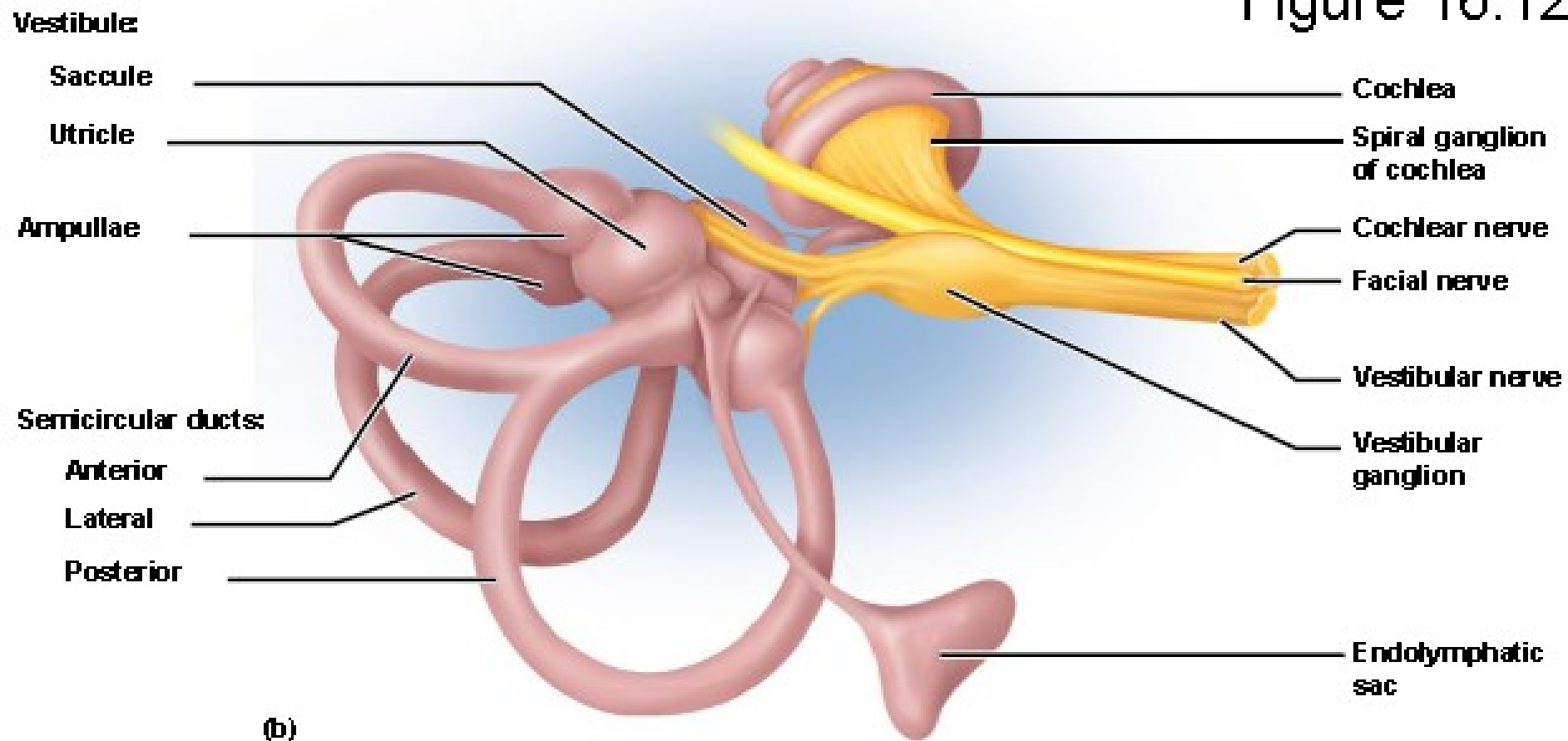
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Cerebral cortex and MGB are responsible for localization of sound.

Innervation of Internal Ear

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Figure 16.12b

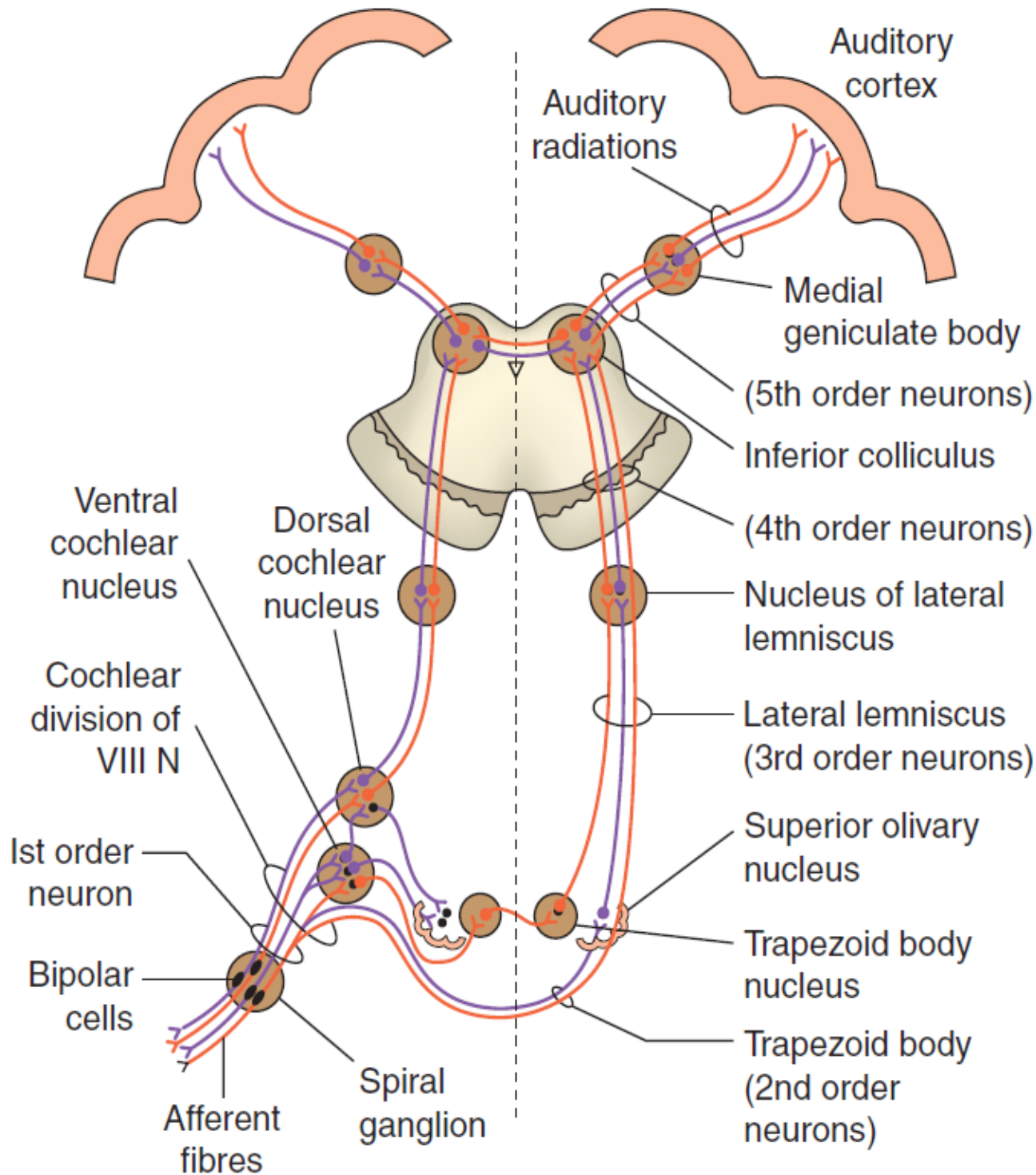


- **vestibular ganglia** - visible lump in vestibular nerve
- **spiral ganglia** - buried in modiolus of cochlea

Auditory Pathway to the Brain

- Impulses from the cochlea pass via the **spiral ganglion** to the cochlear nuclei of upper medulla
- From there, impulses are sent to the:
 - **Superior olivary nucleus** , trapezoid nucleus and via **lateral lemniscus** to
 - **Inferior colliculus** (auditory reflex center in the midbrain)
- From there, impulses pass to the **auditory cortex via MGB**
- Auditory pathways *decussate (cross over)* so that both cortices (temporal lobe) receive input from both ears

Simplified Auditory Pathways



First-order neurons
are the bipolar cells

Cochlear nuclei.

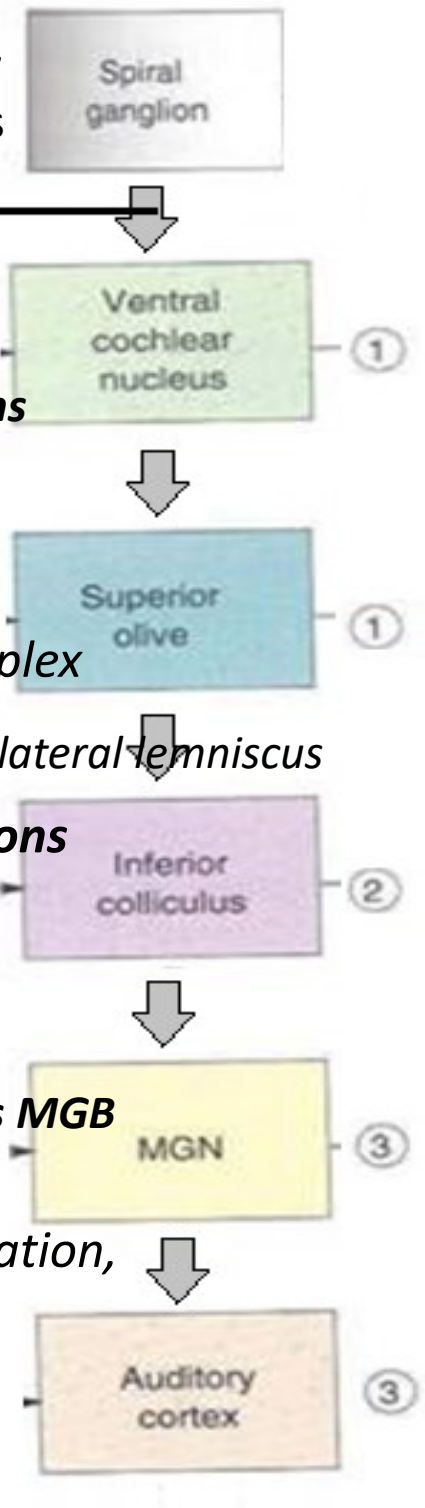
Second-order neurons

Third-order neurons
superior olivary complex

Fourth-order neurons
inferior colliculus,

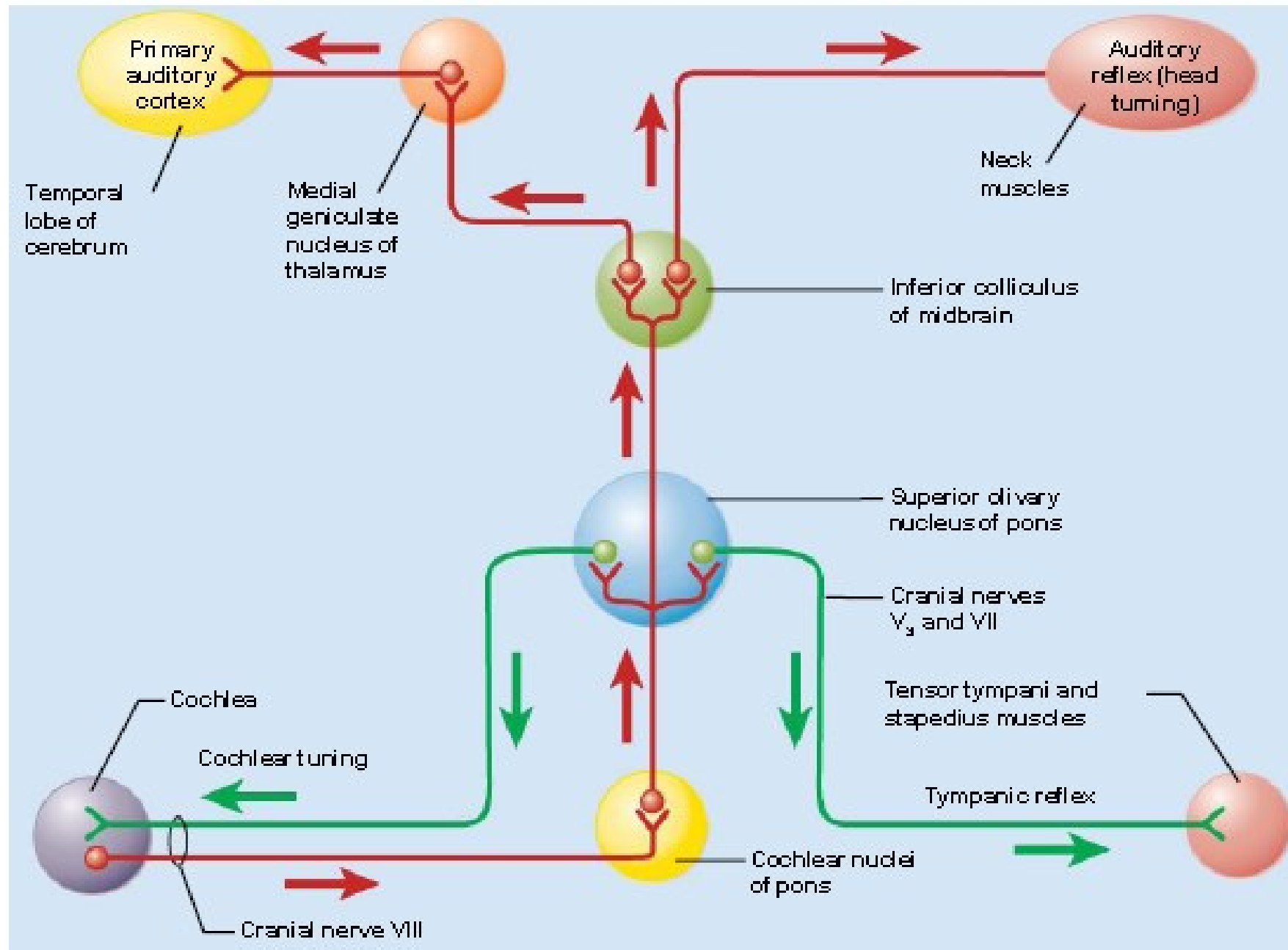
Fifth-order neurons MGB

acoustic radiation,



Cranial Nerve	Fibres	Structures Innervated	Functions	Brainstem Nucleus
I Olfactory	Sensory	Olfactory epithelium (via olfactory bulb)	Olfaction	-----
II Optic	Sensory	Retina	Vision	-----
III Oculomotor	Motor	Superior/middle/inferior rectus, inferior oblique, levator palpebrae.	Movement of eye ball	Oculomotor nucleus
	Parasympathetic	Pupillary constrictor, ciliary muscle of eyeball. Both via the ciliary ganglion	Pupillary constriction and accommodation	Oculomotor nucleus
IV Trochlear	Motor	Superior oblique	Movement of eyeball	Trochlear nucleus
V Trigeminal	Sensory	Face, scalp, cornea, nasal and oral cavities, cranial dura mater.	General sensation	Trigeminal sensory nucleus
	Motor	Muscles of mastication	Opening/closing mouth	Trigeminal Motor nucleus
		Tensor Tympani muscle	Tension of tympanic membrane	Trigeminal Motor nucleus
VI Abducens	Motor	Lateral rectus	Movement of eyeball	Abducens nucleus
VII Facial	Sensory	Anterior 2/3 of tongue	Taste	Nucleus Solitarius
	Motor	Muscles of facial expression	Facial Movement	Facial Motor nucleus
		Stapedius Muscle	Tension of ossicles	Facial Motor Nucleus
	Parasympathetic	Salivary and lacrimal glands via submandibular and pterygopalatine ganglia	Salivation and Lacrimation	Superior Salivatory Nucleus
VIII Vestibulocochlear	Sensory	Cochlea	Hearing	Cochlear Nucleus
		Vestibular apparatus	Proprioception of head, balance.	Vestibular nucleus
IX Glossopharyngeal	Sensory	Eustachian tube, middle ear	General Sensation,	Trigeminal Sensory nucleus
		Carotid Body, and sinus	Chemo/baroreception	
		Pharynx, posterior 1/3 of tongue	Taste	Nucleus Solitarius
	Motor	Styropharyngeous	Swallowing	
	Parasympathetic	Salivary glands via the otic ganglion	Salivation	Inferior Salivatory nucleus
X Vagus	Sensory	Pharynx, larynx, oesophagus, external ear	General Sensation	Trigeminal Sensory nucleus
		Aortic bodies and arch	Chemo/baroreception	
		Thoracic and abdominal viscera	Visceral Sensation	Nucleus Solitarius
	Motor	Soft Palate, larynx, pharynx, upper oesophagus	Speech, swallowing	Nucleus Ambiguus
	Parasympathetic	Cardiovascular, respiratory and gastrointestinal systems.	Control of these systems	Dorsal Motor nucleus of Vagus
XI Accessory	Motor	Sternomastoid, trapezius	Movement of head and shoulders	Nucleus Ambiguus, cranial nerves
XII Hypoglossal	Motor	Intrinsic and extrinsic muscles of tongue	Movement of tongue	Hypoglossal nucleus

Auditory Pathway

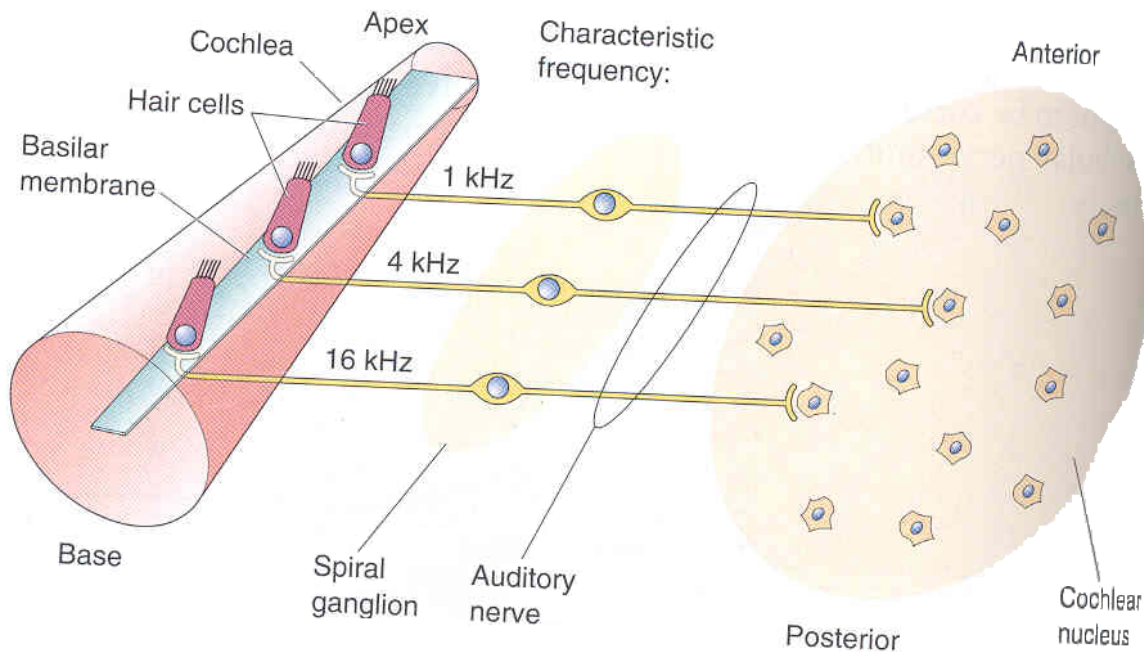


Binaural hearing ~ time/frequency/intensity

- It allows us to
 - (a) ‘map’ the sound in space,
 - (b) pick out soft sounds,
 - (c) pick out distant sound or speech and
 - (d) separate a single voice from surrounding background noise
 - *superior olivary nucleus/cortex*

Sound frequency - Tonotopy

■ Most neurons are frequency sensitive/*characteristic frequency*



Tonotopic maps on the basilar membrane and cochlear nucleus. Going from the base to the apex of the cochlea, the basilar membrane resonates with increasingly lower frequencies. This tonotopy is preserved in the auditory nerve and cochlear nucleus. In the cochlear nucleus, there are bands of cells with similar characteristic frequencies; characteristic frequencies increase progressively from posterior to anterior.

■ a spatial correlation between cells in the sensory organ and specific locations in the primary **auditory cortex: tonotopic organization**
high frequency sounds – posterior ; low frequency sounds - anterior

Air vs. Bone conduction

- Air conduction pathway involves external ear canal, middle ear, and inner ear
- Bone conduction pathway involves direct stimulation of cochlea via vibration of the skull (cochlea is imbedded in temporal bone)
- Reduced hearing may involve:
 - Ossicles (air conduction loss)
 - Cochlea or associated neural pathway (sensory neural loss)

Deafness: hearing loss

Conduction deafness

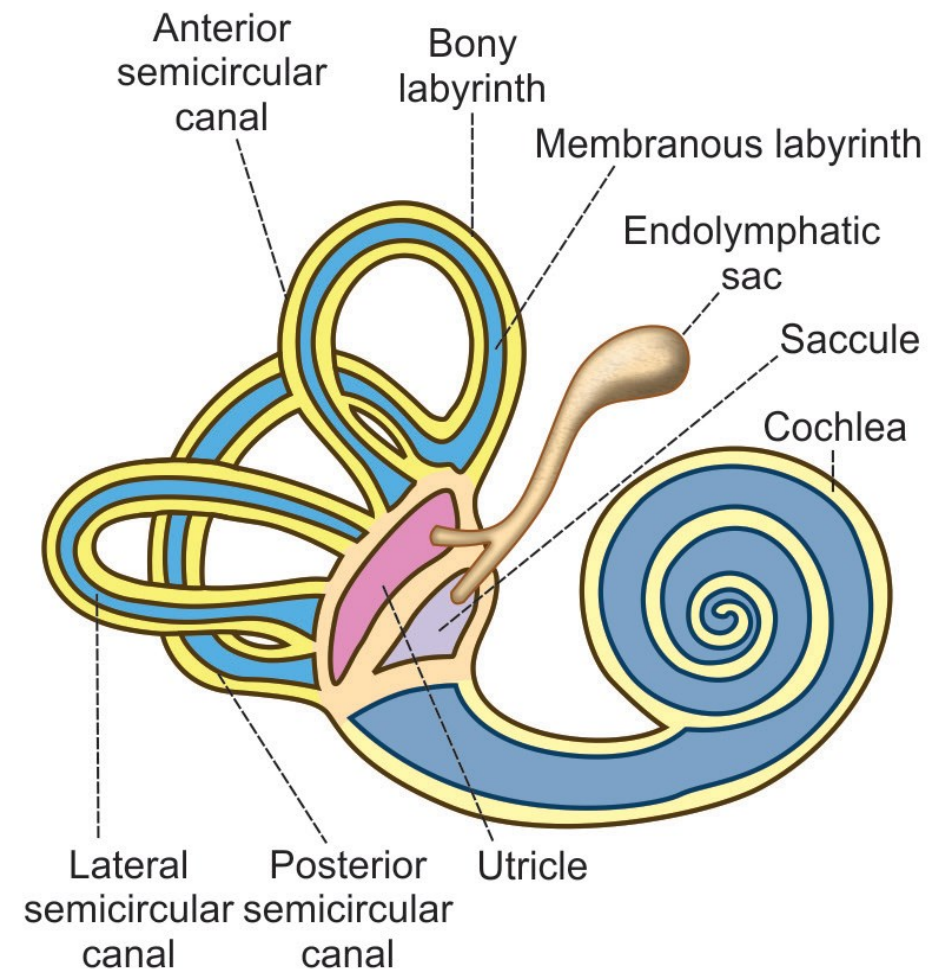
- Deafness due to conditions that interfere with transmission of vibrations to inner ear
 - *impacted earwax*
 - *perforated eardrum*
 - *osteosclerosis of the ossicles*
 - *fluid accumulation from ear infections*
 - *oval window damage*
- *Hearing aids may be useful*

Sensorineural deafness (“nerve deafness”)

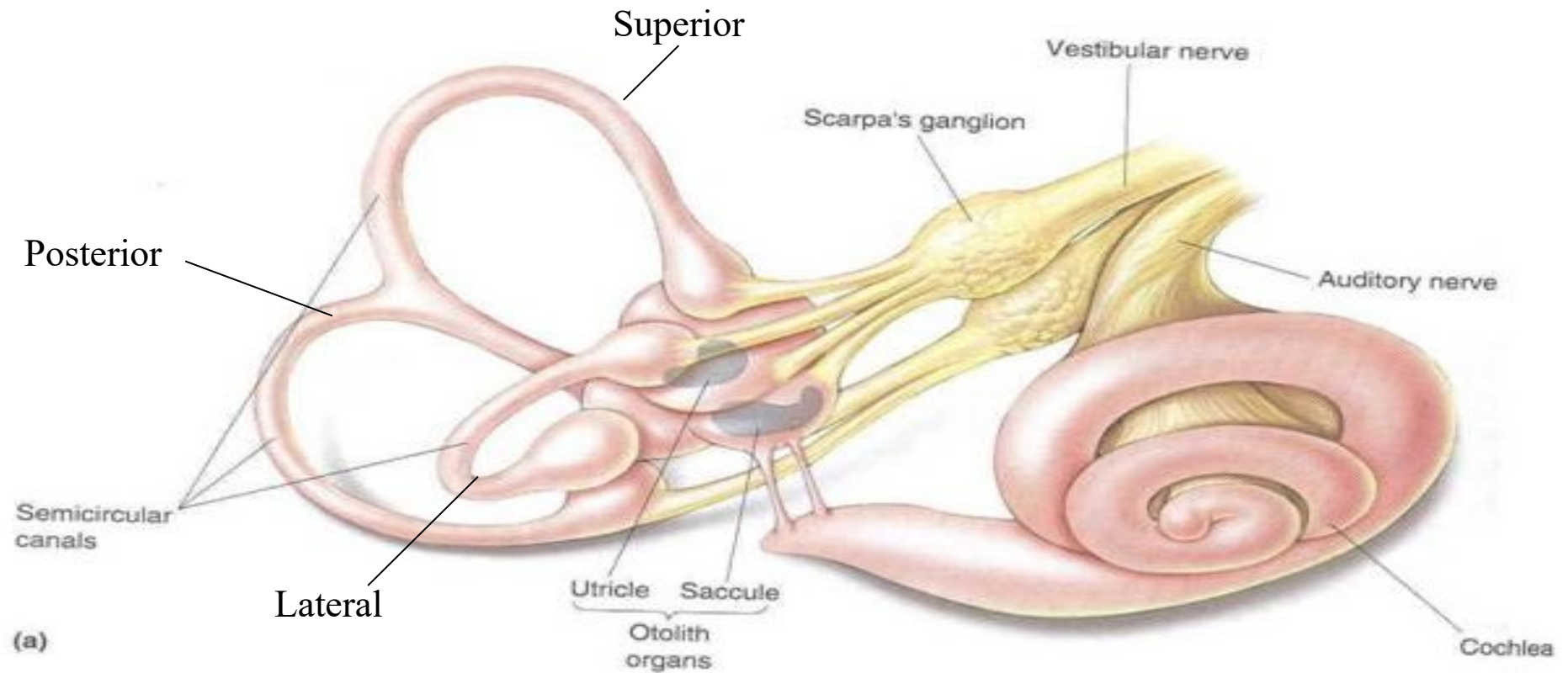
- Caused by damage to the neural structures at any point *from the cochlear hair cells to the auditory cortical cells*
 - Prolonged exposure to high amplitude sound.
- *Hearing aids of little value*

Vestibular system

- Mechanisms of Equilibrium and Orientation
 - Maintains our orientation and balance in space
 - Provides information about position and movement of head in space
- Involves
 - *Vestibular apparatus*
 - *Vestibular nuclei*



Vestibular apparatus



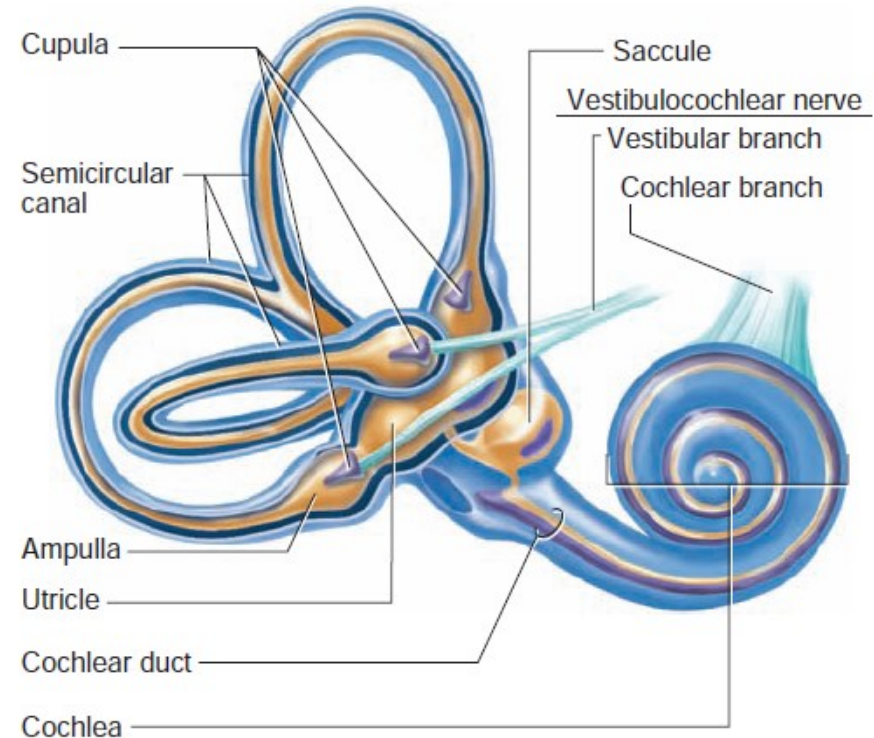
■ Structures in the inner ear

- *2 sacs/otolith organ/vestibule (utricle and saccule) and the 3 semicircular canals*

The Vestibule/ otolith organs

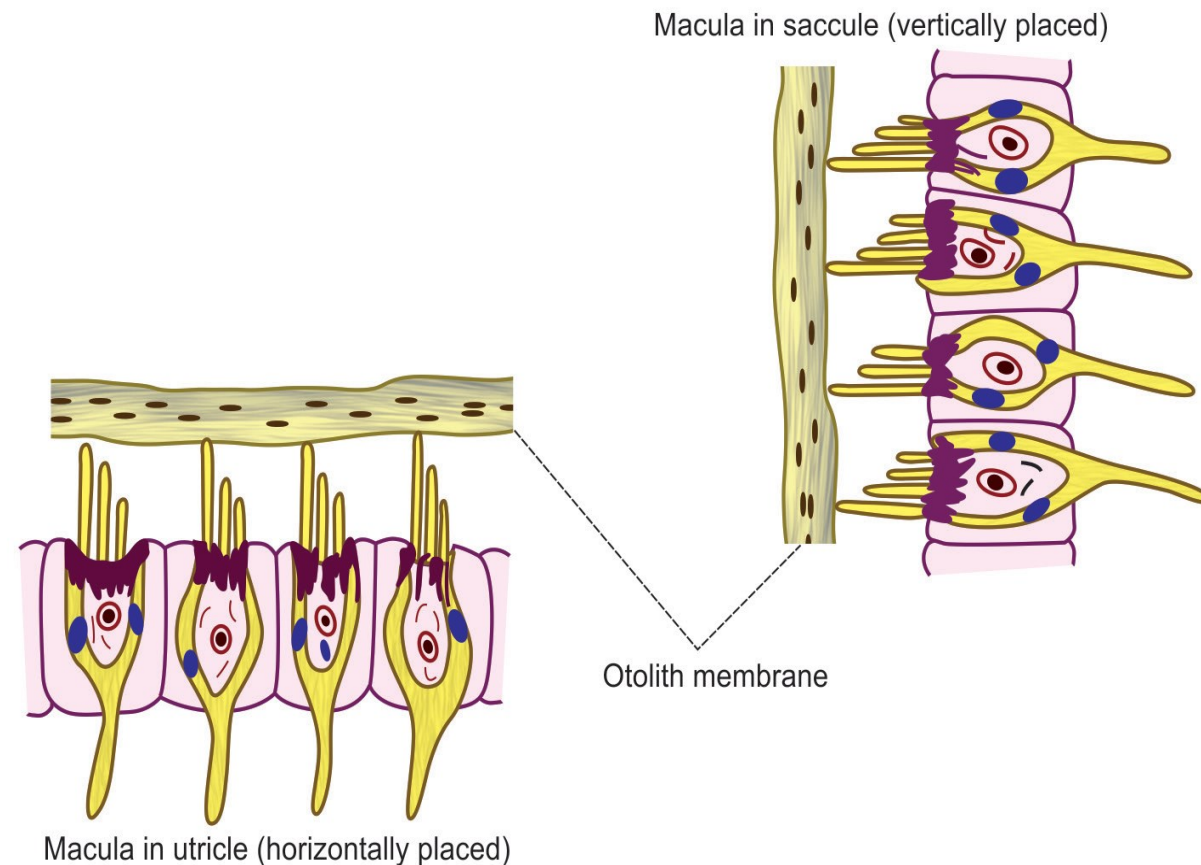
- Central cavity of the bony labyrinth suspended in perilymph
- The **Saccul**e and **Utricle**
- Respond to
 - *static equilibrium*
 - *linear acceleration* - *change in velocity in a straight line/vert/horiz*

Eg. elevator



Maculae/otolith organ of Sacculle and Utricle

- *Macula of utricle lies in the horizontal plane*
 - *Utricular hairs respond to horizontal movement and forward-backward tilt*

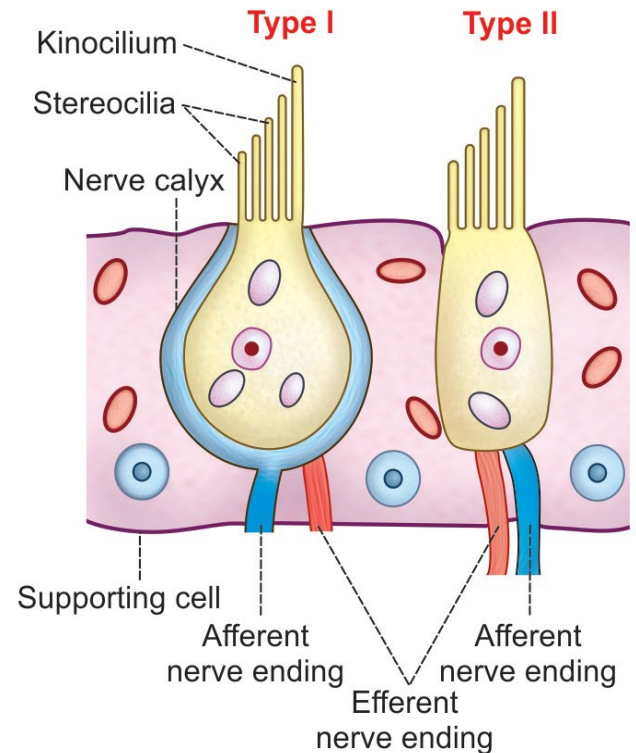


- *Macula of saccule lies in the vertical plane*
 - *Saccular hairs respond to vertical movement*
 - *while climbing up, cilia downwards/stimulates the hair cells.*

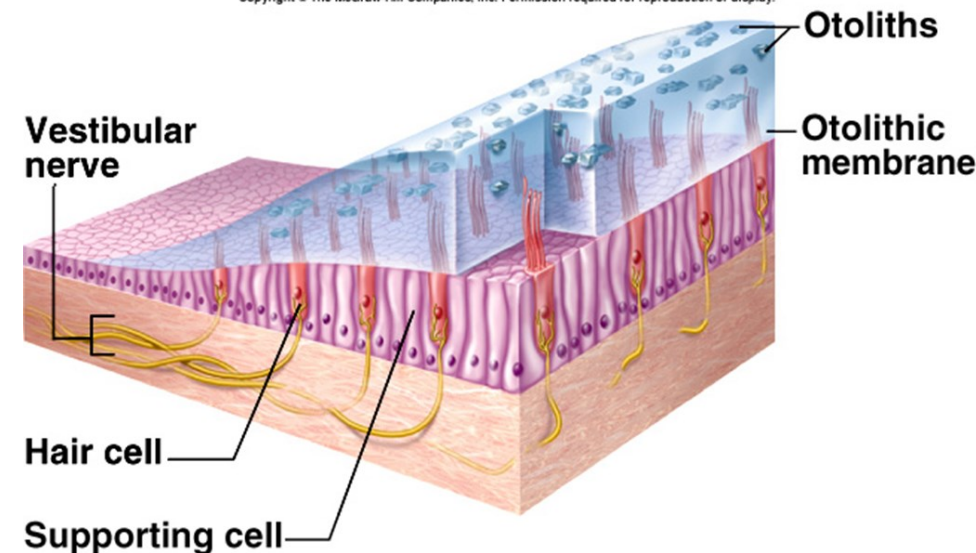
Sacculle and Utricle



- **Maculae** - the sensory receptors for *static equilibrium* (head position/tilting and linear acceleration) and initiate postural reflexes.
 - Each hair cell (also in cristae) has 50-70 moveable cilia(*stereocilia*) and one large stiff cilia (*kinocilium*) embedded in the otolithic membrane
 - **Otolithic membrane** – jellylike mass studded with tiny CaCO_3 stones = *otoliths* (ear stones) = like crosta ampullaris



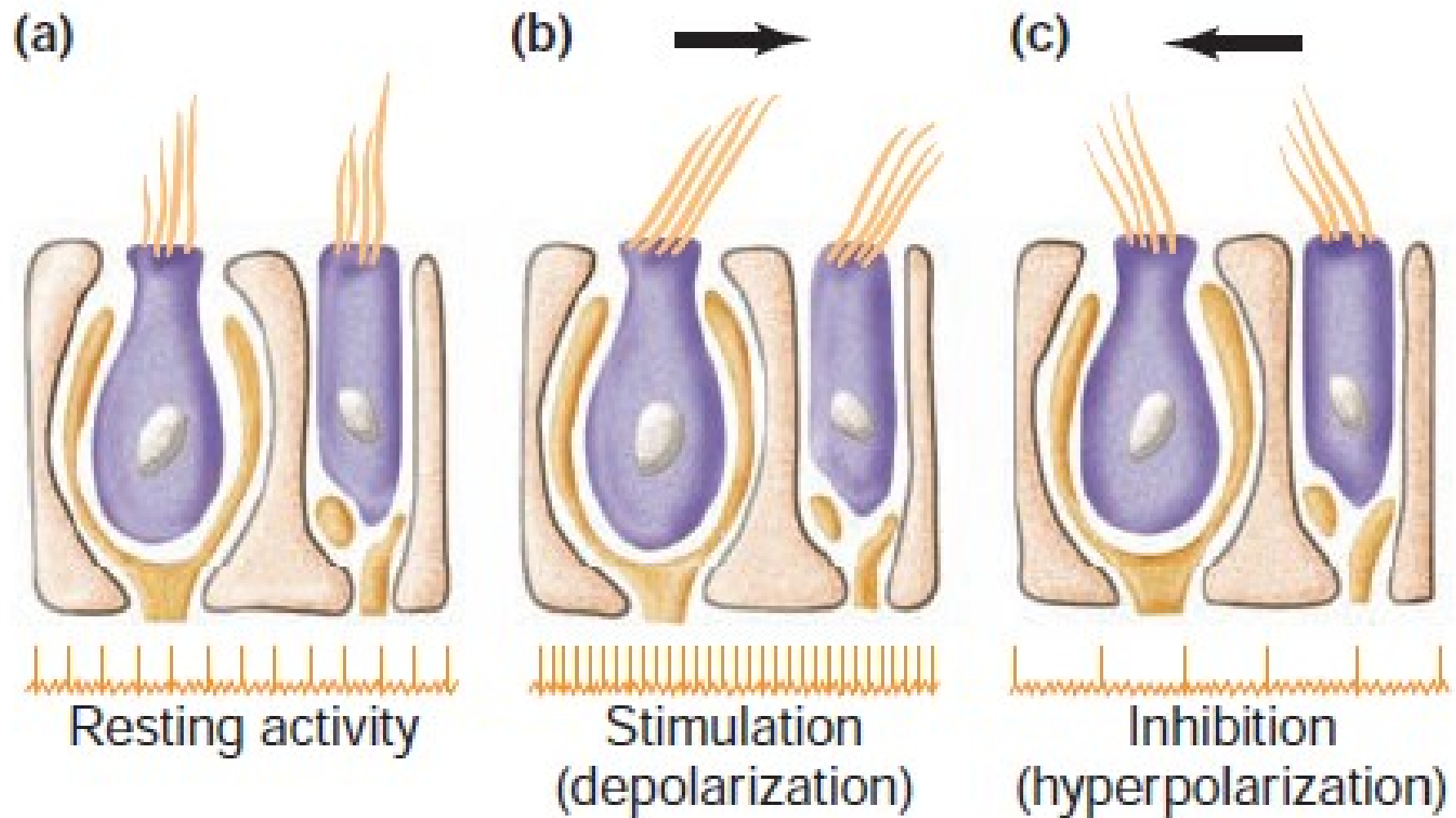
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Effect of Gravity on Utricular Receptor Cells

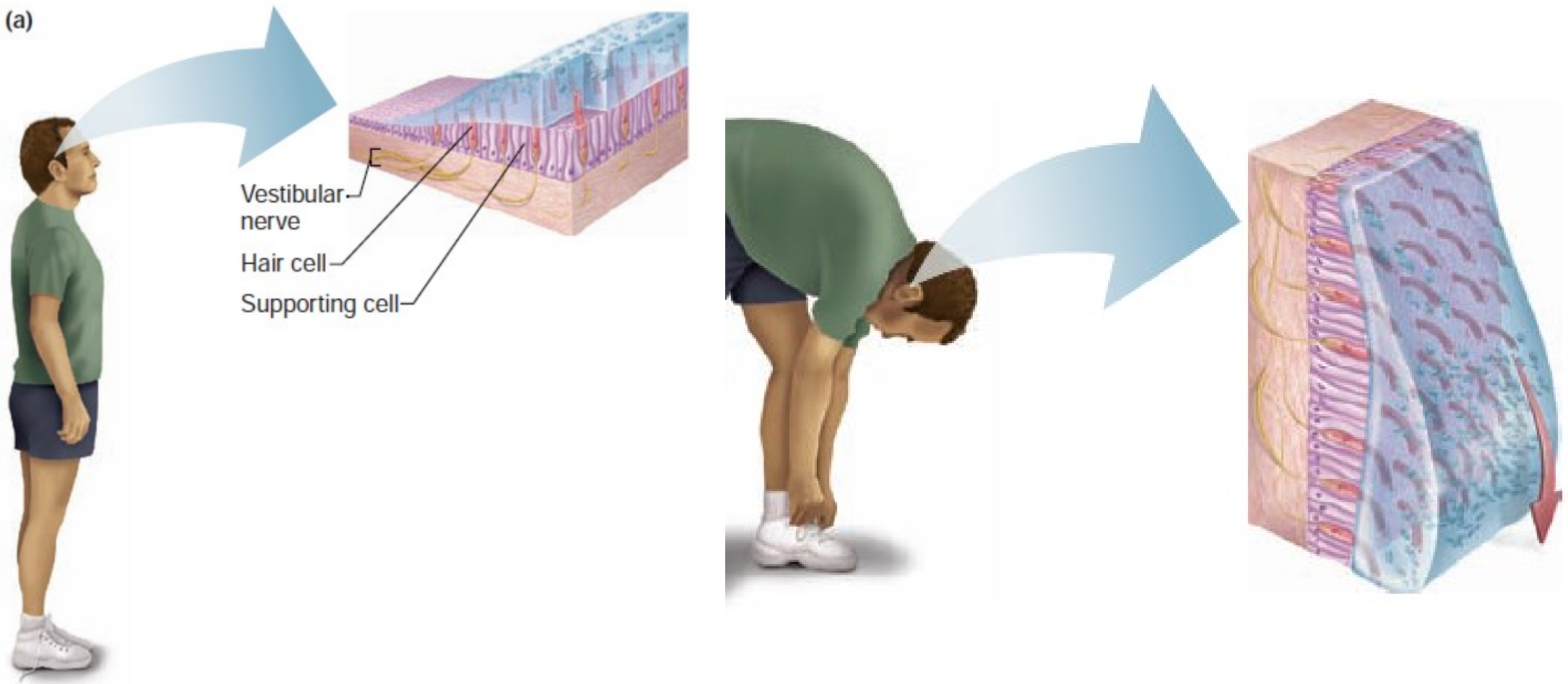
- *Otolithic* movement in the direction of the *kinocilia*
 - *Depolarizes vestibular nerve fibers*
 - *Increases the number of action potentials generated*
- Movement in the opposite direction:
 - *Hyperpolarizes vestibular nerve fibers*
 - *Reduces the rate of impulse propagation*

From this information, the brain is informed of the changing position of the head



Discharge rate of vestibular nerve

(a)



Static
Equilibrium

Vestibule - Sensitive to Tilting of Head
Movements

Macula utriculi
Macula sacculi

(a)

Vestibular
nerve

Otoliths

Otolithic
membrane

(b)

Hair cell

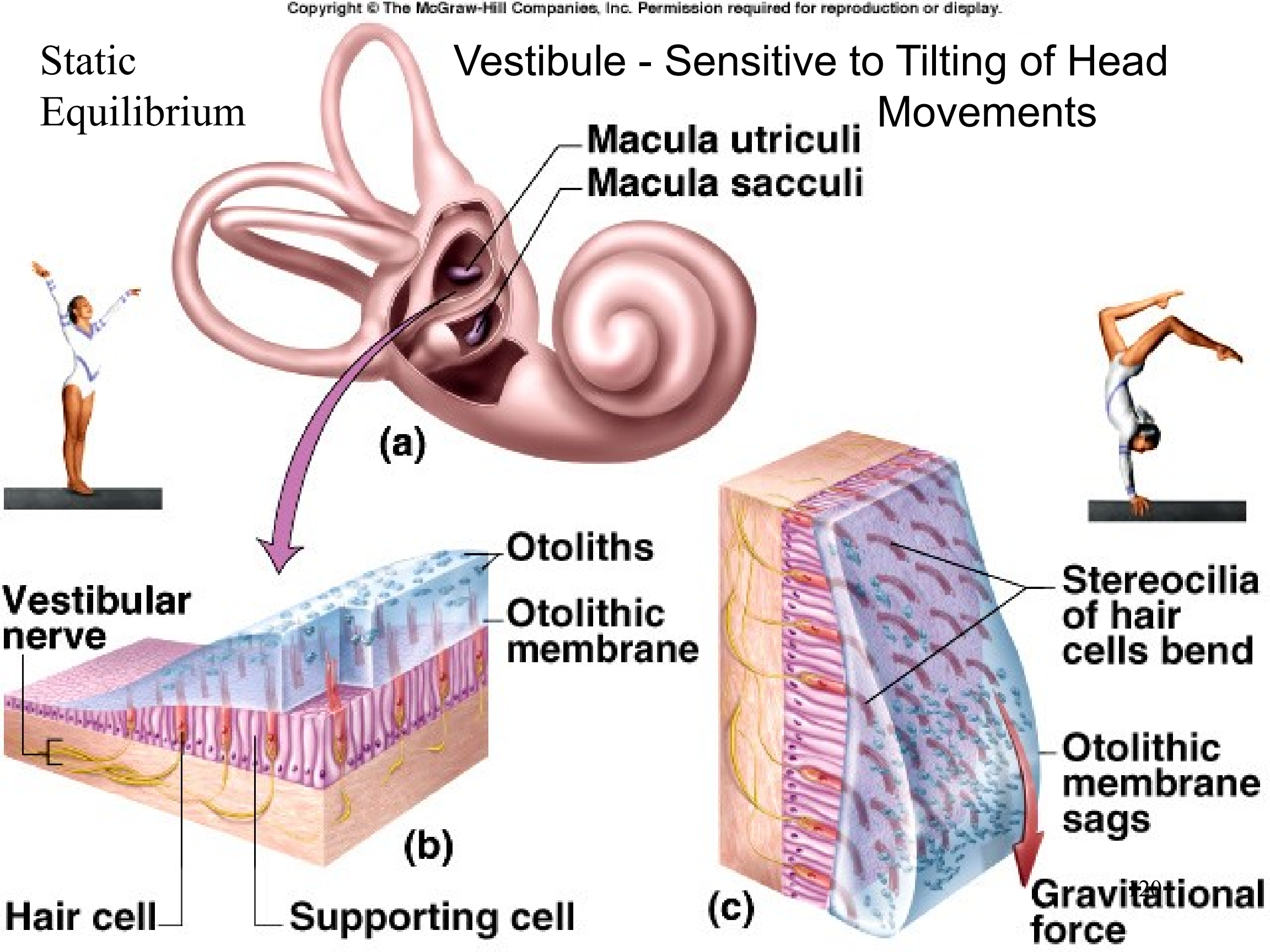
Supporting cell

Stereocilia
of hair
cells bend

Otolithic
membrane
sags

Gravitational
force

(c)



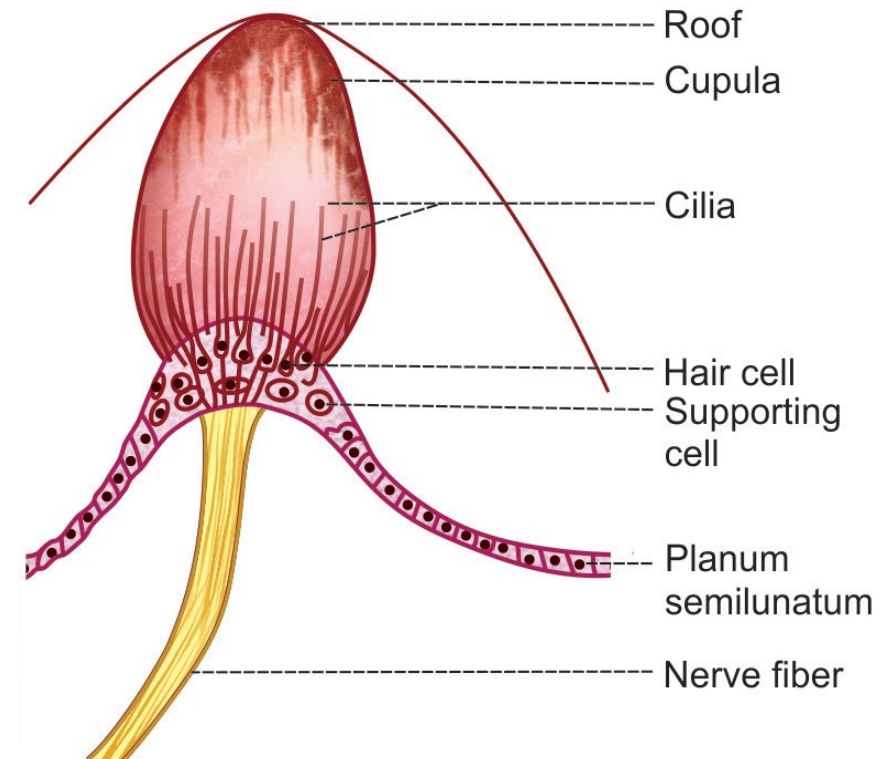
The Semicircular Canals: *lateral, superior, posterior*

- Three canals lie in three planes of space
- Membranous semicircular ducts line each canal and communicate with the utricle by means of five openings.

- Utricle opens into saccule

- **Ampulla**

- *the swollen end of each canal*
 - *houses equilibrium receptors in a region called the **crista ampullaris***
 - *receptors respond to **angular acceleration** of the head*



Dynamic Equilibrium

3 pairs of canals bilaterally at 90° to one another (anterior, horizontal, posterior)

Superior SCC

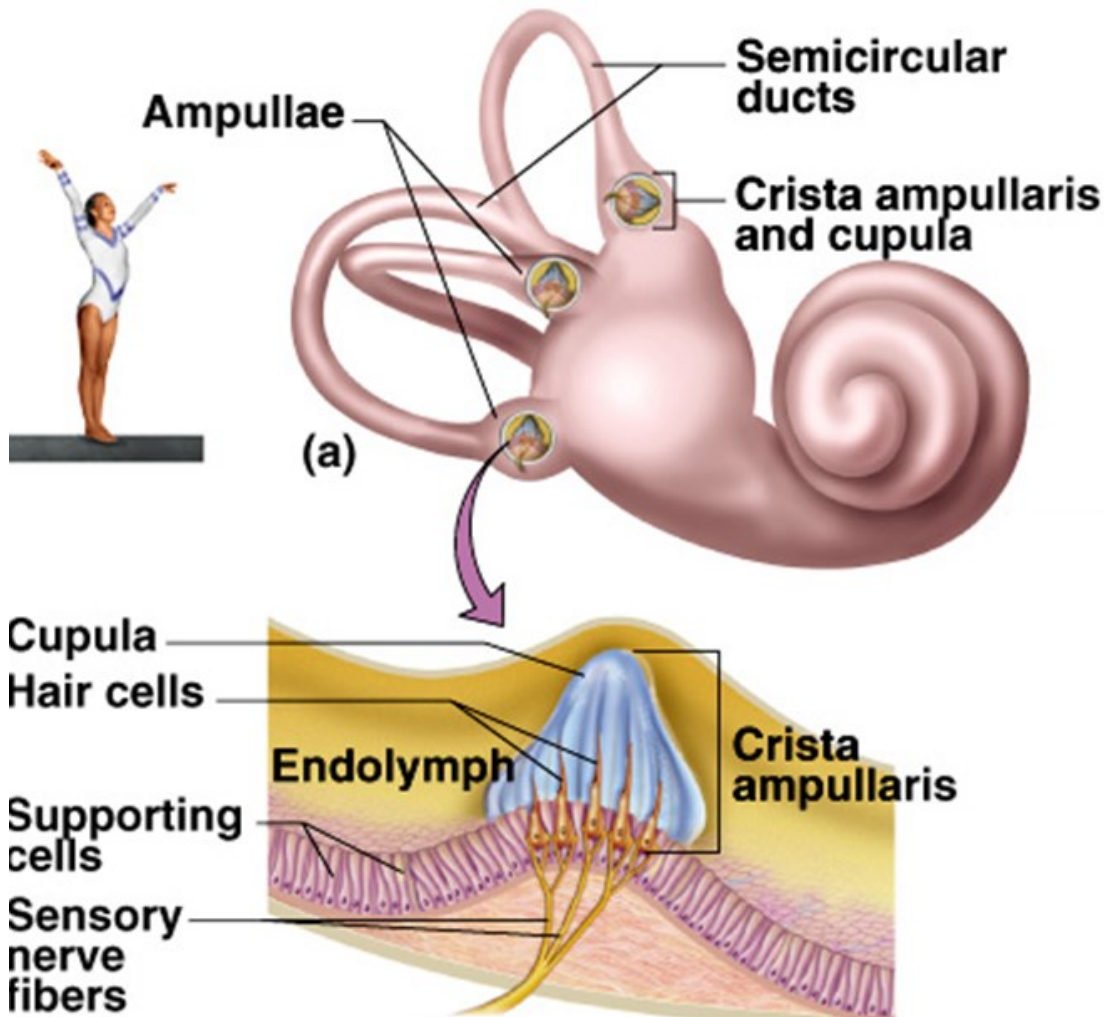
Transverse axis/front to back movements like nodding (yes yes..)

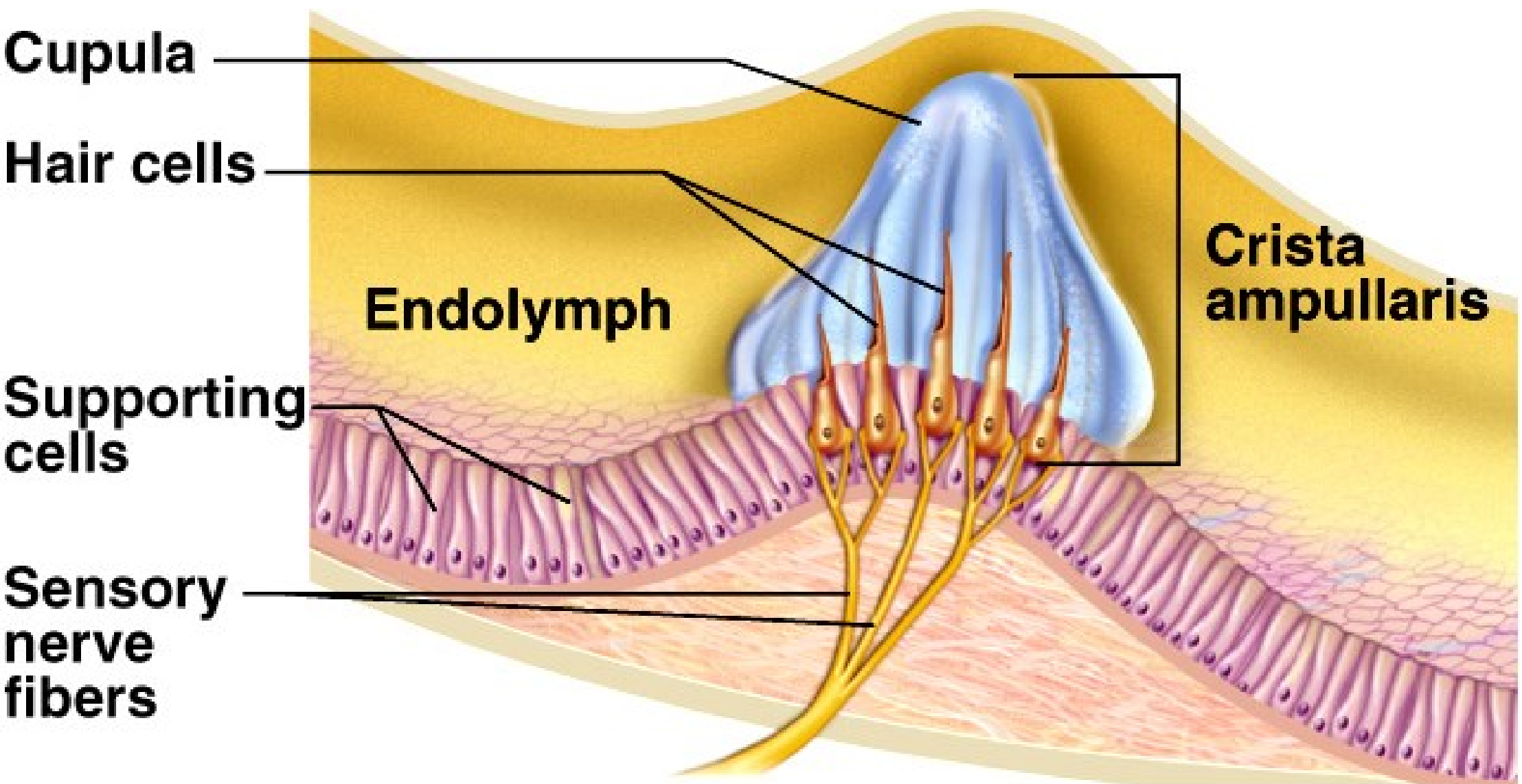
Horizontal SCC

response to rotation in horizontal plane (vertical axis), i.e. side to side movements (left to right or right to left) like shaking the head while saying 'no – no'.

Posterior SCC

response to rotation in the vertical plane /head rotated from shoulder to shoulder.



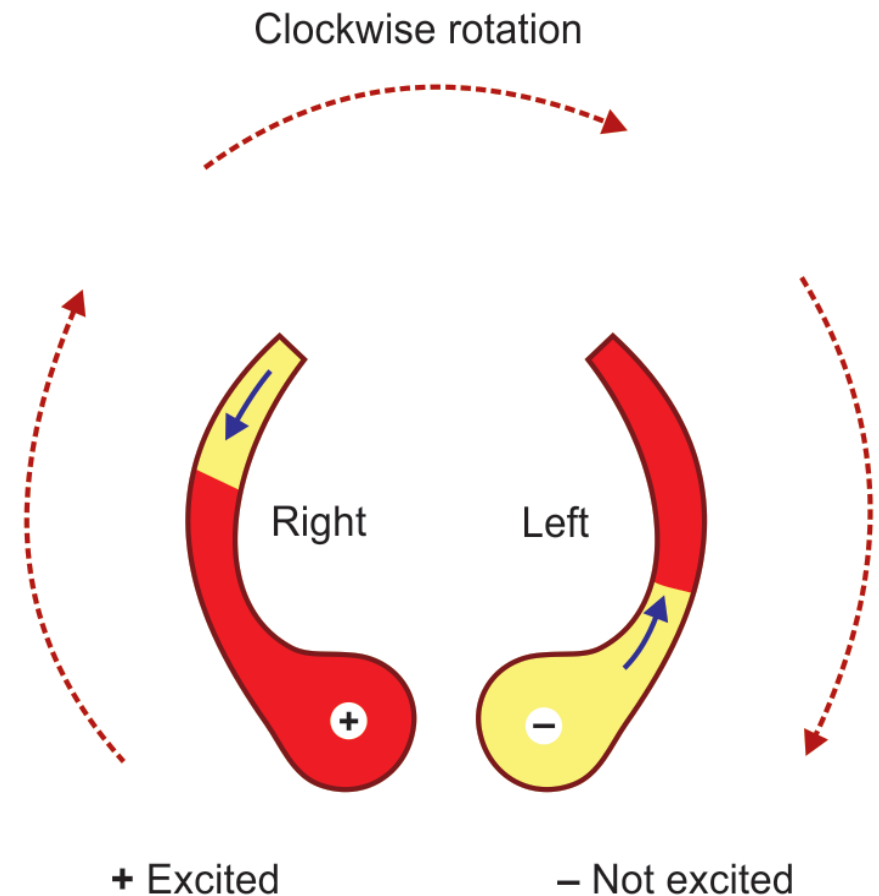


Cupula

loose gelatinous tissue mass on top of crista

Activating Crista Ampullaris Receptors

- Cristae respond to changes in velocity of rotatory movements of the head (stimulated only at the start & end).
- As head begins to rotate clockwise, *endolymph lags behind and bend cupula (anticlockwise)*.
- Directional bending of hair cells in the cristae causes:
 - *Depolarizations/endolymph/cupula to ampulla/ and rapid impulses reach the brain at a faster rate ®*
 - *Hyperpolarization/fewer impulses (L)*
- The result is that the brain is informed of *rotational movements* of the head (acceleration & deceleration)



Semicircular ducts:

Anterior

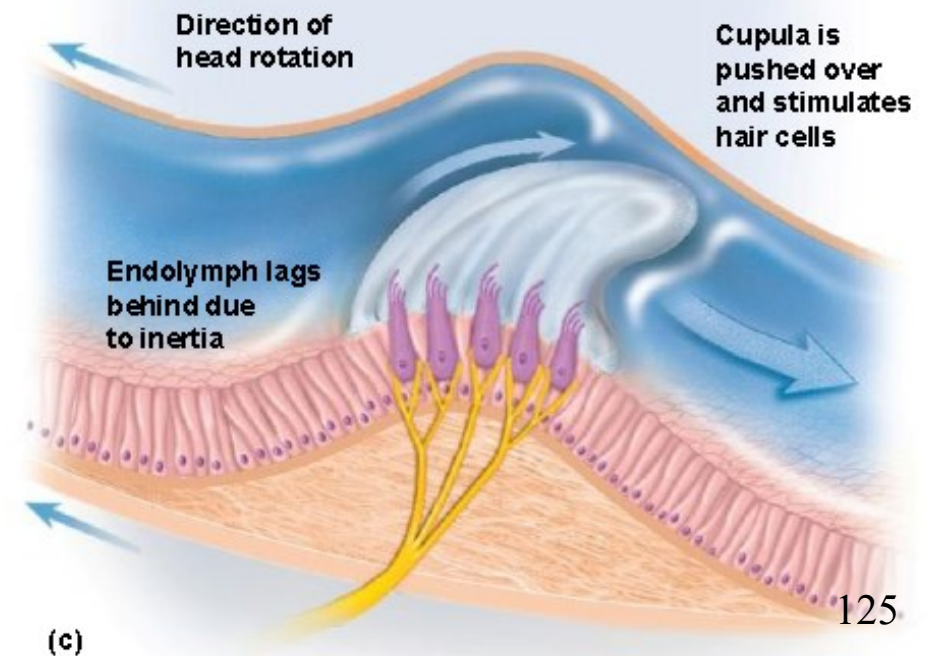
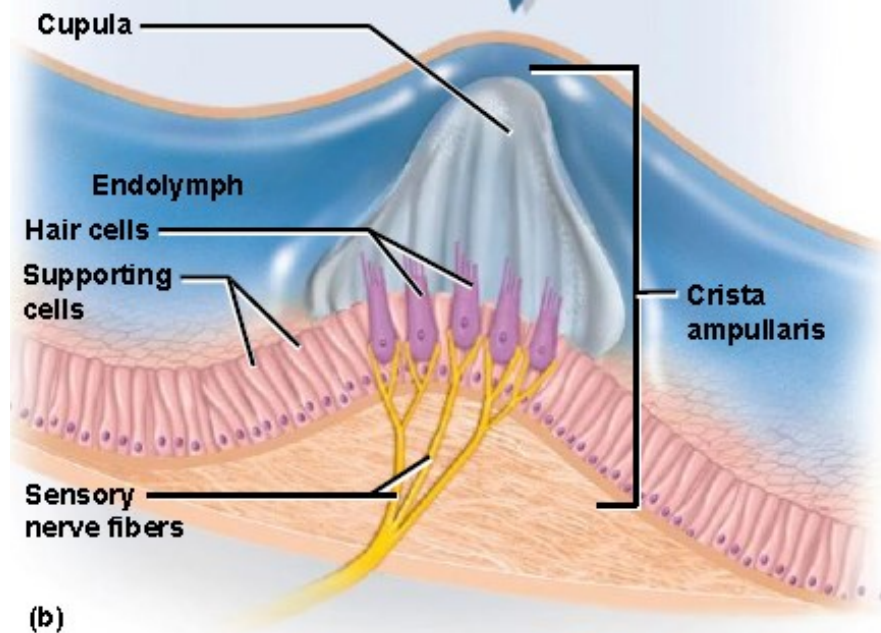
Posterior

Lateral

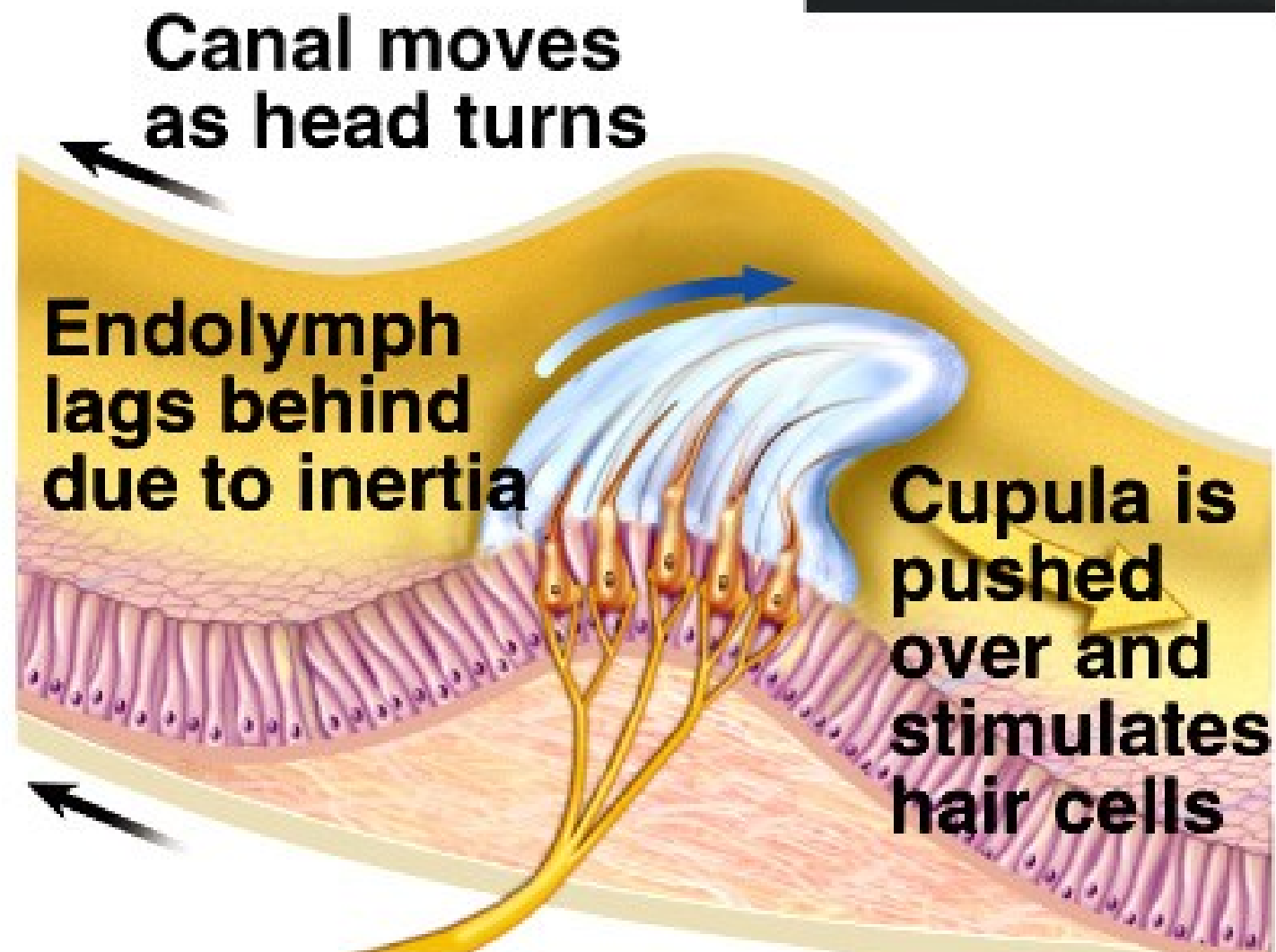
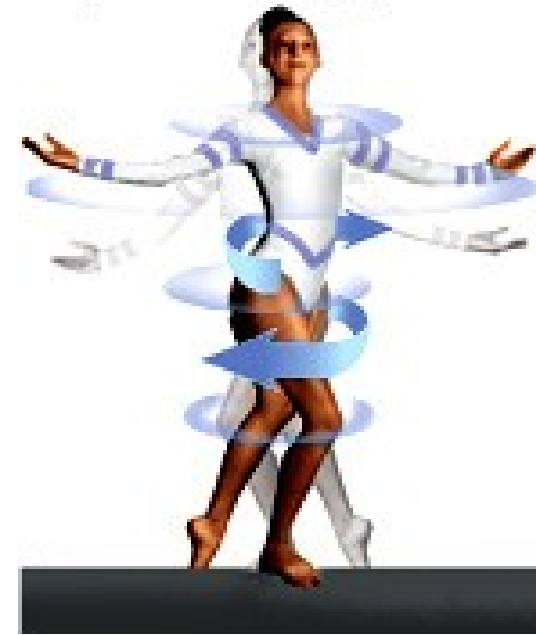
Ampullae

Crista ampullaris
and cupula

(a)

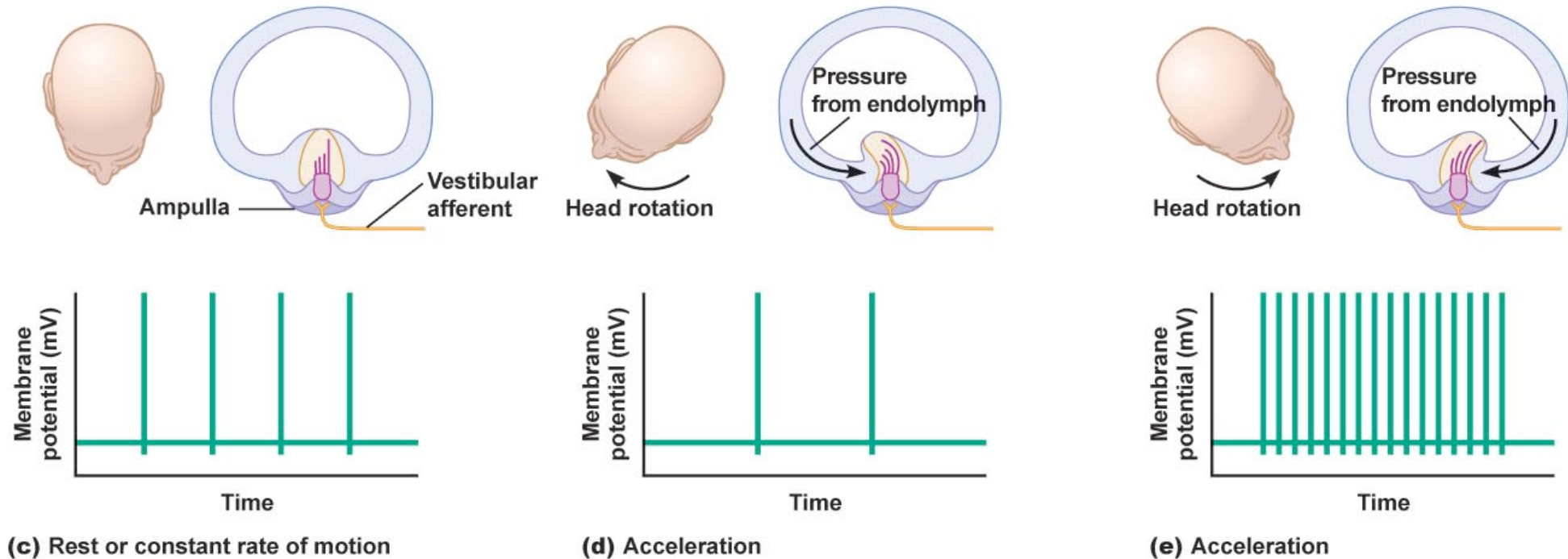


Semicircular Canals Respond to Rotational Movements



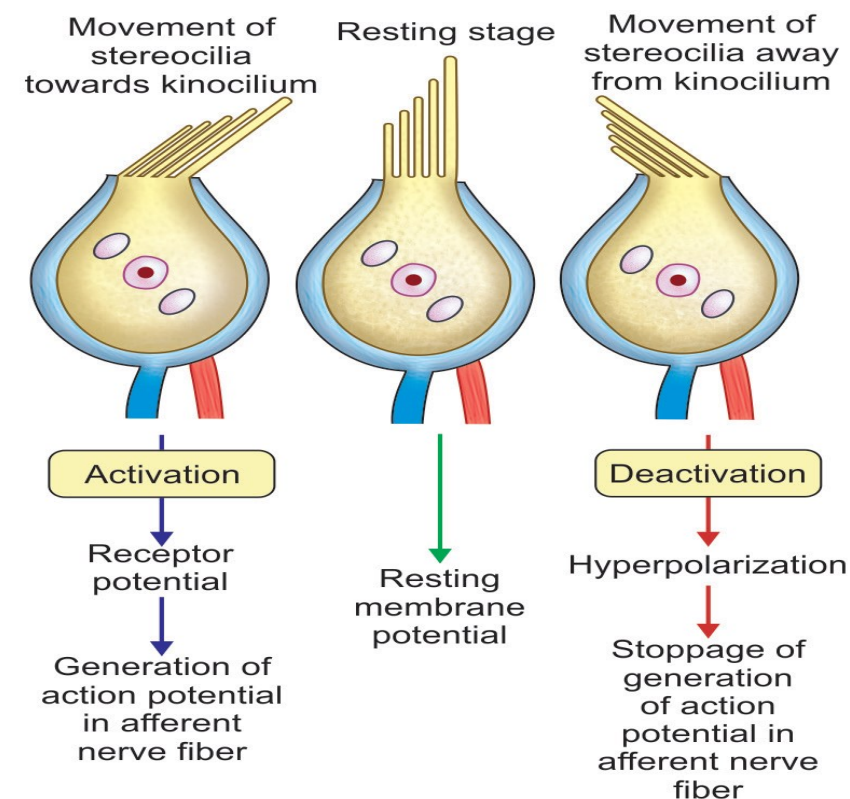
(c)

Electrical Potential in Hair Cells – Mechanotransduction



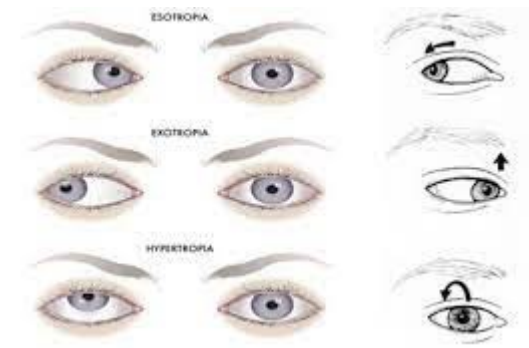
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- At rest, partially depolarized and action potential at a low frequency
- Rotated away from the kinocilium is hyperpolarized
- Rotated toward the kinocilium is depolarized/action potential in VN II (from -60 mV to -50 mV)

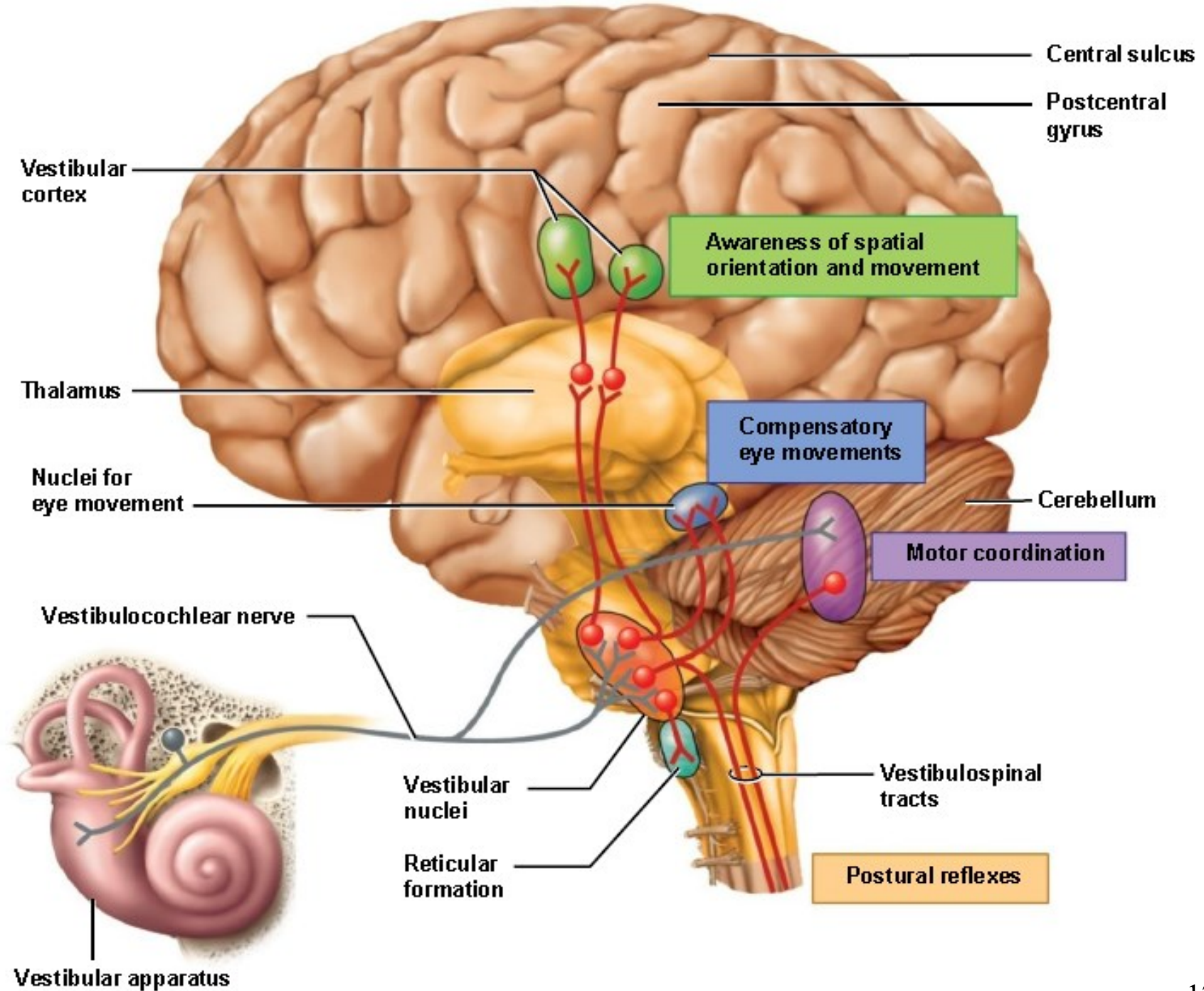


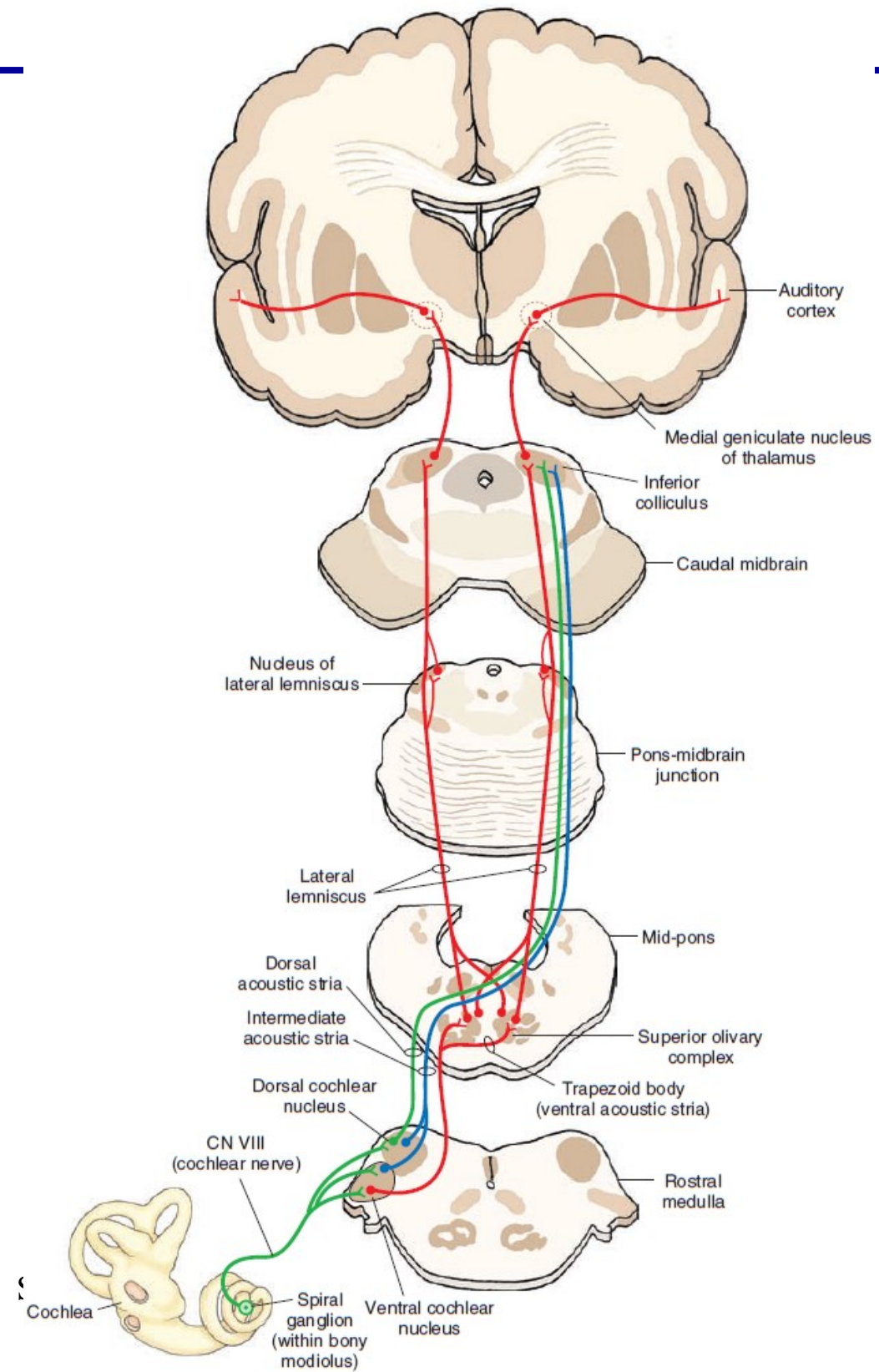
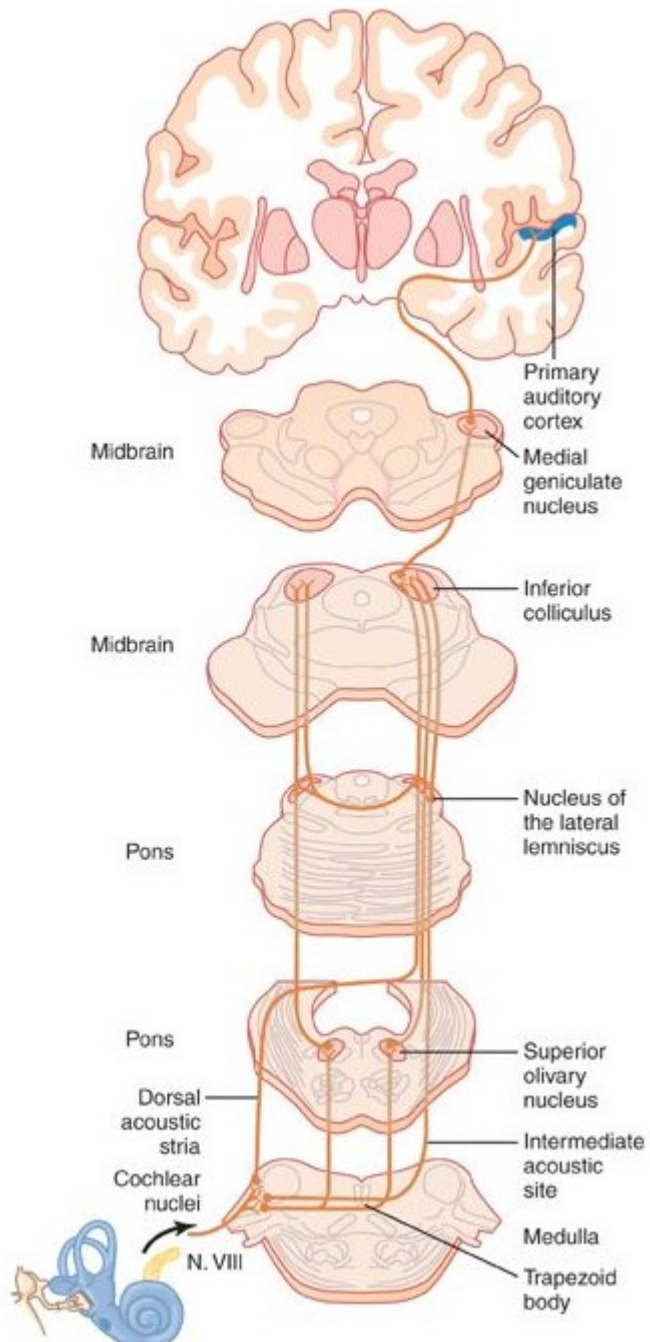
Semicircular Canals

- SCCs are insensitive to angular movement of constant velocity/*only detect changes in motion*
- Generates a receptor potential which alters the firing rate in VIII CN which projects to the vestibular nuclei
- Stimulation of semicircular canals/rotation is associated with
 - *increased extensor tone and nystagmus*
 - *Nystagmus = jerky to—fro/oscillatory movement of eyes*
 - *vestibulo-ocular reflex for visual fixation*



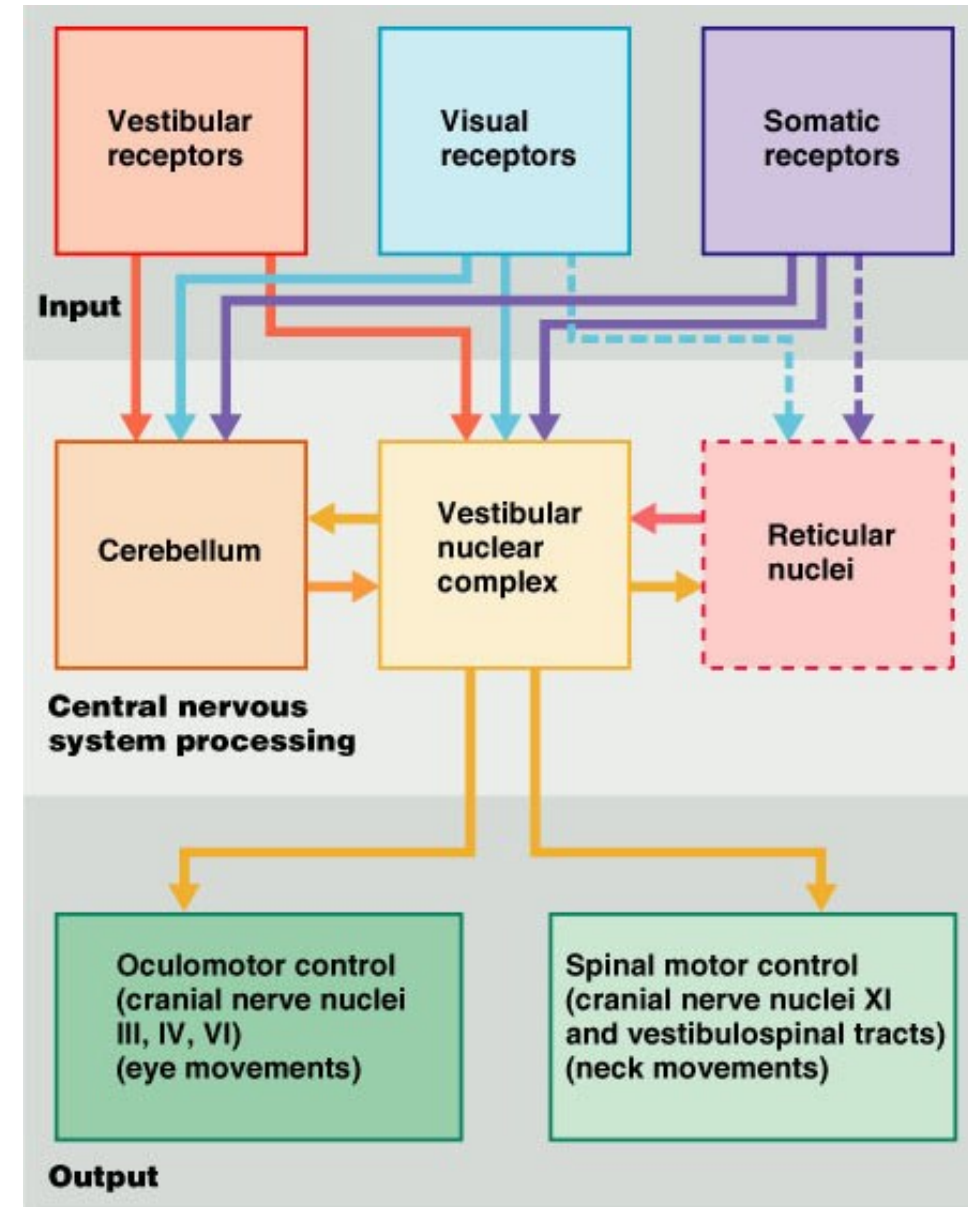
Vestibular Projection Pathways



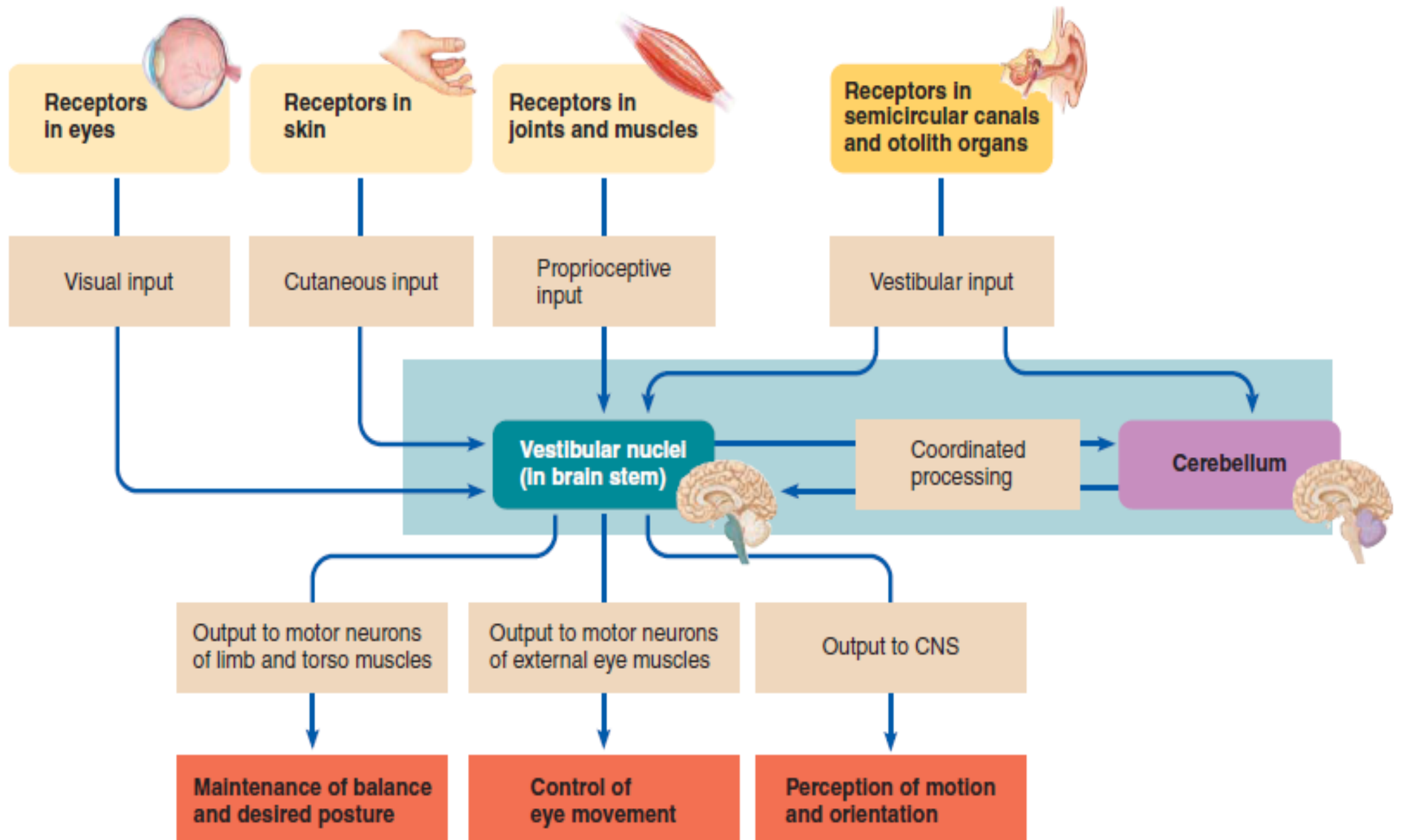


Balance and Orientation Pathways

- Three modes of input for balance and orientation
 - *Vestibular receptors*
 - *Visual receptors*
 - *Somatic receptors*
- These receptors allow our body to respond reflexively.
- Anticipatory corrections
 - *works in close concert with cerebellum flocculonodular lobe*



Reading assignment: Postural reflexes



Input and output of the vestibular nuclei.

Vestibular Projection Pathways

- Information sent to five targets:
 1. **Cerebellum** – integrates vestibular information into its control of *head and eye movements, muscle tone, and posture*
 2. **Nuclei of oculomotor, trochlear, and abducens nerves (CN III, IV, and VI)** to produce *vestibulo-ocular reflex* – keeps your vision fixed on distant object while you are walking
 3. **Reticular formation** – thought to adjust blood circulation and breathing to postural changes
 4. **Spinal cord** – descend through two *vestibulospinal tracts* of spinal cord and innervate extensor (antigravity) muscles
 5. **Thalamus** - thalamic relay to cerebral cortex for *awareness of position*

Vertigo

- Disorders of vestibular function/Illusion of motion occurs.
 - It is either an exaggerated sense of motion when there is no motion or motion in response to a given bodily movement.
 - Persons with vertigo frequently describe a sensation of spinning or tumbling, a “to-and-fro” motion, or falling forward or backward.



Otitis media

- Inflammation of middle ear segment.
- Buildup of fluid and frequently causes an earache. The fluid may or may not be infected.

Motion Sickness

- The endolymph in the SCC becomes 'stirred up', causing confusion between the difference between apparent perceived movement (none or very little), and actual movement
- A form of normal *physiologic vertigo*.
- Caused by repeated rhythmic stimulation of vestibular system
 - E.g. in car, air, or boat travel.
- The principal symptoms - *Vertigo, malaise, nausea, and vomiting*.
- **Autonomic signs**
 - Lowered blood pressure, tachycardia, and excessive sweating
- Usually resolves after the vestibular system becomes accustomed to the stationary influence of being back on land.

Special Senses

1. Vision
2. Hearing
3. Equilibrium
4. Olfaction
5. Taste

Chemical Senses



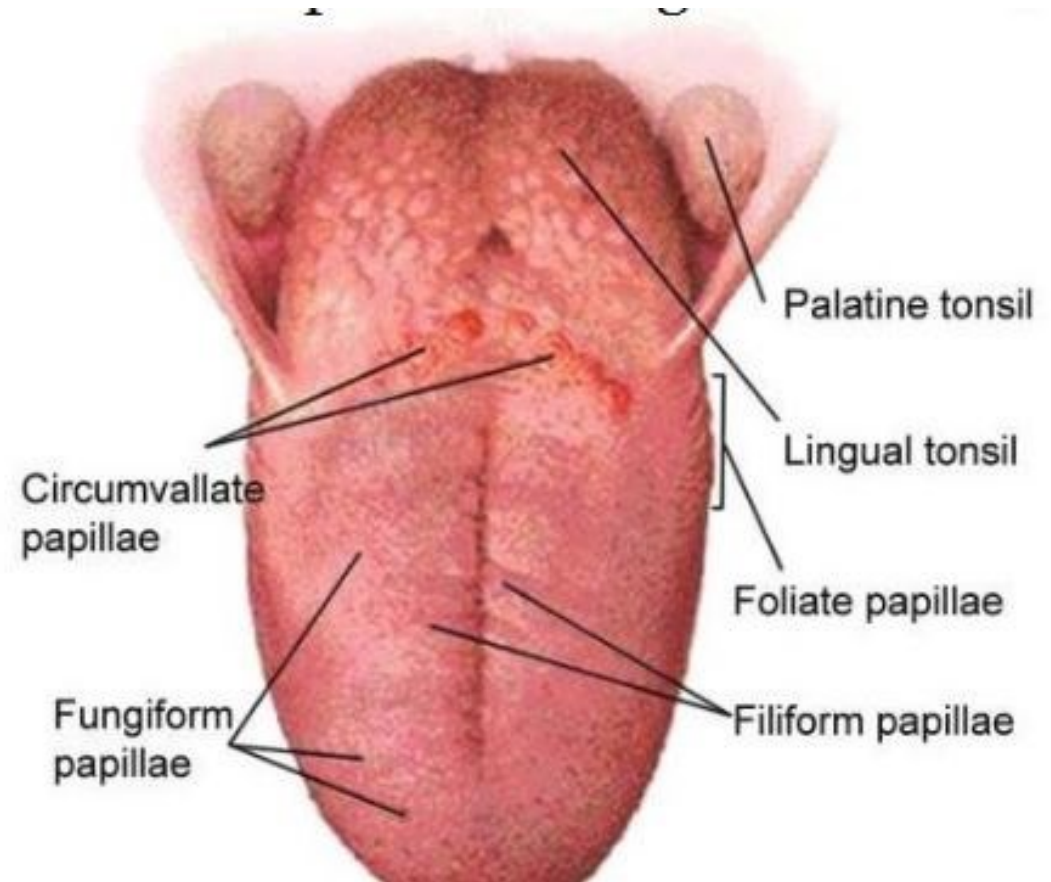
- Chemical senses/*Exteroceptive*
 - Gustation (taste) and Olfaction (smell)
- Chemoreceptors respond to chemicals in aqueous solution
 - Taste – to substances dissolved in saliva
 - Smell – to substances dissolved in fluids of the nasal cavity

Taste – Gustation

- Organs of taste - Taste Buds
 - $> 10,000$ (50 – 70 micron)
 - Most are on *papilla of the tongue (lingual papillae)*
 - Also inside cheeks, palate, pharynx, and epiglottis
- Replaced every 7 - 10 days

Taste Buds

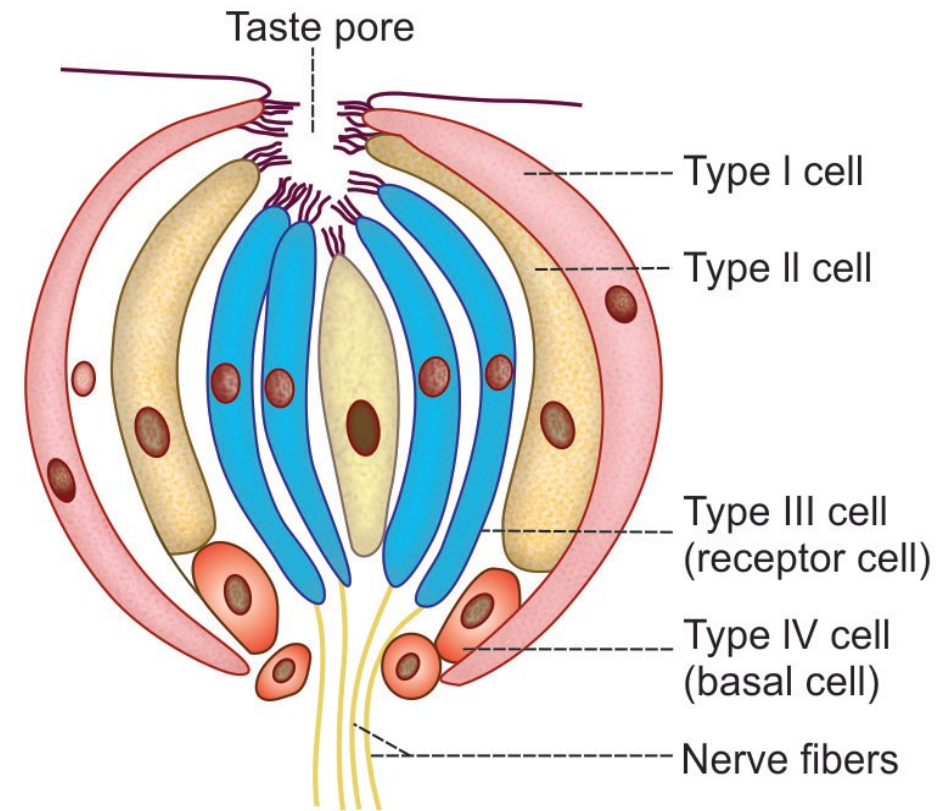
- **Filiform papillae**
 - No/few taste buds
 - important for food texture
- **Foliate papillae**
 - no taste buds
 - weakly developed in humans
- **Fungiform papillae**
 - at tips and sides of tongue
- **Vallate (circumvallate)**
 - at rear of tongue (V shaped)
 - contains 1/2 of all taste buds



Anatomy of a Taste Bud

Each consists of three major cell types

- **Supporting cells (type I)**
 - insulate the receptor
- **Basal cells**
 - stem cells that divide and differentiate into new gustatory cells (every 7 to 10 days)
- **Gustatory cells /taste cells (type III)**
 - have tuft of apical microvilli (**taste hairs**)
 - **receptor surface for taste molecules** —
 - **taste pores** – pit in which the taste hairs project

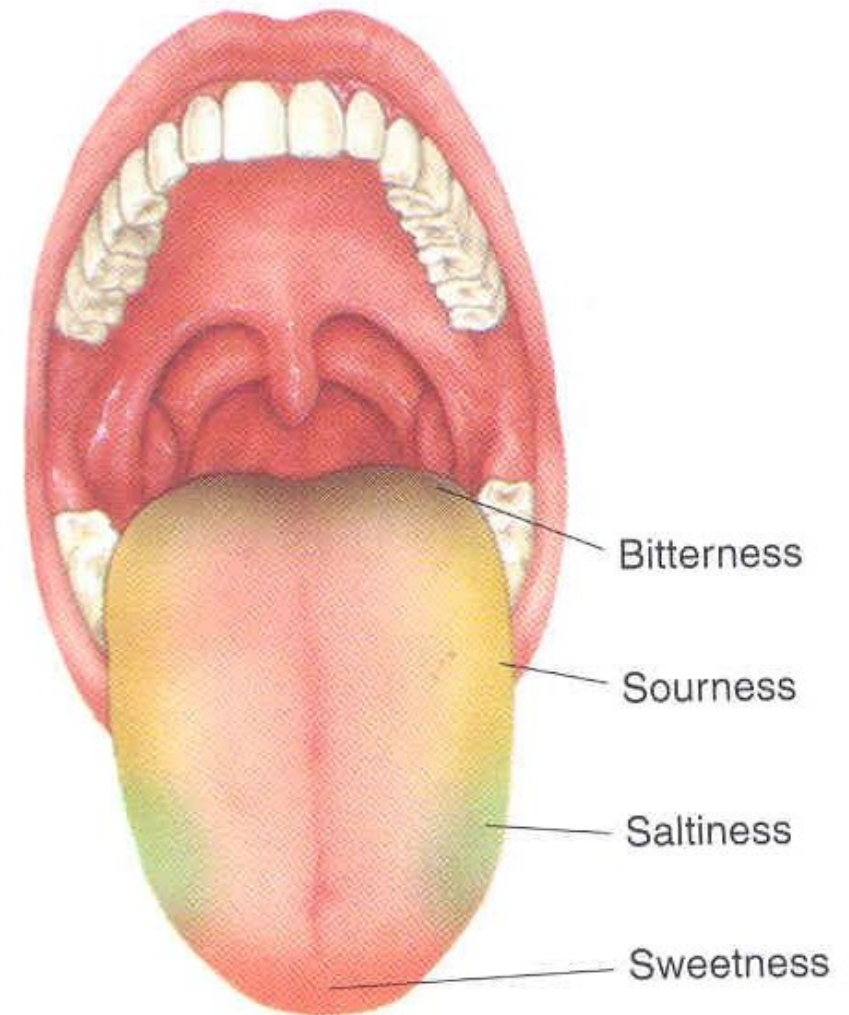


Taste Sensations

- **Five** basic/primary taste sensations
 1. **Sweet** – organic (sugars, saccharin, alcohol, some aminoacids, lead)
 2. **Salt** – metal ions/ionized salts eg. Mainly Na^+
 3. **Sour** – hydrogen ions/acids
 4. **Bitter** – alkaloids (quinine, caffeine, morphine, bile salts, salts of Ca, Mg, NH_4)
 - 1. **Umami** – “beef taste”/delicious ~ Asian food
 - elicited by glutamate, monosodium glutamate.
- Other modalities that affect taste ~ we can perceive 100 variants
- *Temperature (flavor)*
 - *texture*
 - *Pain (ginger)*

Taste Sensations

- Regional differences in taste sensations on tongue
 - *tip is most sensitive to sweet,*
 - *edges to salt and sour*
 - *rear to bitter*



The tongue map: regions of the lowest thresholds for the basic tastes. The center of the tongue lacks taste buds and is relatively insensitive to taste.

When eating a piece of candy, I will use the following to sense

- A) Fungiform papillae
- B) Filiform papillae
- C) Foliate papillae
- D) Circumvallate papillae
- E) All of the above

Physiology of Taste

- In order to be tasted, a chemical:
 - Must be *dissolved in saliva*
 - Must *contact gustatory cells*
- Binding of the food chemical:
 - Depolarizes the taste cell membrane, releasing neurotransmitter
 - Initiates a *generator potential* that elicits an AP

If I have a cold, food may not taste as good to me because

A) The nerve fibrils are not functioning properly

B) Papilla become blocked by mucus and are unable to function

Taste Transduction

Two mechanisms of action

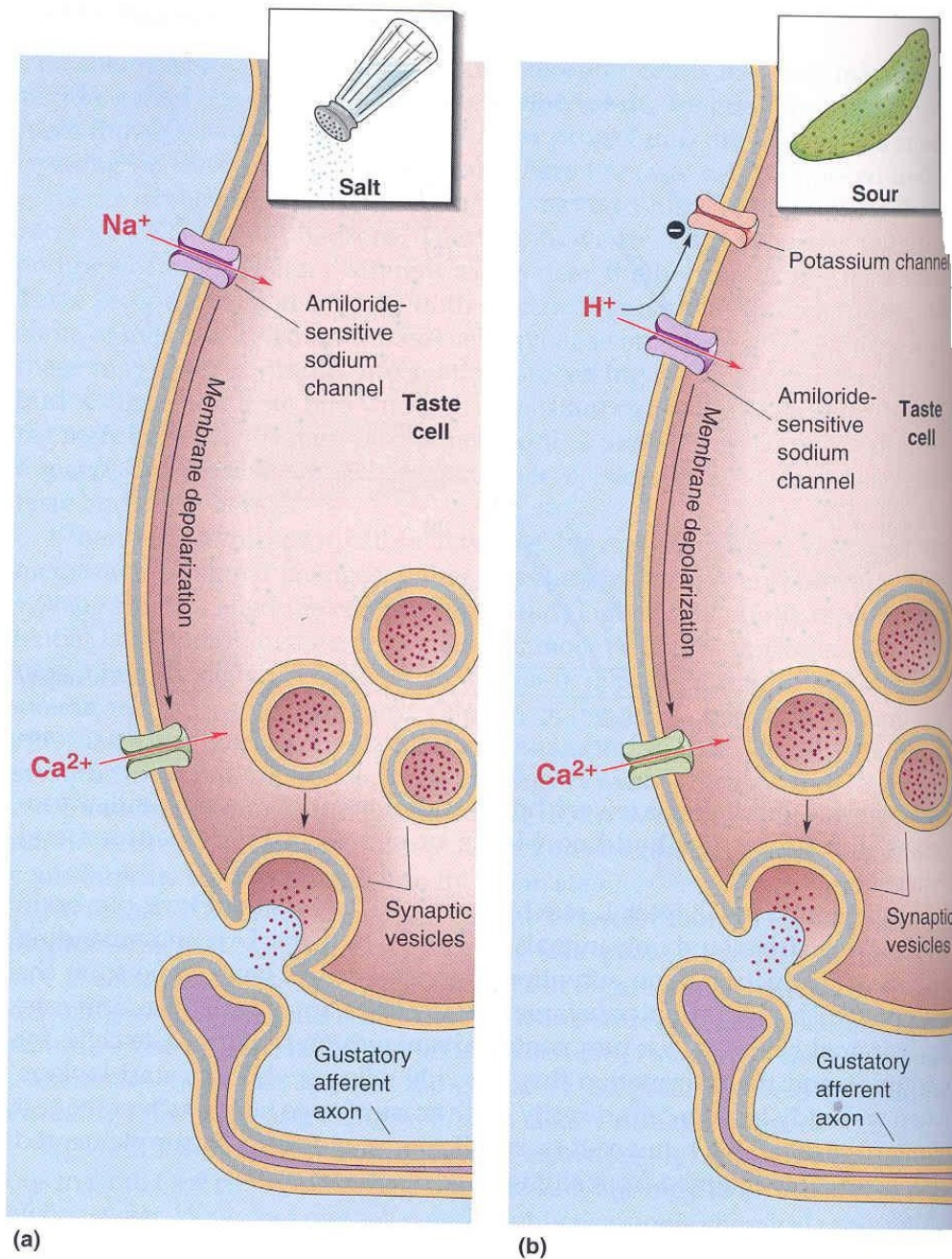
1. Activate 2nd messenger systems

- *Sweet, bitter and Umami* tastes
- sugars, alkaloids, and glutamate bind to receptors which activates *G proteins and second-messenger*

2. Depolarize cells directly

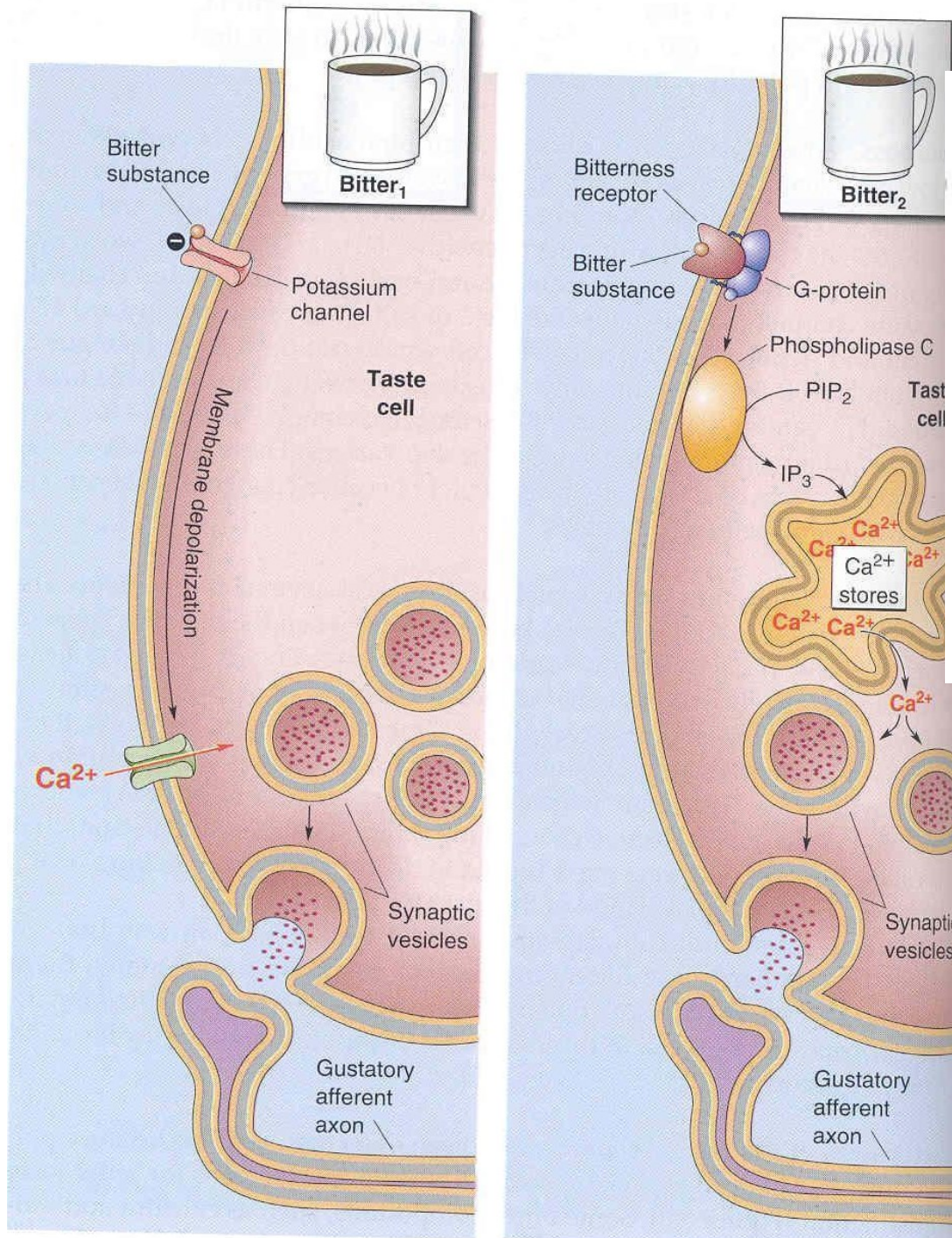
- *Na⁺ influx* in **salty tastes**
- *H⁺* in **sour tastes**

1. Directly pass through ion channels - salt and sour



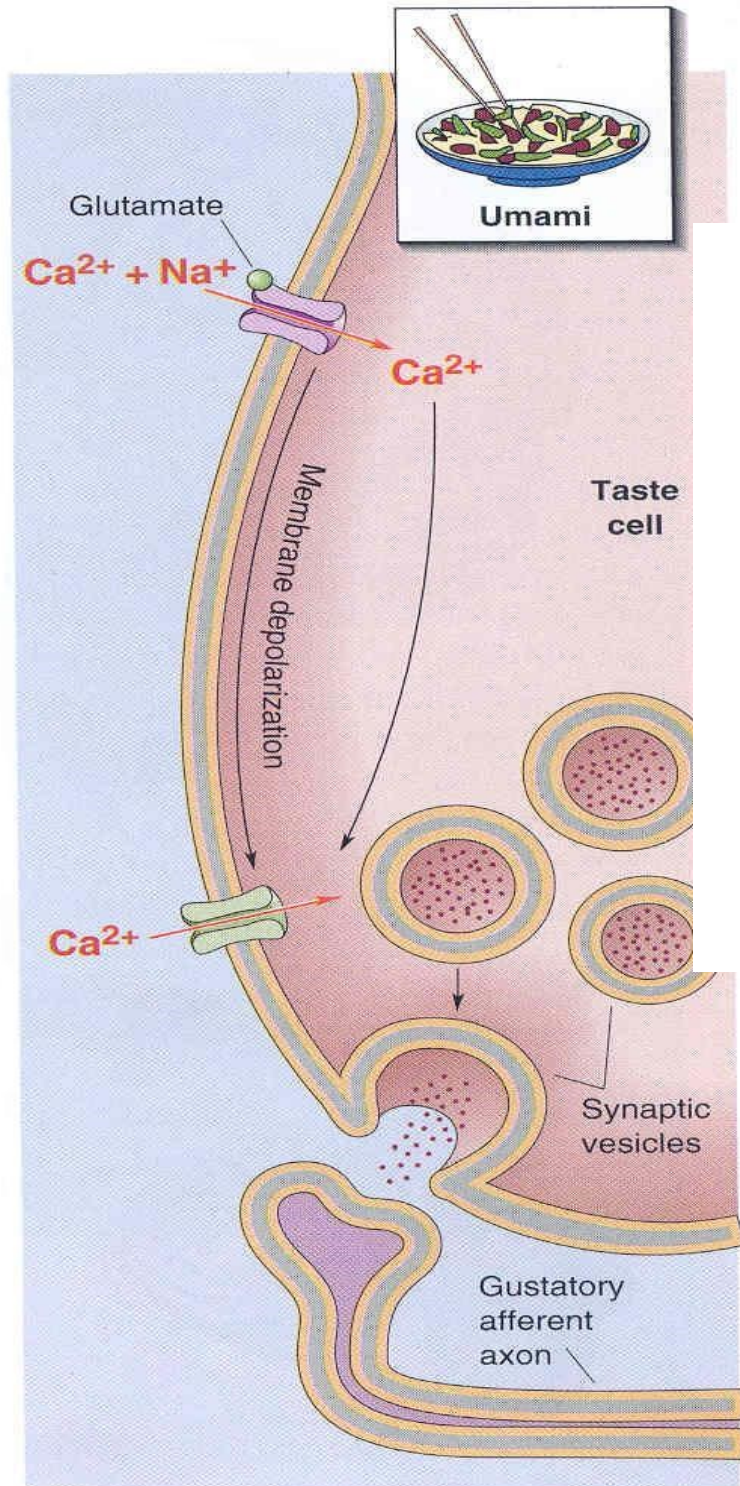
Transduction mechanisms of (a) salt and (b) sour tastants. Tastants can interact directly with ion channels, by either passing through them (Na^+ and H^+) or blocking them (H^+ blocking the potassium channel). Then, the membrane voltage influences calcium channels on the basal membrane, which in turn influence the intracellular $[\text{Ca}^{2+}]$ and transmitter release.

2. Bind to and block ion channels - bitter



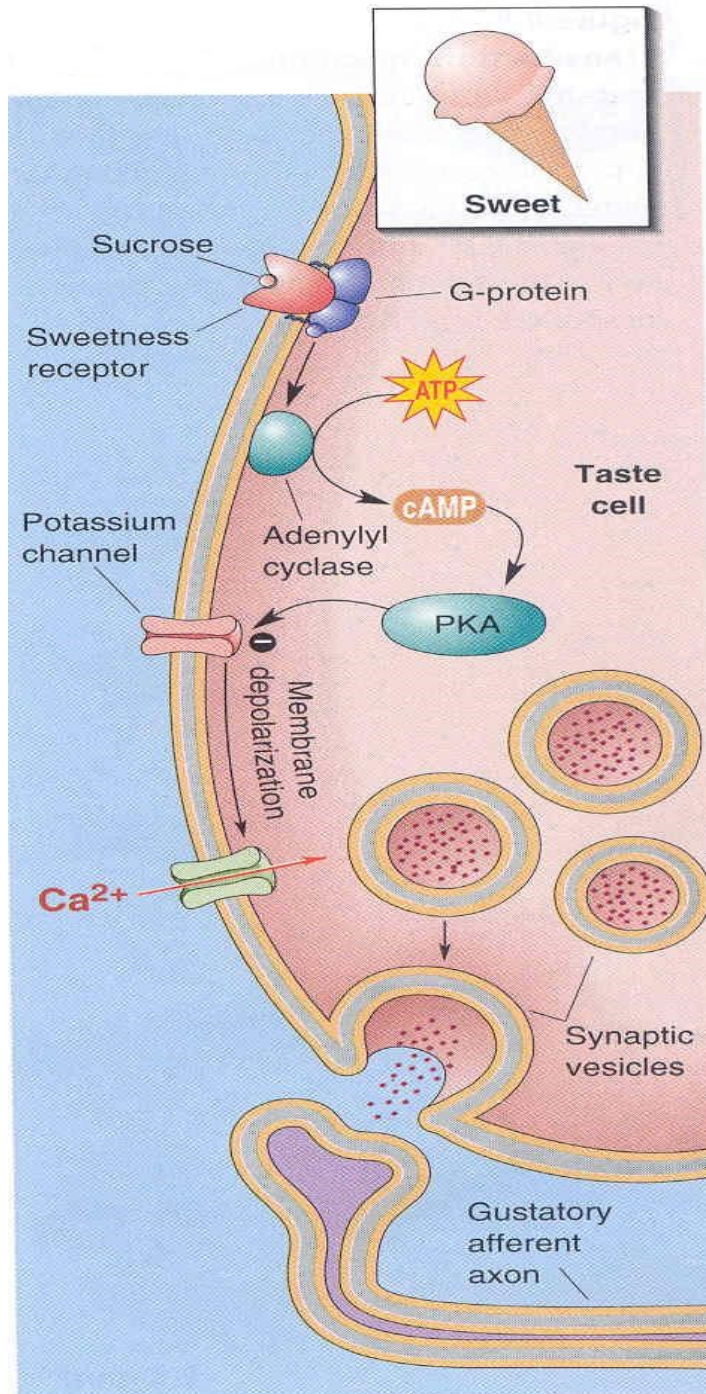
Transduction mechanisms for bitter tastants. Tastants can either block a potassium channel (Bitter₁) or bind directly to a G-protein-coupled membrane receptor (Bitter₂) that triggers IP_3 synthesis and the release of Ca^{2+} from internal storage sites. PIP_2 is the membrane lipid phosphatidylinositol-4,5-bisphosphate.

3. Bind to and open ion channels - some aa



Transduction mechanisms for umami tastants (glutamate). Some amino acid tastants can bind to a cation-permeable channel, leading to a change in membrane current and potential, hence the direct entry of Ca^{2+} .

4. Bind to membrane receptors that activate second messenger systems



Transduction mechanisms for sweet tastants. Tastants bind directly to G-protein-coupled membrane receptors and trigger the synthesis of cAMP, which leads to the blockade of a potassium channel, depolarization, Ca^{2+} entry, and transmitter release.

Gustatory Pathway

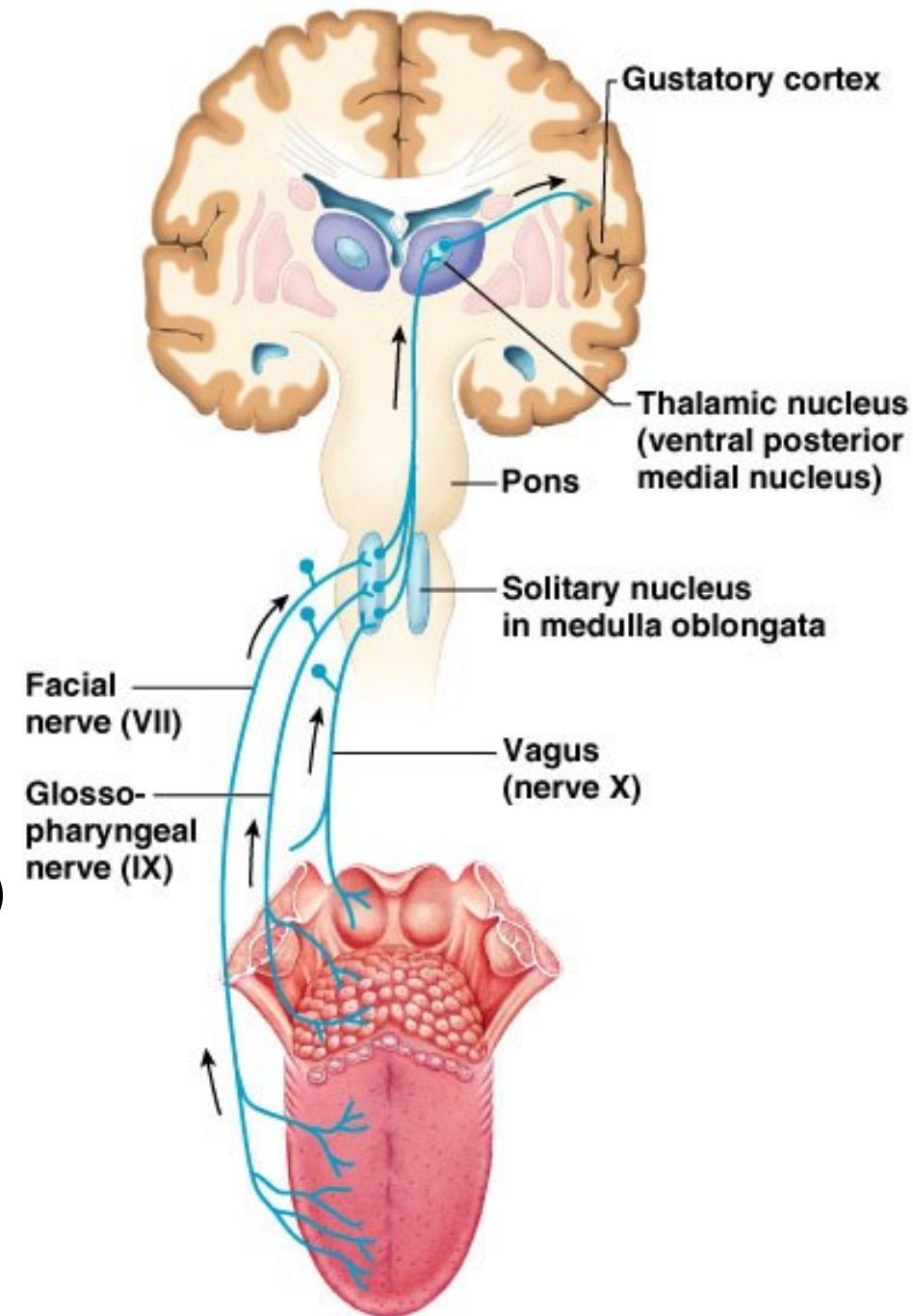
■ First order neurons

- Facial nerves (VII)
 - anterior two-thirds of tongue
- Glossopharyngeal nerve (IX)
 - posterior 1/3rd of tongue
- vagus nerve (X)
 - epiglottis and the lower pharynx

■ All fibers reach **solitary nucleus (NTS)**

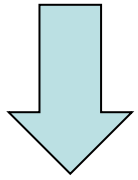


■ Thalamus (VPM)



Gustatory Pathway

- Thalamus



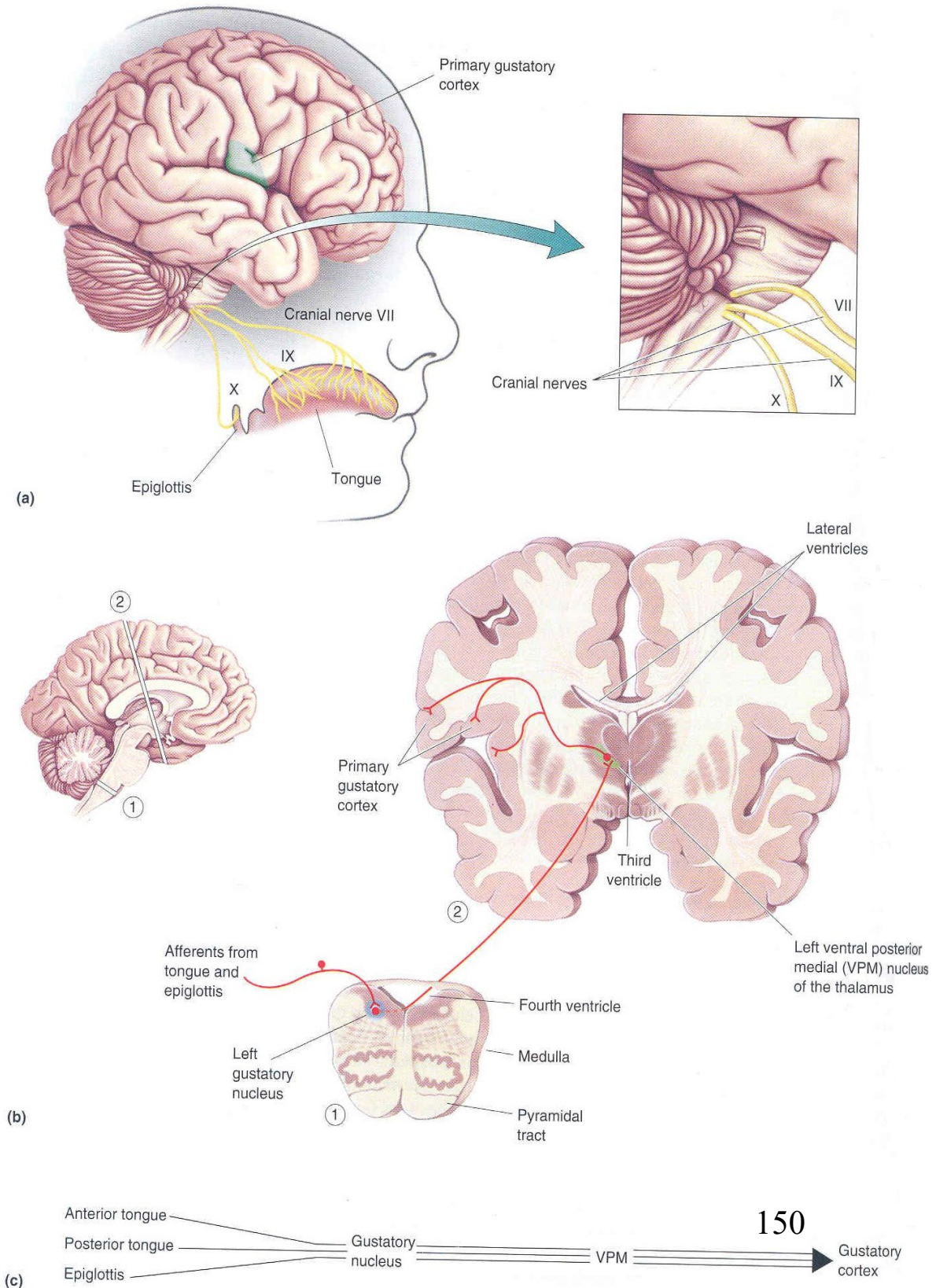
- Gustatory cortex

- parietal lobes

- Hypothalamus and limbic system

- control autonomic reflexes

**Pathways are ipsilateral
= uncrossed pathway**

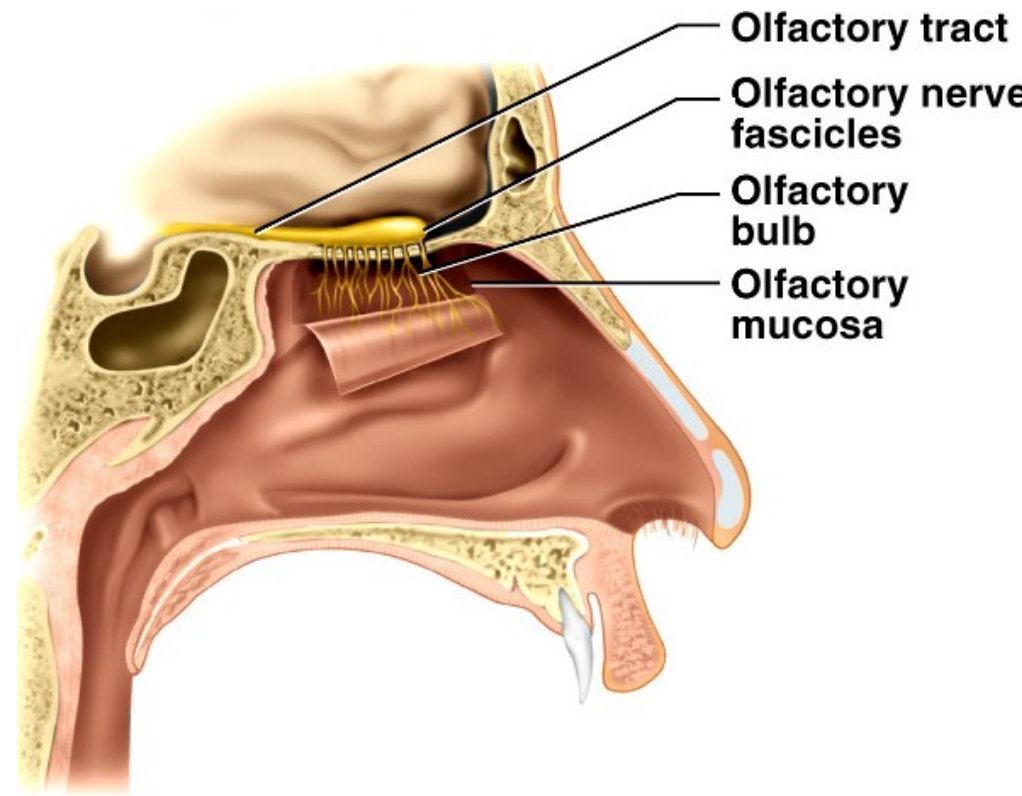


Olfaction –The sense of smell

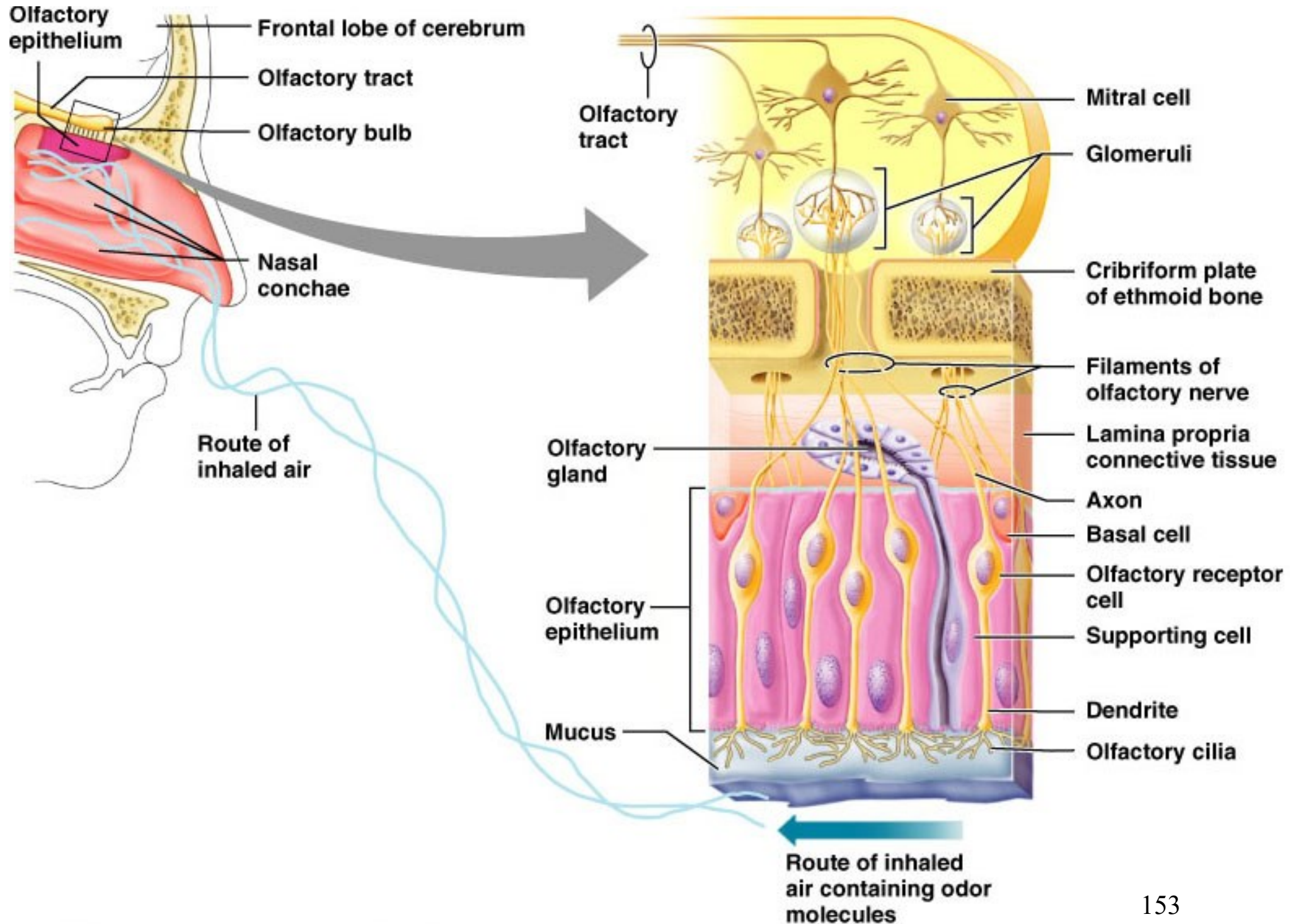
- Significances
 - *Contributes to our emotional life*
 - *Call up memories.*
 - *Helps us avoid consuming spoiled food*
 - *Lets us detect dangerous situations*

Olfaction –The sense of smell

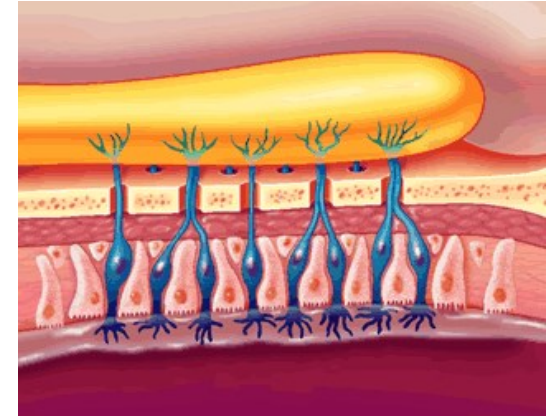
- Olfactory chemoreceptors
 - located in the **olfactory mucosa / olfactory epithelium**
 - *superior concha*
 - *nasal septum*
 - *roof of nasal cavity*
 - Area - 5 cm²
 - contains 10 to 20 million cells
 - Short life span – 40 days
 - Olfactory neurons
 - supporting cells
 - basal stem cells



Olfaction –The sense of smell



Olfaction –The sense of smell



1. Olfactory receptor neurons

- Site of Olfactory transduction
- Replaced every 4 -8 wks by basal cells
 - *The only neuron that can be replaced in adult.*
- Head bears 10 – 20 non motile cilia = *olfactory hairs*
 - Have binding sites for *odorant* molecules

2. Basal cells – source of new receptor neurons

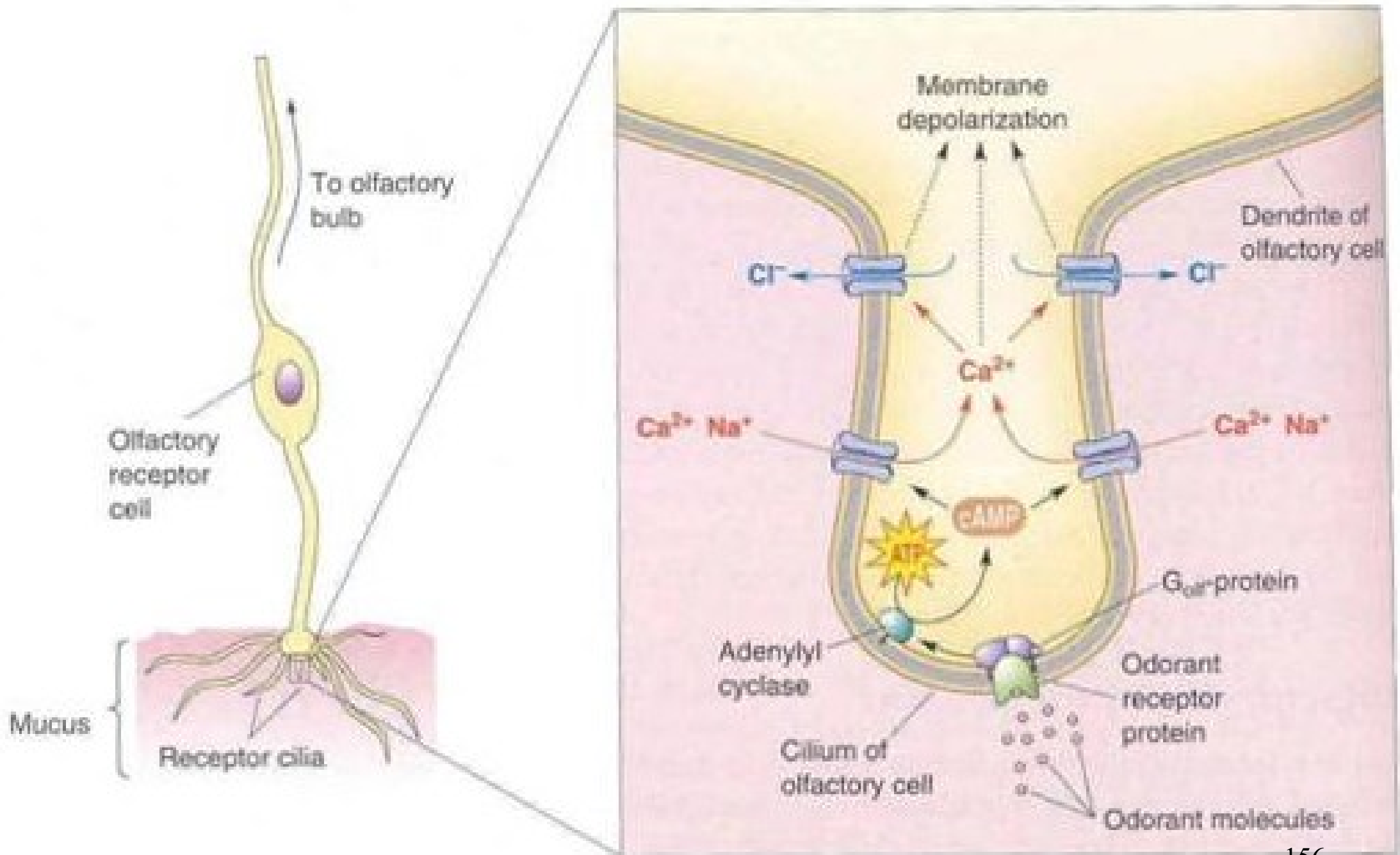
3. Supporting cells

- similar to glia and produce mucus

Physiology of Smell

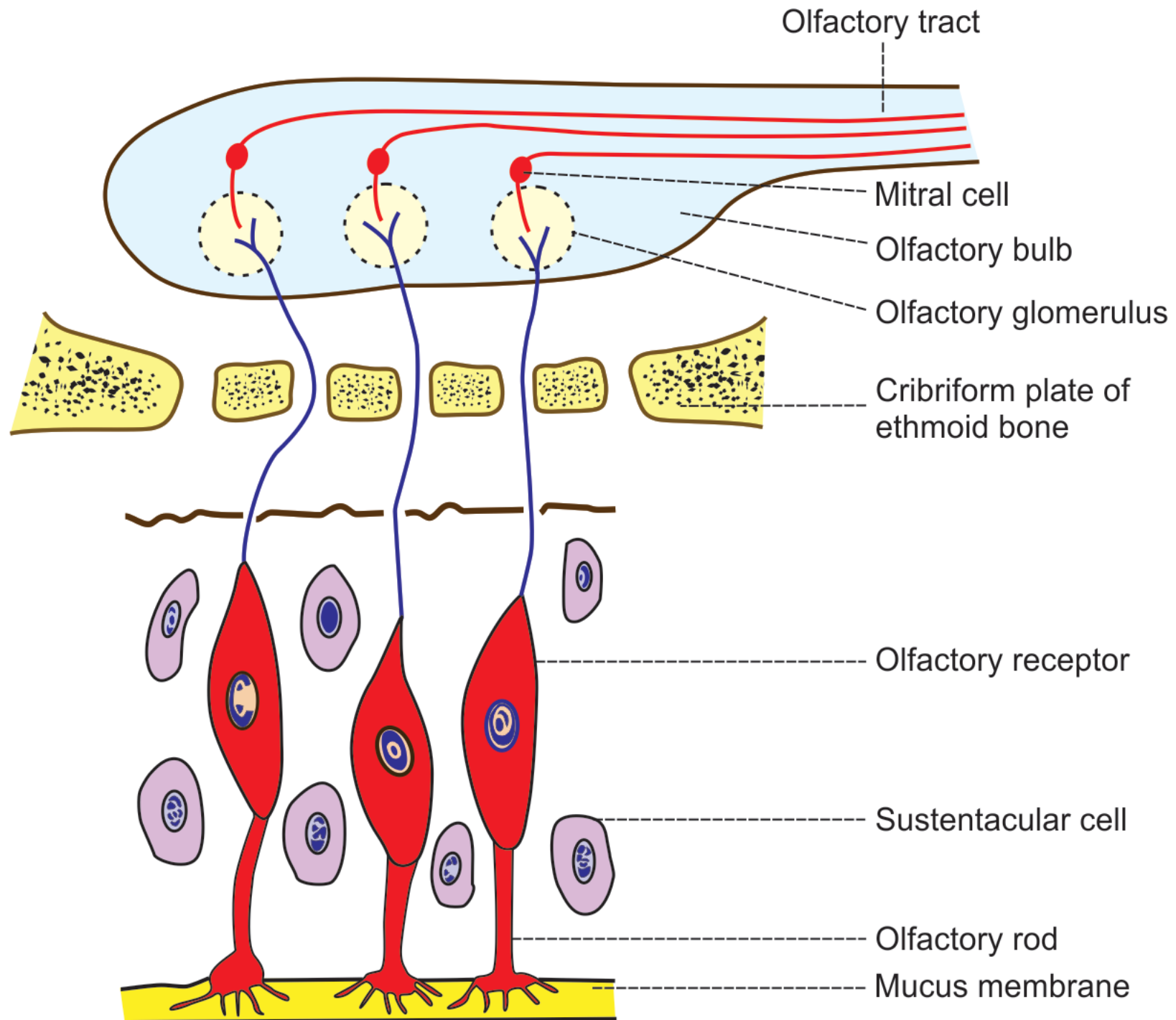
- *2000 to 4000* odors distinguished
- Some odorants act on nociceptors
- Odorants bind with receptor proteins on the hair cells of olfactory neurons/Bipolar cell
- Activation of G protein mechanism - uses *cAMP*
 - cAMP acts directly on PM/ *Na⁺, K⁺ & Ca²⁺ channels*
 - *Ca⁺⁺ influx causes Ca⁺⁺-activated Cl⁻ channels/efflux*
- *Depolarization* of the receptor membrane that then triggers an *action potential* (RMP ~ -55 mv to -30 mv)

Olfactory Transduction Process



Olfactory Pathway

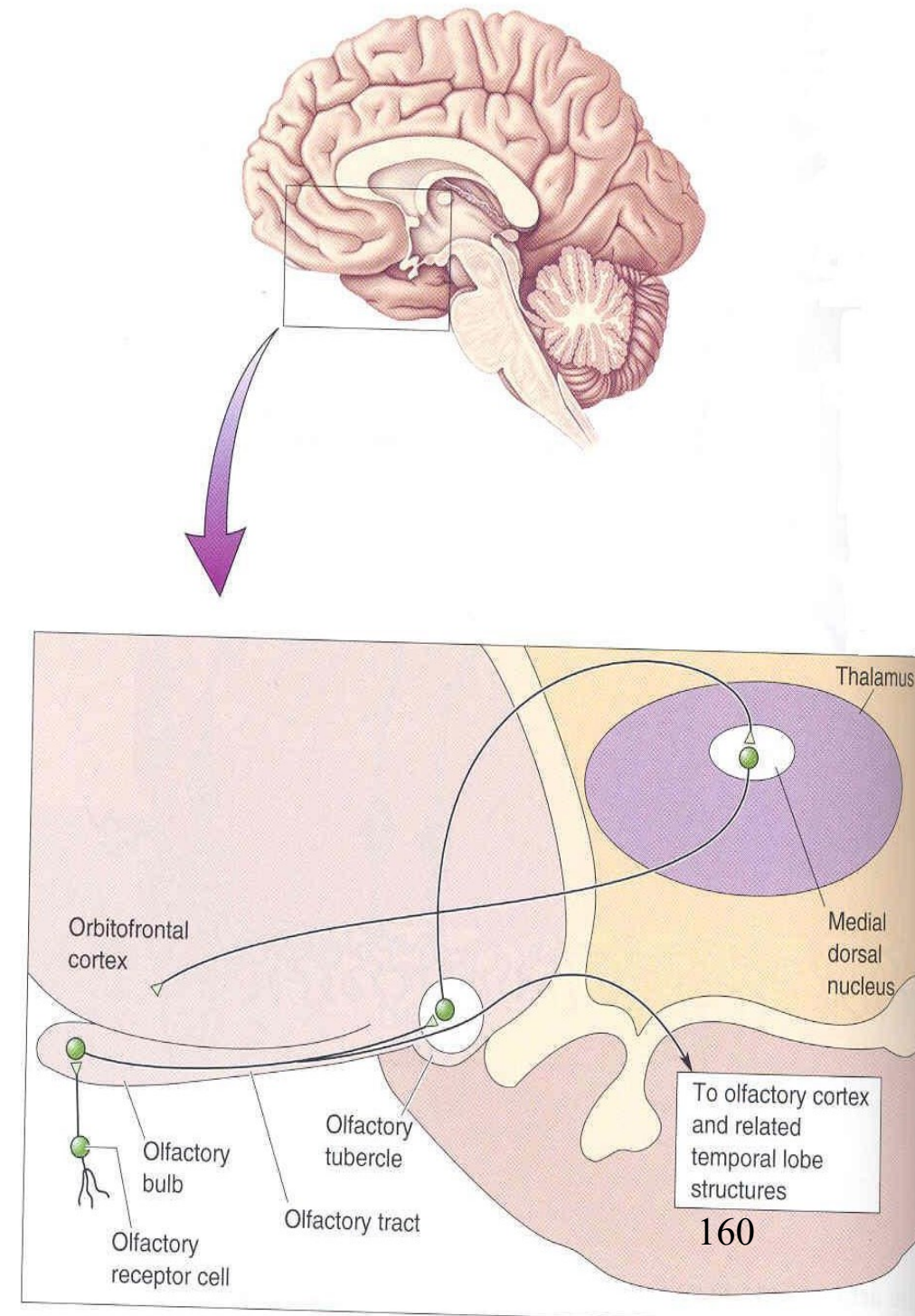
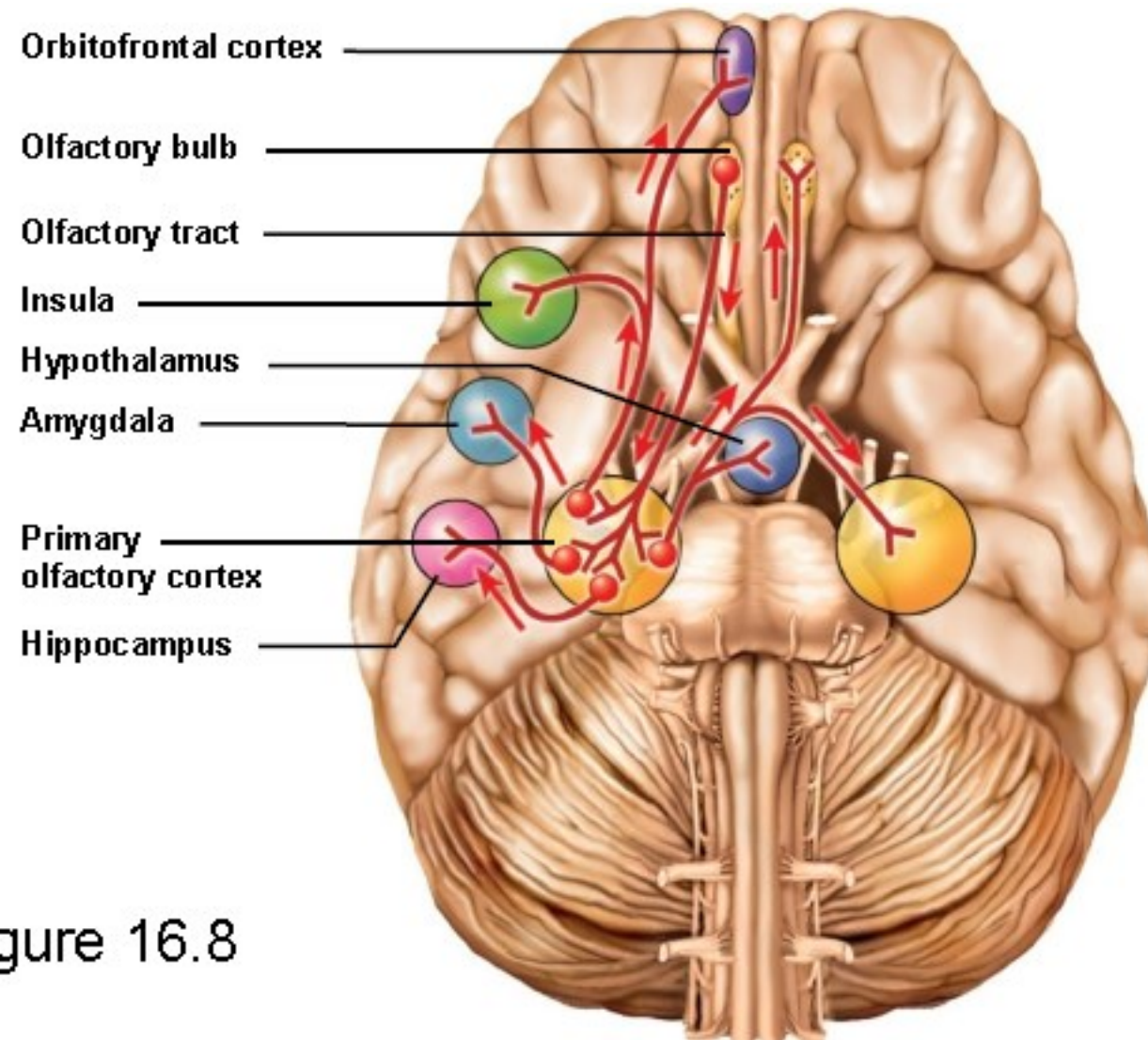
- Axons of each olfactory cell collect into small fascicles (**CN I**) and leave cranial cavity through the **cribriform foramina** in the ethmoid bone.
- Olfactory receptor cells synapse with 2nd order (mitral and Tufted cells) in the **Glomeruli (in the olfactory bulb)**
 - *Mitral cells process odor signals*
 - *Each glomeruli dedicated to single odor*
- Impulses pass through **olfactory tracts in to the brain –**
NB: Newer pathways- passes through the thalamus/DMNT.
(conscious analysis of odor)



Olfactory Pathway

- Olfactory tract reach directly *primary olfactory cortex*
 - inferior surface of the temporal lobe (limbic part)
- **Secondary destinations**
 - *hippocampus, amygdala, hypothalamus, insula, and (via thalamus) orbitofrontal cortex*
 - *Conscious discrimination (orbitofrontal c., better @ right cortex)*
 - *integrate smell with taste*
 - *perceive flavor*
 - *evoke memories and emotional responses (amygdala)*
 - *visceral reactions*
 - *Olfactory memories (via entorhinal c.)*

Olfactory Pathway



Sense of Smell

- *Anosmias* - the absence of the sense of smell
 - Causes
 - head injuries/tumors that tear the olfactory nerves,
 - inflammation of nasal cavity
 - aging.

One-third of all cases of chemical olfactory sense loss is due to a deficiency of the mineral **zinc**.

Summary

<u>Modality</u>	<u>Stimulus</u>	<u>Receptor Types</u>	<u>Receptors</u>
1. Vision	Light	Photoreceptor	Rods, Cones
2. Audition	Sound	Mechanoreceptor	Hair cells (Cochlear)
3. Balance	Head Motion	Mechanoreceptor	Hair cells (semicircular canals)
4. Taste	Chemical	Chemoreceptor	Taste Buds
5. Smell	Chemical	Chemoreceptor	Olfactory sensory N

All the best